

A Rigorous Transition Course on Mathematical Induction

Bridging HKDSE, HKALE, and University Mathematics from First Principles

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Abstract

This series of lecture notes is designed for students with minimal algebraic backgrounds transitioning to advanced, mathematical reasoning. Starting from the absolute basics of human counting, logical systems, and simple language of sets, these notes build the foundational architecture required to comprehend and apply mathematical induction across diverse disciplines.

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1 Preliminaries

1.1 The Paper Work of Mathematics: An Introduction to Axiomatic Systems

To begin our journey, we must examine how mathematical systems are built. Students often view mathematics as a collection of formulas to memorize. However, mathematics behaves more like a game (such as chess). In chess, we do not prove why a Rook moves in straight lines; that is simply a rule we agree on before the game starts.

In mathematics, these fundamental rules are called **axioms**. From these axioms, we build definitions, and use logical deduction to prove larger, undeniable truths called **theorems**.

Definition 1.1.1 (Axiom, Definition, Theorem, Proposition, Lemma). We classify the paper work of modern mathematics as follows:

- An **Axiom** is a statement that is accepted as true without any proof. It serves as an starting rule.
- A **Definition** is an agreement on the precise meaning of a word, symbol, or concept. It does not require proof because it is simply a name we give to something.
- A **Theorem** is a major, highly important mathematical statement that has been proven to be true using logical rules starting from axioms or definitions.
- A **Proposition** is a proven mathematical statement of interest, though typically less central or groundbreaking than a theorem.
- A **Lemma** is a minor, preliminary theorem proven mainly to act as a stepping stone to help prove a larger, primary theorem.

Example 1.1.1 (Practical/Numerical Application). Let us define what an *even number* is, state a proposition about it, and verify it with a simple calculation.

Solution. First, we state our **Definition**: An integer n is *even* if it can be written in the form $n = 2 \times k$ for some integer k . Next, we state our **Proposition**: The sum of any two even integers is also an even integer.

Let us test this with the numbers 6 and 8.

$$6 = 2 \times 3 \quad (\text{so } k = 3 \text{ is the integer})$$

$$8 = 2 \times 4 \quad (\text{so } k = 4 \text{ is the integer})$$

Adding them together gives $6 + 8 = 14$. We can write 14 as 2×7 . Since 7 is an integer, 14 perfectly fits our definition of an even integer. This arithmetic demonstrates the consistency of our definitions. $\square$

Example 1.1.2 (Abstract Applied Mathematics Application). In computers, calculations are limited by memory. Let us define a *modulo-3 arithmetic system* where we only care about the remainder when a number is divided by 3. Show how a basic operation works.

Solution. **Definition**: Let a and b be integers. We write $a \equiv b \pmod{3}$ if the difference $a - b$ is perfectly divisible by 3 (meaning $a - b = 3k$ for some integer k).

Let us verify if $14 \equiv 2 \pmod{3}$. We compute the difference:

$$14 - 2 = 12$$

Since $12 = 3 \times 4$, and 4 is an integer, the difference is indeed a multiple of 3. Thus, the statement $14 \equiv 2 \pmod{3}$ is mathematically correct according to our definition. $\square$

Example 1.1.3 (Generalization to More Variables). Suppose we have a list of n even integers $x_1, x_2, \dots, x_n$. Generalize the proposition about the sum of two even integers to the sum of n even integers.

Solution. Theorem: Let $x_1, x_2, \dots, x_n$ be a set of n even integers. Then their sum $S = x_1 + x_2 + \dots + x_n$ is an even integer.

To understand this, we apply our definition of even numbers to every single variable in the list. Because $x_1, x_2, \dots, x_n$ are even, there must exist integers $k_1, k_2, \dots, k_n$ such that:

$$x_1 = 2k_1, \quad x_2 = 2k_2, \quad \dots, \quad x_n = 2k_n$$

Now, we write the sum:

$$S = 2k_1 + 2k_2 + \dots + 2k_n$$

Factoring out the common number 2 from every term (using basic algebra):

$$S = 2(k_1 + k_2 + \dots + k_n)$$

Because the sum of integers is always another integer, we can let $M = k_1 + k_2 + \dots + k_n$. Thus, $S = 2M$, which matches the exact definition of an even integer. $\square$

Example 1.1.4 (Pure Abstract Discussion). In university abstract mathematics, we define a structure called a *Group*. Let us look at how these definitions are used to establish properties of abstract symbols without assuming any numbers.

Solution. Definition: A Group $(G, *)$ is a set of elements G with an operation $*$ satisfying four axioms:

- (1) Closure: For all $a, b \in G$, $a * b \in G$.
- (2) Associativity: For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
- (3) Identity: There exists an element $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.
- (4) Inverse: For every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

We do not prove these axioms; they are the rules of the game. Using these rules, we can prove the **Theorem:** The identity element e in a group is completely unique. $\square$

Theorem 1.1.1. Let a, b , and c be integers. If a divides b (written as $a \mid b$) and b divides c ($b \mid c$), then a divides c ($a \mid c$).

Example 1.1.5 (Bridging Verification for Theorem 1.1). Let us test this theorem with concrete integers: let $a = 3$, $b = 12$, and $c = 60$.

- Since $12 = 3 \times 4$, we see that $3 \mid 12$ is True.
- Since $60 = 12 \times 5$, we see that $12 \mid 60$ is True.

According to our theorem, 3 must divide 60. Let us check: $60 = 3 \times 20$. Since 20 is an integer, $3 \mid 60$ is verified!

Proof of Theorem 1.1. By definition of divisibility, since $a \mid b$, there exists an integer k_1 such that $b = a \cdot k_1$. Similarly, since $b \mid c$, there exists an integer k_2 such that $c = b \cdot k_2$. Substituting the first equation into the second yields:

$$c = (a \cdot k_1) \cdot k_2 = a \cdot (k_1 \cdot k_2)$$

Since the set of integers is closed under multiplication, let $m = k_1 \cdot k_2 \in \mathbb{Z}$. Thus, $c = a \cdot m$, which proves by definition that $a \mid c$. $\square$

Exercises 1.1

- (1) Explain the fundamental difference between an *Axiom* and a *Theorem*.
- (2) Define what an *odd number* is using algebraic notation with a variable k .
- (3) Using your definition from Exercise 2, prove that the number 17 is an odd number.
- (4) If n is an even integer, show that n^2 must also be an even integer.
- (5) If n is an odd integer, show that n^2 must also be an odd integer.
- (6) The commutative law of addition states that $a + b = b + a$. Is this statement an axiom or a theorem in basic arithmetic? Explain.
- (7) If we define the "divides" relation as: $x \mid y$ if and only if $y = x \cdot c$ for some integer c . Determine if $3 \mid 12$ is true.
- (8) Using the definition in Exercise 7, evaluate if $5 \mid 12$ is true. Show your work.
- (9) Let a be an integer. Explain why $1 \mid a$ is always true using the definition of division.
- (10) Let a be a non-zero integer. Explain why $a \mid 0$ is always true.
- (11) Write down a formal definition of a "prime number" in plain English.
- (12) Determine whether 1 is considered a prime number under standard definitions. Why or why not?
- (13) A student claims that "the product of two odd integers is even." Find a numerical counterexample to show this claim is false.
- (14) Based on Exercise 13, state a correct proposition regarding the product of two odd integers.
- (15) Let the operation $\oplus$ be defined as $x \oplus y = x + y - 1$. Calculate $3 \oplus 4$.
- (16) Using the definition in Exercise 15, is $x \oplus y = y \oplus x$ always true? Show your steps.
- (17) Let S be the sum of 2 odd integers. Express S algebraically and simplify to determine if it is even or odd.
- (18) Why does a mathematical definition not require any proof?
- (19) What is the role of a *Lemma* in mathematical papers?
- (20) Give a real-world analogy of how an Axiomatic System works.

1.2 The Universal and Existential Quantifier

When we want to write mathematical statements precisely, we must be absolutely clear about how many objects satisfy our conditions. For example, the statement "numbers are positive" is confusing. Does it mean *all* numbers are positive, or just *some* numbers? To resolve this, mathematicians use symbols called **quantifiers**.

Definition 1.2.1 (Universal Quantifier ($\forall$)). The symbol $\forall$ is read as "for all", "for every", or "for each". A statement of the form:

$$\forall x \in S, \quad P(x)$$

asserts that the property $P(x)$ is true for *every single element* x belonging to the set S .

Definition 1.2.2 (Existential Quantifier ($\exists$)). The symbol $\exists$ is read as "there exists", "for some", or "there is at least one". A statement of the form:

$$\exists x \in S \quad \text{such that} \quad P(x)$$

asserts that there is *at least one element* x belonging to the set S for which the property $P(x)$ is true.

Example 1.2.1 (Practical/Numerical Application). Translate the following symbolic statement into English and evaluate whether it is True or False:

$$\forall x \in \{1, 2, 3, 4\}, \quad x + 5 > 5$$

Solution. Translation: "For all elements x in the set $\{1, 2, 3, 4\}$, the number $x + 5$ is greater than 5."

Let us test every single element in our set:

- If $x = 1$: $1 + 5 = 6$. Is $6 > 5$? Yes (True).
- If $x = 2$: $2 + 5 = 7$. Is $7 > 5$? Yes (True).
- If $x = 3$: $3 + 5 = 8$. Is $8 > 5$? Yes (True).
- If $x = 4$: $4 + 5 = 9$. Is $9 > 5$? Yes (True).

Since the condition is true for every single element in our specified set, the entire statement is ****True****. □

Example 1.2.2 (Abstract Applied Mathematics Application). In computer engineering, a safety diagnostic system scans a set of N sensors, $S = \{s_1, s_2, \dots, s_N\}$. Let $E(s)$ mean "sensor s reports an error." Write down the condition that "the entire computer system halts if at least one sensor reports an error" using quantifiers.

Solution. We want to state that there exists at least one sensor experiencing an error. Using the existential quantifier, we write:

$$\exists s \in S \quad \text{such that} \quad E(s)$$

If this statement is true, the computer triggers its emergency shutdown protocol. This shows how quantifiers help write clean logic criteria for coding. □

Example 1.2.3 (Generalization to More Variables). Translate and evaluate the following multi-variable statement where x and y are chosen from different lists:

$$\forall x \in \{1, 2\}, \quad \exists y \in \{2, 3, 4\} \quad \text{such that} \quad y = 2 \times x$$

Solution. Translation: "For every number x in $\{1, 2\}$, there exists some number y in $\{2, 3, 4\}$ such that y is exactly equal to 2 times x ."

Let us analyze this step-by-step for each x :

- Let $x = 1$. We must search the list $\{2, 3, 4\}$ for a y that satisfies $y = 2 \times (1) = 2$. We find that 2 is indeed in the list.
- Let $x = 2$. We must search the list $\{2, 3, 4\}$ for a y that satisfies $y = 2 \times (2) = 4$. We find that 4 is indeed in the list.

Since we were successfully able to find a matching y for every choice of x , the statement is ****True****. $\square$

Example 1.2.4 (Pure Abstract Discussion). In mathematical analysis, we define a property called the *Archimedean Property* of real numbers. Let us write this statement formally using a combination of quantifiers.

Solution. The property states that no matter how small a positive real number ϵ we choose, we can always find a sufficiently large positive integer n such that $\frac{1}{n}$ is smaller than ϵ .

Formally, letting $\mathbb{R}^+$ be the set of positive real numbers and $\mathbb{N}$ be the set of positive integers, we write:

$$\forall \epsilon \in \mathbb{R}^+, \quad \exists n \in \mathbb{N} \quad \text{such that} \quad \frac{1}{n} < \epsilon$$

This combination of nested quantifiers ($\forall$ followed by $\exists$) is the standard way mathematicians describe continuous limits. $\square$

Proposition 1.2.1. *Let $\mathbb{R}$ represent the set of all real numbers. The statement:*

$$\exists y \in \mathbb{R} \quad \text{such that} \quad \forall x \in \mathbb{R}, \quad x + y = x$$

asserts the existence of an additive identity element (which we know is 0).

Example 1.2.5 (Bridging Verification for Proposition 1.2). We are asked to find a single magic number y that works for **every** real number x . Let us test $y = 0$:

- If $x = 5$, then $5 + 0 = 5$.
- If $x = -3.14$, then $-3.14 + 0 = -3.14$.

Since $x + 0 = x$ holds for any choice of x , the existential statement is True with $y = 0$.

Proof of Proposition 1.2. Let $y = 0$. By the fundamental arithmetic axioms of real numbers, $0 \in \mathbb{R}$. For any arbitrary real number $x \in \mathbb{R}$, the property of the additive identity guarantees that $x + 0 = x$. Therefore, there exists at least one element $y \in \mathbb{R}$ (specifically, $y = 0$) satisfying $\forall x \in \mathbb{R}, x + y = x$. $\square$

Exercises 1.2

- (1) Translate into plain English: $\forall x \in \{2, 4, 6, 8\}, x$ is divisible by 2.
- (2) Translate into plain English: $\exists x \in \{5, 10, 15\}$ such that $x > 12$.
- (3) Determine the truth value of: $\forall x \in \{-1, 0, 1, 2\}, x^2 \geq 0$. Show your evaluations.
- (4) Determine the truth value of: $\exists x \in \{-2, -1, 0\}, x > 0$. Explain.
- (5) Write using mathematical symbols: "For every real number x , its square is non-negative." (Use $\mathbb{R}$ for real numbers).
- (6) Write using mathematical symbols: "There exists an integer x such that $x + 3 = 7$." (Use $\mathbb{Z}$ for integers).
- (7) Evaluate the truth value of: $\forall x \in \{1, 2\}, \forall y \in \{3, 4\}, x + y \leq 6$.
- (8) Evaluate the truth value of: $\exists x \in \{1, 2\}, \exists y \in \{3, 4\}$ such that $x \cdot y = 6$.

- (9) What is the logical negation of a "for all" ($\forall$) statement? Explain using an everyday example.
- (10) What is the logical negation of a "there exists" ($\exists$) statement?
- (11) Negate the statement: $\forall x \in \mathbb{Z}$, x is even.
- (12) Negate the statement: $\exists x \in \mathbb{R}$ such that $x^2 = -1$.
- (13) Let $S = \{1, 3, 5\}$. Is the statement $\forall x \in S, \exists y \in S$ such that $x + y = 6$ True or False? Prove your answer by testing all cases.
- (14) Let $S = \{1, 2, 3\}$. Is the statement $\exists y \in S$ such that $\forall x \in S, y \geq x$ True or False? Explain.
- (15) Analyze the difference in meaning between:

$$\forall x \in \mathbb{Z}, \quad \exists y \in \mathbb{Z} \quad \text{such that} \quad y > x$$

and

$$\exists y \in \mathbb{Z} \quad \text{such that} \quad \forall x \in \mathbb{Z}, \quad y > x$$

- (16) Evaluate the truth value of both statements in Exercise 15.
- (17) Write the formal definition of an odd integer using the existential quantifier ($\exists$).
- (18) Express the statement "There is no smallest positive real number" using symbolic quantifiers.
- (19) Find a counterexample to show that the following statement is False: $\forall x \in \mathbb{R}, x^2 > x$.
- (20) Why are quantifiers essential when translating colloquial statements into mathematical proofs?

1.3 The Mathematical Logic

In everyday language, we often connect simple assertions to make more complicated claims. For example, "It is raining **and** my shoes are wet." In mathematics, we must analyze such connections with absolute, numerical certainty.

To help students who are weak at algebra, we represent statements as simple switches that can either be "ON" (True, denoted by T) or "OFF" (False, denoted by F). We use a grid called a **truth table** to track every possible combination of these switches.

Definition 1.3.1 (AND ($\wedge$), OR ($\vee$), NOT ($\neg$)). Let P and Q represent mathematical statements.

- **Conjunction** ($P \wedge Q$, read as " P AND Q "): The compound statement is True if and only if **both** P and Q are individually True.
- **Disjunction** ($P \vee Q$, read as " P OR Q "): The compound statement is True if **at least one** of P or Q is True.
- **Negation** ($\neg P$, read as "NOT P "): Reverses the truth value of statement P . If P is True, then $\neg P$ is False; if P is False, then $\neg P$ is True.

Example 1.3.1 (Practical/Numerical Application). Let P be the numerical statement " $10 > 7$ " and Q be " $4 + 4 = 9$ ". Determine the individual truth value of P and Q , and evaluate the compound statement $P \wedge \neg Q$.

Solution. First, we look at each basic statement:

- P is " $10 > 7$ ". This is clearly True (T).
- Q is " $4 + 4 = 9$ ". Since $4 + 4 = 8$, this statement is False (F).

Next, we evaluate $\neg Q$. Since Q is False, its negation $\neg Q$ is True (T).

Now, we look at the compound statement $P \wedge \neg Q$. Substituting our values:

$$P \wedge \neg Q \implies T \wedge T$$

Since both components of the "AND" operator are True, the compound statement $P \wedge \neg Q$ is **True**. □

Example 1.3.2 (Abstract Applied Mathematics Application). A digital safe has two passcode pads, Pad A and Pad B, and a manual key override. The door unlocks if the correct passcode is entered on Pad A **and** Pad B, **or** if the manual override is used. Write the safety logic.

Solution. Let us define three basic statements:

- A : "Correct passcode is entered on Pad A."
- B : "Correct passcode is entered on Pad B."
- K : "The manual key override is used."

The condition for the door unlocking can be written as:

$$(A \wedge B) \vee K$$

If a technician inputs correct keys on Pad A ($A = T$) but forgets Pad B ($B = F$), then $A \wedge B = T \wedge F = F$. If they do not use the override ($K = F$), the master condition becomes $F \vee F = F$, and the safe remains locked. □

Example 1.3.3 (Generalization to More Variables). Suppose we have n statement variables $P_1, P_2, \dots, P_n$. We define a compound statement $S = P_1 \vee P_2 \vee \dots \vee P_n$. Analyze how many truth assignments of the variables will make S evaluate to False.

Solution. By definition, a disjunction ($\vee$) of any number of statements is True if *at least one* statement in the list is True.

Therefore, the only way for the overall statement S to be False is if every single individual variable in the chain is simultaneously False.

$$S \text{ is False} \iff P_1 = F \wedge P_2 = F \wedge \dots \wedge P_n = F$$

For any other scenario out of the 2^n possible combinations in a truth table, S will evaluate to True. $\square$

Example 1.3.4 (Pure Abstract Discussion). Prove that the expression $P \vee \neg P$ is a *tautology* (a statement that is always True, regardless of the inputs) using a truth table.

Solution. To prove this, we construct a truth table with a single variable column P :

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

Looking at the final column, every outcome yields T . Thus, $P \vee \neg P$ is always True, satisfying the Law of Excluded Middle. $\square$

Theorem 1.3.1 (De Morgan's Laws for Logic). *Let P and Q be statements. Then the negation of an "AND" statement is equivalent to the "OR" of the individual negations:*

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

and the negation of an "OR" statement is equivalent to the "AND" of the individual negations:

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Example 1.3.5 (Bridging Verification for Theorem 1.3). Let statement P be "It is Saturday" (T) and Q be "It is raining" (F).

- The compound statement $P \wedge Q$ is $T \wedge F = F$. Its negation $\neg(P \wedge Q)$ is T .
- Now check the right side: $\neg P$ is F and $\neg Q$ is T . Their disjunction $\neg P \vee \neg Q$ is $F \vee T = T$.

Both sides evaluate to True ($T = T$), verifying the law for these specific switch states.

Proof of Theorem 1.3. We construct a complete truth table for all possible truth assignments of P and Q :

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Because the columns for $\neg(P \wedge Q)$ and $\neg P \vee \neg Q$ are identical in every row, the statements are logically equivalent. The proof for $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ follows identically by swapping conjunctions and disjunctions. $\square$

Exercises 1.3

- (1) Construct a complete truth table for the statement $\neg P \wedge Q$.
- (2) Construct a complete truth table for the statement $\neg(P \vee \neg Q)$.
- (3) Let P be True, Q be False, and R be False. Evaluate the truth value of $(P \vee Q) \wedge \neg R$.
- (4) Explain in your own words why the negation of "The box is red ****and**** the ball is blue" is ****not**** "The box is not red ****and**** the ball is not blue."
- (5) Use De Morgan's Laws to simplify the statement $\neg(\neg P \vee Q)$.
- (6) Verify De Morgan's Law $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ using a truth table.
- (7) Let x be an integer. Let P be " $x > 5$ " and Q be " x is even". Write down the meaning of $\neg P \vee \neg Q$ in simple words.
- (8) Find a value of x for Exercise 7 that makes the statement $\neg P \vee \neg Q$ False.
- (9) Show that $(P \wedge \neg P)$ is a contradiction (always False) using a truth table.
- (10) Prove that $P \wedge (Q \vee \neg Q) \equiv P$ using truth tables.
- (11) Evaluate the statement $\neg(\neg(\neg P))$ when $P = T$.
- (12) Let A and B be integers. Negate the statement: " $A > 0$ or $B \leq 3$."
- (13) Let $x = 4$. Determine the truth value of $(x + 1 = 5) \vee (x - 1 = 10)$.
- (14) Let $x = 4$. Determine the truth value of $(x + 1 = 5) \wedge (x - 1 = 10)$.
- (15) Construct a truth table for the three-variable statement $(P \wedge Q) \vee R$. How many rows are needed?
- (16) Find the truth values of P and Q if the statement $\neg P \vee Q$ is known to be False.
- (17) Simplify the expression $\neg(\neg P \wedge \neg Q)$ using De Morgan's laws.
- (18) If $P \wedge \neg Q$ is True, what are the individual truth values of P and Q ?
- (19) Show that the statement $(P \wedge Q) \vee (\neg P \vee \neg Q)$ is always True using a truth table.
- (20) A light bulb turns on if Switch 1 is closed ($P = T$) and Switch 2 is open ($Q = F$). Express the bulb being ON as a symbolic logic expression.

1.4 The Implies Logic

Most mathematical theorems are written in the conditional form: "If statement P is true, then statement Q must be true." This is written symbolically as $P \implies Q$.

Students often struggle to understand why $P \implies Q$ is considered True when P is False. To convince readers, we compare this to a contract. If a promise condition is not triggered, the contract is never violated (making it "vacuously" true).

Definition 1.4.1 (Logical Implication ($\implies$)). The statement $P \implies Q$ (read as " P implies Q ") is a compound statement. It is defined to be logically equivalent to:

$$\neg P \vee Q$$

As a result, $P \implies Q$ is only False when the hypothesis P is True and the conclusion Q is False. If the hypothesis P is False, the statement is automatically True.

Example 1.4.1 (Practical/Numerical Application). Evaluate the truth value of the numerical statement: "If $2 + 2 = 5$, then $10 + 10 = 20$."

Solution. Let us break this down into components:

- Hypothesis P : " $2 + 2 = 5$ ". This is False (F).
- Conclusion Q : " $10 + 10 = 20$ ". This is True (T).

Using the definition of implication:

$$P \implies Q \equiv \neg P \vee Q \implies \neg(F) \vee T \implies T \vee T \implies T$$

Because the starting premise P is False, the implication $P \implies Q$ is ****True**** (vacuously). $\square$

Example 1.4.2 (Abstract Applied Mathematics Application). An automatic sprinkler system operates under the safety rule: "If smoke is detected (S), then the water valves must open (W).". Express this rule, and analyze what happens if no smoke is detected.

Solution. The rule is written as $S \implies W$. Let us check the behavior when no smoke is detected ($S = F$). Under our logic:

$$S \implies W \equiv \neg S \vee W \equiv \neg(F) \vee W \equiv T \vee W \equiv T$$

The safety rule is evaluated as True regardless of whether the water valves are open ($W = T$) or closed ($W = F$). The contract is only breached (False) if smoke is detected ($S = T$) but the water valves fail to open ($W = F$). $\square$

Example 1.4.3 (Generalization to More Variables). Let us evaluate a chain of implications with n variables: $P_1 \implies P_2 \implies \dots \implies P_n$. Show how a single False value at the beginning affects the entire chain.

Solution. Suppose we are evaluating the nested implication $P_1 \implies (P_2 \implies P_3)$ and we know that $P_1 = F$.

By the definition of implication, since the leading hypothesis P_1 is False, the entire expression $P_1 \implies X$ evaluates immediately to True, regardless of the truth value of $X = (P_2 \implies P_3)$. This generalizes to any length of nested implications: if the primary assumption is False, the entire logical chain is vacuously valid. $\square$

Example 1.4.4 (Pure Abstract Discussion). Prove that the implication $P \implies Q$ is logically equivalent to its contrapositive $\neg Q \implies \neg P$ using a truth table.

Solution. Let us construct the comparative truth table columns:

P	Q	$P \implies Q$	$\neg Q$	$\neg P$	$\neg Q \implies \neg P$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Comparing columns 3 and 6, the truth values match exactly in every row. Thus, $(P \implies Q) \equiv (\neg Q \implies \neg P)$. $\square$

Theorem 1.4.1. *The implication statement $P \implies Q$ is logically equivalent to $\neg P \vee Q$.*

Example 1.4.5 (Bridging Verification for Theorem 1.4). Consider the contract: "If you get an A (P), I will give you \$10 (Q).". This contract is equivalent to saying: "Either you do NOT get an A ($\neg P$), OR I give you \$10 (Q).". If you get a B ($P = F$), the statement $\neg P \vee Q$ becomes $T \vee Q = T$ (contract unbroken).

Proof of Theorem 1.4. We evaluate both expressions across all four rows of a standard truth table:

P	Q	$P \implies Q$	$\neg P$	$\neg P \vee Q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The identical truth values in columns 3 and 5 confirm that $(P \implies Q) \equiv (\neg P \vee Q)$. $\square$

Exercises 1.4

- (1) Construct a truth table for $(P \implies Q) \wedge P$.
- (2) Let P be False and Q be False. What is the truth value of $P \implies Q$?
- (3) Let P be True and Q be False. What is the truth value of $P \implies Q$?
- (4) Write down the converse of the statement: "If x is divisible by 4, then x is even."
- (5) Write down the contrapositive of the statement: "If x is divisible by 4, then x is even."
- (6) Write the negation of $P \implies Q$ in terms of AND and NOT operators. (Hint: Negate $\neg P \vee Q$).
- (7) Explain why the statement "If $3 = 5$, then $7 = 9$ " is mathematically True.
- (8) Evaluate the truth value of $(P \implies Q) \implies P$ when $P = F$ and $Q = T$.
- (9) Let x be an integer. Is the statement "If $x > 10$, then $x > 5$ " True for all possible values of x ? Explain.
- (10) Let x be an integer. Is the statement "If $x > 5$, then $x > 10$ " True for all possible values of x ? If not, find a counterexample.
- (11) Show that $(P \implies Q) \equiv \neg(P \wedge \neg Q)$ using a truth table.
- (12) Translate into symbols: "If n is prime and $n > 2$, then n is odd."
- (13) Use the contrapositive rule to rewrite: "If n^2 is odd, then n is odd."
- (14) Prove that $P \implies P$ is always True (a tautology) using a truth table.

- (15) Explain the logical error in assuming that because $P \implies Q$ is True and Q is True, P must also be True. (This is called the fallacy of affirming the consequent).
- (16) Find a numerical counterexample to the converse of the statement: "If $x = -5$, then $x^2 = 25$."
- (17) Let $P \iff Q$ mean $(P \implies Q) \wedge (Q \implies P)$. Construct the truth table for $P \iff Q$.
- (18) Show that $P \iff Q$ is True if and only if P and Q have the exact same truth values.
- (19) Negate the statement: "If it is Saturday, I do not wake up early."
- (20) Under what exact conditions is the contract statement "If you clean your room (C), I will give you \$100 (H)" broken? Write this broken condition using symbols.

1.5 The Trichotomy Law on Equality

When we compare any two numbers on a flat line, there are only a limited number of ways they can relate. This basic reality of numbers is called the **Trichotomy Law**.

To help students with weak algebra skills, think of this like a physical scale: when you weigh two blocks a and b , the scale must tip left, balance perfectly, or tip right. There is no other physical possibility.

Definition 1.5.1 (Trichotomy Law). For any two real numbers a and b on the number line, exactly one of the following relations holds true:

$$a < b, \quad a = b, \quad \text{or} \quad a > b$$

Example 1.5.1 (Practical/Numerical Application). Let $a = 2 \times 3$ and $b = 10 - 4$. Evaluate which of the three trichotomy relations holds between these two numbers.

Solution. Let us calculate each side using basic arithmetic:

- $a = 2 \times 3 = 6$
- $b = 10 - 4 = 6$

Now, we test our three candidate relations:

- (1) Is $a < b$? Is $6 < 6$? No (False).
- (2) Is $a > b$? Is $6 > 6$? No (False).
- (3) Is $a = b$? Is $6 = 6$? Yes (True).

Exactly one relation ($a = b$) is True. □

Example 1.5.2 (Abstract Applied Mathematics Application). In computer databases, sorted search indices are built using comparative blocks. If a system compares an incoming user age record A to a benchmark age $B = 18$, show how trichotomy guarantees a safe path.

Solution. By the Trichotomy Law, the input age A must satisfy exactly one of:

$$A < 18, \quad A = 18, \quad \text{or} \quad A > 18$$

The system routes users to three separate pages based on these exclusive outcomes:

- If $A < 18$: Redirect to parental restriction filter.
- If $A = 18 \vee A > 18$: Grant full access permissions.

Because the three relations are mutually exclusive and cover all real values, the search engine will never experience an unhandled state or infinite loop. □

Example 1.5.3 (Generalization to More Variables). Suppose we have three real numbers x, y , and z . Show how the Trichotomy Law, combined with the assumption that $x < y$ and $y < z$, allows us to write a single ordered chain of inequalities.

Solution. By the Trichotomy Law, the relationship between x and z must fall into exactly one category: $x < z$, $x = z$, or $x > z$.

Let us analyze using basic intervals:

- If $x = z$: Since $y < z$, we can substitute x for z to get $y < x$. However, we are already given that $x < y$. This contradicts the Trichotomy Law (since $x < y$ and $y < x$ cannot both be true).

- If $x > z$: Since $y < z$, this would imply $y < x$. This again directly contradicts our given statement $x < y$.

Thus, the only logically surviving possibility is $x < z$. We can combine this into the continuous order chain: $x < y < z$. $\square$

Example 1.5.4 (Pure Abstract Discussion). In university algebra, we say a set X is *totally ordered* if it has a relation $\leq$ satisfying reflexivity, transitivity, and antisymmetry, along with the comparability axiom. Discuss how this compares to the Trichotomy Law.

Solution. The comparability axiom states: $\forall a, b \in X, a \leq b \vee b \leq a$. If we define $a < b$ to mean $(a \leq b) \wedge (a \neq b)$, we can use the properties of our order relations to prove that for any two elements, exactly one of $a < b$, $a = b$, or $b < a$ holds. Thus, the Trichotomy Law is the defining feature of any linear or total coordinate line. $\square$

Proposition 1.5.1. *Let $a, b \in \mathbb{R}$. If it is false that $a < b$, and it is also false that $a > b$, then $a = b$.*

Example 1.5.5 (Bridging Verification for Proposition 1.5). Suppose we compare a mystery number x with $b = 10$. We are told:

- (1) It is FALSE that $x < 10$.
- (2) It is FALSE that $x > 10$.

By the process of elimination on our physical number scale, the scale neither tips left nor right. Thus, x must balance perfectly at $x = 10$.

Proof of Proposition 1.5. By the Trichotomy Law, for any two real numbers $a, b \in \mathbb{R}$, the compound disjunction $(a < b) \vee (a = b) \vee (a > b)$ is always True, and the three cases are mutually exclusive. We are given that $\neg(a < b)$ is True and $\neg(a > b)$ is True. By applying the logical rule of disjunctive syllogism twice, eliminating the false branches $a < b$ and $a > b$, the only surviving relation is $a = b$. $\square$

Exercises 1.5

- (1) Let $a = -5$ and $b = -2$. Identify which of the three trichotomy relations holds.
- (2) Write the negation of the statement " $x < 5$ " using the Trichotomy Law.
- (3) Let x be a real number. If $x \neq 0$, show that either $x > 0$ or $x < 0$.
- (4) Let a and b be real numbers. If we know that $a \leq b$ (meaning $a < b \vee a = b$), what relation is completely ruled out by the Trichotomy Law?
- (5) Show that it is impossible for $x > 3$ and $x < 1$ to be True at the same time.
- (6) Let x be an integer. If $x^2 = 9$, use the Trichotomy Law to analyze the possible relationships between x and 0.
- (7) Prove that if $x < y$, then $y - x > 0$.
- (8) Let $a = 0.5$ and $b = 1/2$. Determine which trichotomy relation holds.
- (9) Let x be a real number. Can x satisfy $x > S(x)$? (Assume $S(x) = x + 1$). Explain.
- (10) If $a < b$ and $c < 0$, prove that $ac > bc$.
- (11) Let $S = \{x \in \mathbb{R} : x < 10\}$. Does $10 \in S$? Explain using the Trichotomy Law.
- (12) Let $x, y \in \mathbb{R}$. If $x \leq y$ and $y \leq x$, show that $x = y$.

- (13) Translate using symbols: "The number x is strictly between -2 and 3 ."
- (14) Why is the statement " $a < b$ and $a = b$ " always False?
- (15) Let $x = t^2 + 1$ and $y = 0$. For any real number t , which trichotomy relation holds? (Recall that $t^2 \geq 0$).
- (16) Suppose we compare complex numbers $a = 1 + i$ and $b = 2 + i$. Can we use the standard Trichotomy Law directly? Why or why not?
- (17) Prove that if $x > 1$, then $x^2 > x$.
- (18) Prove that if $0 < x < 1$, then $x^2 < x$.
- (19) Let $a, b > 0$. Show that if $a^2 > b^2$, then $a > b$.
- (20) Draw a diagram of the real number line and show how the Trichotomy Law helps us partition the line around a fixed point c .

1.6 The Language of Sets

Modern mathematics is written in the language of **sets**. A set is simply a collection of objects, which we call "elements". Think of a set like a bucket, and the elements as toys inside the bucket.

To help students with weak algebraic skills, we emphasize that we only care about whether an item is inside ($\in$) or outside ($\notin$) our bucket.

Definition 1.6.1 (Set, Subset, Set Equality). We define basic set concepts as follows:

- We write $x \in A$ if x is an element of set A . If not, we write $x \notin A$.
- A set B is a **subset** of A (written $B \subseteq A$) if and only if **every** element in B is also inside A .

$$B \subseteq A \iff (\forall x, \quad x \in B \implies x \in A)$$

- Two sets A and B are **equal** ($A = B$) if and only if they contain the exact same elements. That is:

$$A = B \iff (A \subseteq B \wedge B \subseteq A)$$

Example 1.6.1 (Practical/Numerical Application). Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4\}$. Determine if $B \subseteq A$ and if $B = A$.

Solution. To test if $B \subseteq A$, we check every single element in B :

- Is $2 \in A$? Yes.
- Is $4 \in A$? Yes.

Since every element in B is also in A , the statement $B \subseteq A$ is **True**.

Now we check if $B = A$. For $B = A$ to be True, we must also have $A \subseteq B$. Let us check if $1 \in B$. Looking at $B = \{2, 4\}$, we see that $1 \notin B$. Since there is an element in A that is not in B , $A \subseteq B$ is False. Therefore, $B \neq A$. $\square$

Example 1.6.2 (Abstract Applied Mathematics Application). In relational computer databases, let the set U represent all registered users. Let A be the set of users who paid their fees, and let B be the set of active subscribers.

Solution. If the system has a design rule stating "Only paid users are allowed to be active subscribers," then the database must guarantee at all times that:

$$B \subseteq A$$

If the database engine detects a user ID u such that $u \in B$ but $u \notin A$, then $B \subseteq A$ is violated, representing a system error. $\square$

Example 1.6.3 (Generalization to More Variables). Let $A_1, A_2, \dots, A_n$ be a chain of sets such that $A_1 \subseteq A_2, A_2 \subseteq A_3, \dots, A_{n-1} \subseteq A_n$. Prove that $A_1 \subseteq A_n$ using logical implication.

Solution. To prove $A_1 \subseteq A_n$, we must show that for any arbitrary element x , if $x \in A_1$, then $x \in A_n$. Let $x \in A_1$.

- Since $A_1 \subseteq A_2$, we have $x \in A_1 \implies x \in A_2$.
- Since $A_2 \subseteq A_3$, we have $x \in A_2 \implies x \in A_3$.
- Repeating this chain of logical implications, we eventually get $x \in A_{n-1} \implies x \in A_n$.

By the transitivity of logical implications, $x \in A_1 \implies x \in A_n$. Thus, $A_1 \subseteq A_n$. $\square$

Example 1.6.4 (Pure Abstract Discussion). Let $\emptyset$ represent the empty set (a set with no elements). Prove that $\emptyset$ is a subset of any set S .

Solution. By definition, $\emptyset \subseteq S$ is written as:

$$\forall x, \quad x \in \emptyset \implies x \in S$$

Let us analyze the truth value of the hypothesis " $x \in \emptyset$ ". Because the empty set has no elements, the statement " $x \in \emptyset$ " is False for any x .

Recall from Subsection 1.4 that any logical implication with a False hypothesis is automatically True (vacuously). Therefore, the implication $x \in \emptyset \implies x \in S$ is always True. Consequently, $\emptyset \subseteq S$ is True for any set S . $\square$

Theorem 1.6.1. Let A, B , and C be sets. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Example 1.6.5 (Bridging Verification for Theorem 1.6). Let $A = \{2\}$, $B = \{2, 4\}$, and $C = \{1, 2, 3, 4\}$.

- We see that $A \subseteq B$ because $2 \in B$.
- We see that $B \subseteq C$ because both 2 and 4 are in C .

Checking A against C , we see that the element 2 is indeed in C , confirming $A \subseteq C$.

Proof of Theorem 1.6. By definition of subset, we must prove that for any element x , if $x \in A$, then $x \in C$. Assume $x \in A$. Since $A \subseteq B$, the definition implies that $x \in B$. Now, since $x \in B$ and we are given $B \subseteq C$, the definition again implies that $x \in C$. Therefore, $x \in A \implies x \in C$, proving $A \subseteq C$. $\square$

Exercises 1.6

- (1) Let $A = \{1, 2, 3\}$. List all possible subsets of A . (There should be exactly 8 subsets).
- (2) Let $A = \{1, 2\}$ and $B = \{1, 2, \{3\}\}$. Is $3 \in B$? Is $\{3\} \in B$?
- (3) Let $S = \{x \in \mathbb{Z} : 0 \leq x < 4\}$. Write S in roster form (by listing its elements).
- (4) Let $A = \{2, 4, 6\}$ and $B = \{x \in \mathbb{Z} : x \text{ is an even integer}\}$. Show that $A \subseteq B$.
- (5) Let $A = \{1, 3\}$ and $B = \{3, 1, 3\}$. Are these two sets equal? Explain.
- (6) True or False: $\{5\} \subseteq \{5, 6, 7\}$. Explain.
- (7) True or False: $\{5\} \in \{5, 6, 7\}$. Explain.
- (8) Let $\emptyset$ be the empty set. Find the number of elements in $\emptyset$.
- (9) Let $A = \{x \in \mathbb{R} : x^2 = -4\}$. If we only consider real numbers, show that $A = \emptyset$.
- (10) Write down the set of prime numbers less than 10 using set-builder notation.
- (11) List the elements of the set $A = \{n \in \mathbb{N} : n^2 \leq 16\}$.
- (12) Let $A \subseteq \emptyset$. What can you conclude about the set A ?
- (13) If a set has n elements, it has 2^n subsets. How many subsets does $S = \{a, b, c, d\}$ have?
- (14) Let $A = \{x \in \mathbb{R} : x > 5\}$ and $B = \{x \in \mathbb{R} : x \geq 5\}$. Is $A \subseteq B$? Is $B \subseteq A$?
- (15) Let $A = \{2, 3\}$. Write down the *Power Set* of A (the set of all subsets of A).

- (16) Show that if $A \subseteq B$ and $B \subseteq A$, then $A = B$ is guaranteed.
- (17) Why is the empty set $\emptyset$ different from the set $\{\emptyset\}$?
- (18) Let $A = \{1, 2\}$. Is it true that $\emptyset \in A$? Is it true that $\emptyset \subseteq A$?
- (19) Let $S = \{x \in \mathbb{Z} : x \text{ is a multiple of } 4\}$. Let $T = \{x \in \mathbb{Z} : x \text{ is even}\}$. Is $S \subseteq T$?
- (20) Explain how set language avoids confusion when dealing with large lists of numbers.

1.7 Proofs

A mathematical proof is an airtight, logical demonstration that a statement must be True under all possible scenarios. In secondary school, students often think that showing a few working examples is a "proof". It is not.

We introduce four basic methods of proving mathematical claims.

Definition 1.7.1 (Direct, Contrapositive, Contradiction, Counterexample). We define the major modes of mathematical proofs:

- **Direct Proof:** Assume hypothesis P is True, and use direct, logical steps to show conclusion Q must be True.
- **Proof by Contrapositive:** Prove $P \implies Q$ by instead proving the equivalent statement $\neg Q \implies \neg P$.
- **Proof by Contradiction:** Assume the hypothesis P is True and the conclusion Q is False ($P \wedge \neg Q$). Show that this leads to a logical impossibility (such as $1 = 0$ or $x \in \emptyset$).
- **Disproof by Counterexample:** To show a universal statement $\forall x \in S, P(x)$ is False, we only need to find a single, specific element $c \in S$ for which $P(c)$ is False.

Example 1.7.1 (Practical/Numerical Application). Prove directly that if n is an even integer, then n^2 is an even integer.

Solution. We write a formal ****Direct Proof****:

- (1) Assume the hypothesis: n is an even integer.
- (2) By definition of even integers, there exists an integer k such that:

$$n = 2k$$

- (3) Now, we calculate n^2 :

$$n^2 = (2k)^2 = 4k^2$$

- (4) We rewrite this expression using basic arithmetic:

$$n^2 = 2 \times (2k^2)$$

- (5) Let $m = 2k^2$. Because k is an integer, k^2 is an integer, and $2k^2$ is also an integer. Thus, m is an integer.
- (6) We have $n^2 = 2m$. By definition, this means n^2 is an even integer.

This completes our direct proof. □

Example 1.7.2 (Abstract Applied Mathematics Application). In safety critical software, a program checks if a variable x satisfies: "If x^2 is odd, then x is odd." Prove this statement by contrapositive so the programmer can optimize the check.

Solution. Let P be " x^2 is odd" and Q be " x is odd". We want to prove $P \implies Q$. We use ****Proof by Contrapositive****, which means we prove $\neg Q \implies \neg P$:

- (1) State the negations:
 - $\neg Q$: " x is not odd" $\implies$ " x is even".
 - $\neg P$: " x^2 is not odd" $\implies$ " x^2 is even".

(2) We must prove: "If x is even, then x^2 is even."

(3) Let x be even. Then $x = 2k$ for some integer k .

(4) Calculate x^2 :

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

(5) Since $2k^2$ is an integer, x^2 is even.

Since we have proven $\neg Q \implies \neg P$, the original equivalent statement $P \implies Q$ is proven. $\square$

Example 1.7.3 (Generalization to More Variables). Disprove the following universal statement: "For all real numbers x and y , $(x + y)^2 = x^2 + y^2$."

Solution. To disprove a statement of the form $\forall x, \forall y, P(x, y)$, we use a **Counterexample**. We need to find just one pair of values (x, y) that violates the equation.

Let $x = 1$ and $y = 1$. Let us evaluate both sides:

- Left Hand Side (LHS): $(x + y)^2 = (1 + 1)^2 = 2^2 = 4$.
- Right Hand Side (RHS): $x^2 + y^2 = 1^2 + 1^2 = 1 + 1 = 2$.

Comparing the two results, we see that $4 \neq 2$. Since we found a specific case where $\text{LHS} \neq \text{RHS}$, the universal statement is proven False. $\square$

Example 1.7.4 (Pure Abstract Discussion). Prove by contradiction that there is no greatest integer.

Solution. We write a formal **Proof by Contradiction**:

- (1) Assume the statement is False. That means: "There exists a greatest integer, say G ."
- (2) This means that for any integer n , $n \leq G$.
- (3) Now, consider the number $G + 1$.
- (4) Since G is an integer, $G + 1$ is also an integer.
- (5) Using basic arithmetic, we know that $G + 1 > G$.
- (6) This contradicts our assumption that G is the greatest integer (since $G + 1$ is an integer and is strictly larger than G).
- (7) Because our assumption leads to a direct contradiction, the assumption must be False.

Therefore, there is no greatest integer. $\square$

Theorem 1.7.1. The real number $\sqrt{2}$ is irrational (it cannot be written as a fraction of integers).

Example 1.7.5 (Bridging Verification for Theorem 1.7). Let us test why simple decimals fail to represent $\sqrt{2}$:

- Try $1.4 = \frac{14}{10} = \frac{7}{5}$. Squaring gives $(\frac{7}{5})^2 = \frac{49}{25} = 1.96 \neq 2$.
- Try $1.414 = \frac{1414}{1000} = \frac{707}{500}$. Squaring gives $\frac{499849}{250000} = 1.999396 \neq 2$.

No matter how large the numerator and denominator become, their square will never land exactly on 2.

Proof of Theorem 1.7. We proceed by contradiction. Assume $\sqrt{2}$ is rational, meaning there exist coprime integers $p, q \in \mathbb{Z}$ ($q \neq 0$) with $\gcd(p, q) = 1$ such that $\sqrt{2} = \frac{p}{q}$. Squaring both sides yields $2 = \frac{p^2}{q^2}$, or $p^2 = 2q^2$. This implies p^2 is even. Since the square of an odd number is odd, p itself must be even. Let $p = 2k$ for some integer k . Substituting $p = 2k$ into our equation gives:

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies 2k^2 = q^2$$

This implies q^2 is even, so q must also be even. We have concluded that both p and q are even, meaning they share a common factor of 2. This contradicts our initial assumption that $\gcd(p, q) = 1$. Thus, $\sqrt{2}$ is irrational. $\square$

Exercises 1.7

- (1) Prove directly that the sum of two odd integers is even.
- (2) Prove directly that the product of two odd integers is odd.
- (3) Let n be an integer. Prove that if $3n + 2$ is odd, then n is odd, using contrapositive.
- (4) Prove by contradiction that the sum of a rational number and an irrational number is irrational.
- (5) Disprove the statement: "For all integers n , $n^2 + n + 11$ is a prime number."
- (6) What type of proof is best suited to disprove a general statement?
- (7) Prove that if x is even and y is odd, then $x + y$ is odd.
- (8) Let x be an integer. Prove that if x^2 is even, then x is even, using contrapositive.
- (9) Prove by contradiction that there is no smallest positive real number.
- (10) Disprove: "For all real numbers x , $x^2 \geq x$."
- (11) Let $a, b \in \mathbb{R}$. Prove directly that if $a > 0$ and $b > 0$, then $a \times b > 0$.
- (12) Prove by contrapositive: "If xy is even, then x is even or y is even."
- (13) Prove by contradiction that $\sqrt{3}$ is irrational.
- (14) Disprove: "All prime numbers are odd."
- (15) Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$ for integers a, b, c .
- (16) Explain why we cannot prove the statement "All dogs have four legs" by showing a list of 100 four-legged dogs.
- (17) Find a counterexample to the statement: "For all real numbers x , $\sqrt{x^2} = x$."
- (18) Prove that the product of any integer n and its successor $n + 1$ is always even.
- (19) Why is proof by contradiction so powerful in mathematical logic?
- (20) Discuss the difference between a direct proof and a proof by contrapositive.

1.8 More About Mathematical Structure

Now that we understand sets and logic, we can combine them to build more complex mathematical structures. We will show how set operations (like union, intersection, and complement) are exactly identical to our logical operators ($\vee, \wedge, \neg$).

To help students with weak algebraic backgrounds, think of this as overlapping regions: the logical operator tells us the rule for an element, while the set notation lists the collection of elements satisfying that rule.

Definition 1.8.1 (Intersection, Union, Complement). Let U represent the universal set containing all possible elements under consideration. Let A and B be subsets of U .

- **Intersection** ($A \cap B$, read as " A intersect B "): The set of all elements that are in **both** A and B .

$$A \cap B = \{x \in U : (x \in A) \wedge (x \in B)\}$$

- **Union** ($A \cup B$, read as " A union B "): The set of all elements that are in **at least one** of A or B .

$$A \cup B = \{x \in U : (x \in A) \vee (x \in B)\}$$

- **Complement** (A^c or A' , read as " A complement"): The set of all elements in U that are **not** in A .

$$A^c = \{x \in U : \neg(x \in A)\}$$

Example 1.8.1 (Practical/Numerical Application). Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, and $B = \{3, 4, 5\}$. Find $A \cap B$, $A \cup B$, and A^c .

Solution. Using our set operations:

- $A \cap B$: We find elements common to both sets. Only 3 is in both. Thus, $A \cap B = \{3\}$.
- $A \cup B$: We list all elements from both sets, combining them without repeating duplicates. Thus, $A \cup B = \{1, 2, 3, 4, 5\}$.
- A^c : We look at the universal set U and remove any elements belonging to A . Thus, $A^c = \{4, 5, 6\}$.

□

Example 1.8.2 (Abstract Applied Mathematics Application). In a computer database, Table A lists "active users" and Table B lists "unpaid accounts." The billing software needs to find "active users who have not paid." Write this as a set operation.

Solution. We translate the database query requirements:

- "Active users": Elements belonging to set A .
- "Unpaid accounts": Elements belonging to set B .
- "Active users **and** not paid": We need elements x such that $(x \in A) \wedge (x \notin B)$.

Using set operations, this is the intersection of A and the complement of B :

$$A \cap B^c$$

This is also commonly written as the set difference $A \setminus B$.

□

Example 1.8.3 (Generalization to More Variables). Let $A_1, A_2, \dots, A_n$ be sets. Define their generalized intersection and union, and explain how they relate to the universal ($\forall$) and existential ($\exists$) quantifiers.

Solution. We generalize our operations to n sets:

- ****Generalized Intersection**** ($\bigcap_{i=1}^n A_i$): Elements that belong to ****all**** of the sets.

$$\bigcap_{i=1}^n A_i = \{x \in U : \forall i \in \{1, 2, \dots, n\}, \quad x \in A_i\}$$

- ****Generalized Union**** ($\bigcup_{i=1}^n A_i$): Elements that belong to ****at least one**** of the sets.

$$\bigcup_{i=1}^n A_i = \{x \in U : \exists i \in \{1, 2, \dots, n\} \text{ such that } x \in A_i\}$$

This shows how quantifiers provide the foundation for infinite set systems. $\square$

Example 1.8.4 (Pure Abstract Discussion). Prove that $(A \cup B)^c = A^c \cap B^c$ using logic equivalence rules.

Solution. We want to prove that an element $x \in (A \cup B)^c \iff x \in A^c \cap B^c$:

$$\begin{aligned} x \in (A \cup B)^c &\iff \neg(x \in A \cup B) && \text{(by definition of complement)} \\ &\iff \neg(x \in A \vee x \in B) && \text{(by definition of union)} \\ &\iff \neg(x \in A) \wedge \neg(x \in B) && \text{(by logical De Morgan's Law)} \\ &\iff (x \in A^c) \wedge (x \in B^c) && \text{(by definition of complement)} \\ &\iff x \in A^c \cap B^c && \text{(by definition of intersection)} \end{aligned}$$

Because each step is connected by " $\iff$ " (logical equivalence), the sets contain identical elements. Thus, $(A \cup B)^c = A^c \cap B^c$. $\square$

Theorem 1.8.1 (De Morgan's Laws for Sets). *Let A and B be subsets of a universal set U . Then:*

$$(A \cap B)^c = A^c \cup B^c$$

Example 1.8.5 (Bridging Verification for Theorem 1.8). Let $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$.

- Intersection: $A \cap B = \{2\}$. The complement is $(A \cap B)^c = \{1, 3, 4\}$.
- Individual complements: $A^c = \{3, 4\}$ and $B^c = \{1, 4\}$.
- Union of complements: $A^c \cup B^c = \{1, 3, 4\}$.

Both methods yield the exact same set $\{1, 3, 4\}$.

Proof of Theorem 1.8. We prove set equality by showing mutual inclusion via logical equivalences for any element $x \in U$:

$$\begin{aligned} x \in (A \cap B)^c &\iff \neg(x \in A \cap B) && \text{(by definition of complement)} \\ &\iff \neg(x \in A \wedge x \in B) && \text{(by definition of intersection)} \\ &\iff \neg(x \in A) \vee \neg(x \in B) && \text{(by logical De Morgan's Law)} \\ &\iff (x \in A^c) \vee (x \in B^c) && \text{(by definition of complement)} \\ &\iff x \in A^c \cup B^c && \text{(by definition of union)} \end{aligned}$$

Since the chain of bidirectional equivalences holds, $(A \cap B)^c = A^c \cup B^c$. $\square$

Exercises 1.8

- (1) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 6, 8, 10\}$, and $B = \{1, 2, 3, 4, 5\}$. Find $A \cap B$.
- (2) Find $A \cup B$ for the sets in Exercise 1.
- (3) Find A^c and B^c for the sets in Exercise 1.
- (4) Find $A \cap B^c$ for the sets in Exercise 1.
- (5) Show that $A \cap A^c = \emptyset$ for any set A .
- (6) Show that $A \cup A^c = U$ for any set A with universal set U .
- (7) Simplify the expression $(A^c)^c$.
- (8) Prove that $A \subseteq B \iff A \cap B = A$.
- (9) Prove that $A \subseteq B \iff A \cup B = B$.
- (10) Draw a Venn Diagram representing three sets A , B , and C such that $A \cap B \neq \emptyset$, but $A \cap C = \emptyset$.
- (11) Let $A = \{x \in \mathbb{R} : x \geq 2\}$ and $B = \{x \in \mathbb{R} : x < 5\}$. Find $A \cap B$ using interval notation.
- (12) Let A and B be sets. Prove that $A \cap B \subseteq A$.
- (13) Let A and B be sets. Prove that $A \subseteq A \cup B$.
- (14) Simplify the set expression $A \cap (A \cup B)$ using Venn diagrams or logical reasoning.
- (15) Simplify the set expression $A \cup (A \cap B)$.
- (16) Let A and B be disjoint sets ($A \cap B = \emptyset$). If $x \in A$, show that $x \in B^c$.
- (17) Prove that $(A \cap B)^c = A^c \cup B^c$ using logical steps.
- (18) Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Find $A \cap B$.
- (19) Let A, B , and C be sets. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
- (20) Discuss how logical operations $(\wedge, \vee, \neg)$ are isomorphic to set operations $(\cap, \cup, c)$.

2 The Natural Number System

2.1 A Naive Way to Define Natural Numbers

Before we can understand advanced mathematical induction, we must look closely at our building blocks: the numbers we use to count. To an absolute beginner, numbers seem like physical objects. We see three apples and say the number "3" is physically there.

Historically, humans counted using tally marks (|, ||, |||, ||||). Over time, different civilizations developed symbols (like 1, 2, 3 in the Arabic system, or I, II, III in Roman numerals) as a shorthand convention to record these tallies. This intuitive, everyday approach is called the **naive definition of natural numbers**.

Definition 2.1.1 (Naive Natural Numbers ($\mathbb{N}$ and $\mathbb{N}_0$)). The set of **natural numbers** (or counting numbers), denoted by $\mathbb{N}$, is the infinite collection of integers starting from 1:

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots\}$$

If we include the number 0, representing the complete absence of items, we denote the set as $\mathbb{N}_0$ (or non-negative integers):

$$\mathbb{N}_0 = \{0, 1, 2, 3, 4, \dots\}$$

Example 2.1.1 (Practical/Numerical Application). A shepherd wants to keep track of his sheep. On Monday, he has no sheep. On Tuesday, he buys 4 sheep. On Wednesday, he sells 2 sheep. Show how these states are represented in our naive counting systems.

Solution. We map the physical count of sheep directly to the elements of our set $\mathbb{N}_0$:

- Monday (no sheep): Represented by $0 \in \mathbb{N}_0$.
- Tuesday (4 sheep): Represented by $4 \in \mathbb{N}_0$, which corresponds to the tally marks ||||.
- Wednesday (2 sheep left after selling 2): Represented by $2 \in \mathbb{N}_0$.

This direct mapping of physical quantities to symbols shows the practical, naive use of natural numbers. $\square$

Example 2.1.2 (Abstract Applied Mathematics Application). In computer programming, a loop counter variable i tracks how many times a code segment is run. Let us write down the values that i takes when a loop runs N times, starting from 0.

Solution. If a computer program is instructed to run a task N times, the loop index variable i is typically initialized at 0. For each iteration, the system increments i by 1. The list of index states is:

$$I = \{0, 1, 2, \dots, N - 1\}$$

Since I is a collection of non-negative integers up to $N - 1$, $I \subseteq \mathbb{N}_0$. This demonstrates how computer memory uses naive natural number segments to control repetitive actions. $\{0, 1, 2, \dots, N-1\} \subseteq \mathbb{N}_0$ $\square$

Example 2.1.3 (Generalization to More Variables). Suppose we want to define a coordinate system on a digital image screen. The screen is made up of a grid of pixels. Let the screen width be W pixels and the height be H pixels. Represent the coordinates of all pixels using natural numbers.

Solution. Every individual pixel can be located by an ordered pair of grid coordinates, (x, y) . Because computer pixels are discrete steps, both x and y must be whole counting numbers:

- The horizontal position satisfies $x \in \{0, 1, 2, \dots, W - 1\}$.
- The vertical position satisfies $y \in \{0, 1, 2, \dots, H - 1\}$.

The set of all pixel locations on the screen is the set of all pairs (x, y) such that $x \in \mathbb{N}_0$ and $y \in \mathbb{N}_0$. This generalizes our simple one-dimensional counting numbers into a two-variable grid. $\square$

Example 2.1.4 (Pure Abstract Discussion). Explain why the naive definition of natural numbers using the trailing dots " $\dots$ " is not considered mathematically rigorous, and explain why this leads to logical doubt.

Solution. The symbol " $\dots$ " (ellipsis) means "and so on, following the obvious pattern."

However, in rigorous mathematics, what is "obvious" to a human is not a formal logical rule. For instance, if we write the list $\{3, 5, 7, \dots\}$, one reader might assume the next number is 9 (the next odd number), while another might assume it is 11 (the next prime number).

Because the " $\dots$ " relies on human intuition rather than an explicit algebraic rule, the naive definition cannot be used to prove statements about infinity. We must find a way to generate numbers without assuming the reader can "guess" the pattern. $\square$

Exercises 2.1

- (1) List the first 8 elements of the set $\mathbb{N}$ of natural numbers.
- (2) Explain the symbolic difference between the sets $\mathbb{N}$ and $\mathbb{N}_0$.
- (3) If $n \in \mathbb{N}$, write down the algebraic expression representing the next consecutive natural number.
- (4) Find all elements of the set $S = \{x \in \mathbb{N} : x < 7\}$.
- (5) Find all elements of the set $T = \{x \in \mathbb{N}_0 : x \leq 3\}$.
- (6) Is the sum of any two natural numbers always a natural number? Check with $a = 5$ and $b = 12$.
- (7) Is the product of any two natural numbers always a natural number? Check with $a = 3$ and $b = 4$.
- (8) Is the difference of two natural numbers always a natural number? Find a counterexample.
- (9) Is the quotient of two natural numbers always a natural number? Find a counterexample.
- (10) Let $S = \{x \in \mathbb{N} : x \text{ is a multiple of } 5\}$. Write down the first five elements of S .
- (11) Let $T = \{x \in \mathbb{N} : x \text{ is prime and } x \leq 15\}$. List all elements of T .
- (12) Explain why we cannot count the total number of elements in $\mathbb{N}$ using a finite integer.
- (13) If $a, b \in \mathbb{N}$ and we are given that $a > b$, is $a - b$ guaranteed to be a natural number? Explain with a simple calculation.
- (14) Define the set of all square numbers $\{1, 4, 9, 16, \dots\}$ using set-builder notation.
- (15) Let A be a subset of $\mathbb{N}$ containing exactly 5 elements. Does A have a maximum element? Explain.
- (16) Let B be a non-empty subset of $\mathbb{N}$. Does B always have a minimum element? Discuss.
- (17) Write down the Roman numeral representation for the numbers 5, 10, and 50.
- (18) Why did ancient civilizations find tally marks insufficient as their societies and economies grew larger?
- (19) Let $x \in \mathbb{N}_0$. If $x + 3 = 3$, find the value of x .
- (20) Why is the number 0 considered a revolutionary concept in the history of number systems?

2.2 A Systematic Way to Understand Natural Numbers

To remove the logical doubt of the naive trailing dots "...", we need a systematic engine to generate numbers.

Think of this like climbing a ladder: we do not need to visualize the entire infinite ladder all at once. We only need two things: a ground floor to stand on, and a rule that tells us how to take a single step up from our current rung. We abstract this "single step up" as a mathematical function called the **successor map**.

Definition 2.2.1 (Successor Map). We define 0 as our starting origin. We then assume the existence of a function S (called the ****successor map****) that takes any number and outputs the next consecutive number:

$$S : \mathbb{N}_0 \rightarrow \mathbb{N}_0$$

We define our standard numbers as repeated applications of this successor map:

- $1 = S(0)$
- $2 = S(1) = S(S(0))$
- $3 = S(2) = S(S(S(0)))$

In general, for any number n , its successor is $S(n) = n + 1$.

Example 2.2.1 (Practical/Numerical Application). Using only the successor map S and the origin 0, write down the representation of the number 4, and verify its value.

Solution. To represent the number 4, we apply the successor map S four times starting from our origin 0:

$$4 = S(S(S(S(0))))$$

Let us track the value step-by-step:

- $S(0) = 1$
- $S(S(0)) = S(1) = 2$
- $S(S(S(0))) = S(2) = 3$
- $S(S(S(S(0)))) = S(3) = 4$

This rigorous notation replaces the naive tally marks |||| with nested function applications. □

Example 2.2.2 (Abstract Applied Mathematics Application). In computer science, a *Singly Linked List* is a chain of data structures where each data node contains a pointer, **next**, pointing to the next node in the memory. Show how this matches our successor map.

Solution. Let the first node in memory (the head) be called **Node_0**. We can define a pointer dereference function:

$$\text{NextNode}(\text{Node}_n) = \text{Node}_n \rightarrow \text{next}$$

By calling this function repeatedly, we traverse the memory:

- $\text{NextNode}(\text{Node}_0) = \text{Node}_1$
- $\text{NextNode}(\text{NextNode}(\text{Node}_0)) = \text{Node}_2$

The **next** pointer acts exactly like our successor map S . It establishes a systematic path to navigate the list node-by-node. □

Example 2.2.3 (Generalization to More Variables). Any infinite sequence of numbers $a_1, a_2, a_3, \dots$ can be formally defined as a function $a : \mathbb{N} \rightarrow \mathbb{R}$. Use the successor map to define how we step from one term of a sequence to the next.

Solution. Let $n \in \mathbb{N}$ represent the index of our sequence. The value of the n -th term is denoted by $a(n)$.

To find the very next term in the sequence, we do not write $n + \text{"dots"}$. Instead, we apply the successor map to our index variable n :

$$\text{Next term} = a(S(n)) = a(n + 1)$$

For example, if the sequence is defined by the formula $a(n) = n^2$, then:

$$a(S(n)) = (S(n))^2 = (n + 1)^2 = n^2 + 2n + 1$$

This shows how the successor map formally controls sequence steps for any variable n . □

Example 2.2.4 (Pure Abstract Discussion). Discuss how the successor map S guarantees that the set of natural numbers is infinite, without ever using the word "infinity" or the symbol ∞ .

Solution. We can prove the set is infinite by showing that we can never run out of new numbers. We establish two properties of S :

- (1) The map S is *injective* (one-to-one): if $S(a) = S(b)$, then $a = b$.
- (2) The starting element 0 is not in the range of S : there is no element $x \in \mathbb{N}_0$ such that $S(x) = 0$.

Suppose the set of numbers was finite, containing N elements. If we apply S repeatedly, we must eventually loop back to an element we have already visited.

However, we cannot loop back to 0 (by property 2), and we cannot loop back to any other visited element y because that would require two different elements to have the same successor (violating the injectivity of property 1).

Therefore, the chain can never loop or end, proving that the set generated by S must contain an endless collection of unique elements. □

Exercises 2.2

- (1) If 0 is the origin, write down the successor representation of the number 5.
- (2) Evaluate the expression $S(S(S(S(S(S(0))))))$. What standard number does this represent?
- (3) Let us define addition of 1 as: $x + 1 = S(x)$. Evaluate $6 + 1$ using successor notation.
- (4) Let us define the operation of adding 2 as: $x + 2 = S(S(x))$. Evaluate $3 + 2$.
- (5) If a sequence is defined by $a_1 = 4$ and $a_{S(n)} = a_n + 3$, find the values of a_2, a_3 , and a_4 .
- (6) Show that $S(n) \neq n$ for $n = 0$ and $n = 1$.
- (7) If $S(a) = 8$, find the value of a .
- (8) If $S(S(b)) = 10$, find the value of b .
- (9) Let $f(n) = 2^n$. Express $f(S(n))$ in terms of $f(n)$.
- (10) Let $f(n) = n!$ (factorial). Express $f(S(n))$ in terms of $f(n)$ and n .
- (11) Explain in your own words why 0 cannot be the successor of any natural number.

- (12) Let $S(x) = y$. Is it possible for x and y to be the same number? Explain.
- (13) Why does the successor map prevent the natural numbers from forming a closed circle (like hours on a clock)?
- (14) If we have a function P such that $P(S(n)) = n$, what is the physical meaning of P ?
- (15) Find $P(4)$ using your definition from Exercise 14.
- (16) Why is the successor map considered an "active" way of building numbers?
- (17) Write down the successor representations for the numbers 1, 2, and 3.
- (18) Let $g(n) = 3n + 1$. Calculate $g(S(2))$.
- (19) If $S(x) = S(y)$, what can you conclude about x and y ?
- (20) How does the successor map replace the need for the human mind to "imagine" infinity?

2.3 Peano Axioms

In 1889, the Italian mathematician Giuseppe Peano published a set of five simple, elegant axioms. These axioms serve as the official "constitution" of the natural numbers. Everything we do in arithmetic—from simple addition to the complex proofs of mathematical induction—is derived directly from these five rules.

Definition 2.3.1 (Peano Axioms). The natural numbers $\mathbb{N}_0$ is a set containing a distinguished element 0 and a successor function $S : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfies the following five axioms:

- (1) **Axiom 1:** $0 \in \mathbb{N}_0$. (There is a starting point.)
- (2) **Axiom 2:** If $x \in \mathbb{N}_0$, then $S(x) \in \mathbb{N}_0$. (We can always take a step.)
- (3) **Axiom 3:** For all $x, y \in \mathbb{N}_0$, if $S(x) = S(y)$, then $x = y$. (Different steps lead to different destinations.)
- (4) **Axiom 4:** There is no $x \in \mathbb{N}_0$ such that $S(x) = 0$. (We can never step backward into our starting point.)
- (5) **Axiom 5:** Let K be a subset of $\mathbb{N}_0$. If $0 \in K$, and $(\forall n \in K \implies S(n) \in K)$, then $K = \mathbb{N}_0$. (The Principle of Mathematical Induction.)

Example 2.3.1 (Practical/Numerical Application). Prove that $1 \neq 2$ using only the Peano Axioms.

Solution. Recall our definitions of these numbers:

$$1 = S(0), \quad 2 = S(1) = S(S(0))$$

Assume, for the sake of contradiction, that $1 = 2$. Substituting our successor representations:

$$S(0) = S(S(0))$$

By Peano **Axiom 3** (injectivity of S), we can strip the outer S from both sides of the equation:

$$0 = S(0)$$

However, Peano **Axiom 4** states that 0 is not the successor of any natural number (meaning $S(x) \neq 0$ for all x). Thus, $0 = S(0)$ is a direct contradiction of Axiom 4.

Therefore, our assumption must be False. We conclude that $1 \neq 2$. □

Example 2.3.2 (Abstract Applied Mathematics Application). In programming language compiler design, we must verify that our recursive arithmetic functions do not crash. Suppose we define addition recursively as:

$$\begin{aligned} a + 0 &= a \\ a + S(b) &= S(a + b) \end{aligned}$$

Verify the calculation $2 + 1$ using this formal definition.

Solution. Let us compute $2 + 1$. Recall that $1 = S(0)$. Thus, our calculation is:

$$2 + S(0)$$

Using the recursive addition rule $a + S(b) = S(a + b)$, where $a = 2$ and $b = 0$:

$$2 + S(0) = S(2 + 0)$$

Now, using the base rule $a + 0 = a$, we know $2 + 0 = 2$. Substituting this back:

$$S(2 + 0) = S(2) = 3$$

Thus, $2 + 1 = 3$. This formal addition is completely controlled by our successor rules, meaning the computer can run arithmetic without human intuition. □

Example 2.3.3 (Generalization to More Variables). Using the recursive definition of addition:

$$\begin{aligned} a + 0 &= a \\ a + S(b) &= S(a + b) \end{aligned}$$

Prove that for any three variables a, b , and c , the associative law holds: $(a + b) + c = a + (b + c)$.

Solution. To prove this associative property for all variables, we apply Peano **Axiom 5** (induction) on the variable c . Let K be the set of all numbers c for which the equation holds:

$$K = \{c \in \mathbb{N}_0 : (a + b) + c = a + (b + c)\}$$

Step 1 (Base Case): We test if $0 \in K$.

- Left Hand Side: $(a + b) + 0 = a + b$ (by addition base rule).
- Right Hand Side: $a + (b + 0) = a + b$ (by addition base rule).

Since LHS = RHS, $0 \in K$ is True.

Step 2 (Inductive Step): We assume $k \in K$ (meaning $(a + b) + k = a + (b + k)$) and must show $S(k) \in K$. We calculate:

$$(a + b) + S(k) = S((a + b) + k) \quad (\text{using addition recursive rule})$$

Using our assumption to substitute the inner term:

$$= S(a + (b + k))$$

Now, look at the right side of our target equation for $S(k)$:

$$a + (b + S(k)) = a + S(b + k) = S(a + (b + k))$$

Both sides match perfectly. Thus, $S(k) \in K$.

By Peano Axiom 5, $K = \mathbb{N}_0$, proving associativity for all variables. $\square$

Example 2.3.4 (Pure Abstract Discussion). Explain why Peano Axiom 5 is necessary. Show a mathematical system that satisfies Axioms 1, 2, 3, and 4, but is NOT the natural numbers.

Solution. Axioms 1-4 only state that we have a starting point 0 and an endless line of successors. However, they do not prevent "extra" disconnected elements from leaking into our system.

Consider a system $X = \mathbb{N}_0 \cup \{\alpha, \beta\}$, where α and β are extra abstract symbols. Let the successor function S act on the natural numbers as usual, and let:

$$S(\alpha) = \beta, \quad S(\beta) = \alpha$$

Let us check if this system satisfies Axioms 1-4:

- Axiom 1: $0 \in X$ (True).
- Axiom 2: Successor of any element is in X (True).
- Axiom 3: S is injective (True; no two elements have the same successor).
- Axiom 4: $S(x) \neq 0$ for all x (True; 0 has no predecessor).

This system contains our natural numbers, plus a disconnected loop of symbols $\{\alpha, \beta\}$ that is completely unreachable from 0.

Axiom 5 (Induction) saves us. If we choose a subset $K \subseteq X$ starting at 0 and closed under successor, K will contain all of $\mathbb{N}_0$, but it can never reach α or β . Thus, $K \neq X$.

Axiom 5 asserts that any such valid set K **must** equal the entire space, which logically outlaws the existence of any unreachable loops like $\{\alpha, \beta\}$. Thus, Axiom 5 guarantees our numbers are only those reachable from 0. $\square$

Theorem 2.3.1. For all $a \in \mathbb{N}_0$, $0 + a = a$.

Example 2.3.5 (Bridging Verification for Theorem 2.3). Let us verify $0 + a = a$ using our nested successor map for $a = 2$:

- By definition, $2 = S(S(0))$.
- We want to calculate $0 + 2 = 0 + S(S(0))$.
- Using the recursive addition rule $x + S(y) = S(x + y)$, we get:

$$0 + S(S(0)) = S(0 + S(0)) = S(S(0 + 0))$$

- Since the base rule states $x + 0 = x$, we have $0 + 0 = 0$. Substituting back gives $S(S(0)) = 2$.

Proof of Theorem 2.3. We proceed by Peano induction on the variable a . Let $K = \{a \in \mathbb{N}_0 : 0 + a = a\}$.

Base Case ($a = 0$): By the base definition of recursive addition ($x + 0 = x$), setting $x = 0$ yields $0 + 0 = 0$. Thus, $0 \in K$. **Inductive Step:** Assume $k \in K$, which means $0 + k = k$. We must show $S(k) \in K$, meaning $0 + S(k) = S(k)$. Using the recursive rule of addition:

$$0 + S(k) = S(0 + k)$$

By our inductive hypothesis, we substitute $0 + k = k$ into the right side:

$$0 + S(k) = S(k)$$

Thus, $S(k) \in K$. By Peano Axiom 5, $K = \mathbb{N}_0$, proving the theorem for all natural numbers. $\square$

Exercises 2.3

- (1) State the five Peano Axioms in your own words.
- (2) Which Peano Axiom prevents our numbers from wrapping around in a circle back to 0?
- (3) Which Peano Axiom prevents two different numbers from having the exact same successor?
- (4) Using the recursive addition rules, calculate $1 + 1$ step-by-step.
- (5) Prove that $S(a) = a + 1$ using the formal definitions of addition.
- (6) Show that $a + 2 = S(S(a))$ using the addition rules.
- (7) If we define multiplication recursively as:

$$\begin{aligned} a \cdot 0 &= 0 \\ a \cdot S(b) &= (a \cdot b) + a \end{aligned}$$

Calculate $3 \cdot 1$ showing every step.

- (8) Using the multiplication rules in Exercise 7, show that $2 \cdot 2 = 4$.
- (9) Prove that $0 \cdot a = 0$ for all $a \in \mathbb{N}_0$ using induction on a .
- (10) Prove that $a + S(b) = S(a) + b$ using induction on b .
- (11) Why does the system of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ violate Peano Axiom 4?
- (12) Why does the system of rational numbers $\mathbb{Q}$ violate Peano Axiom 2 if we define $S(x)$ as the "next fraction"?

- (13) Let $K = \{n \in \mathbb{N}_0 : n + 0 = n\}$. Show that $0 \in K$.
- (14) Continuing Exercise 13, show that if $k \in K$, then $S(k) \in K$. Conclude using Axiom 5.
- (15) Explain how Peano Axiom 5 acts as a filter that blocks fractional values from entering our set of natural numbers.
- (16) What is the role of the starting element 0 in the axioms? Could we build the axioms starting with 1?
- (17) Show that the successor function is not surjective (meaning its range is not the entirety of $\mathbb{N}_0$).
- (18) Let $S(x) = S(y)$. If we know $S(x) = 5$, find the values of x and y .
- (19) Discuss why Peano Axioms are considered the foundation of arithmetic.
- (20) Explain how the "domino effect" of mathematical induction is modeled by Axiom 5.

2.4 The Set Counter: Natural Numbers as Sets

In modern advanced mathematics, especially set theory, mathematicians do not like to assume that "numbers" exist. They prefer to build everything out of a single, basic logical tool: the **empty set** $\emptyset$.

In this view, we build the natural numbers out of nothing, using sets to count other sets. This elegant construction was developed by the mathematician John von Neumann.

Definition 2.4.1 (von Neumann Ordinals). We define our natural numbers as nested sets starting from the empty set:

$$\begin{aligned} 0 &= \emptyset \\ 1 &= \{0\} = \{\emptyset\} \\ 2 &= \{0, 1\} = \{\emptyset, \{\emptyset\}\} \\ 3 &= \{0, 1, 2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \end{aligned}$$

In general, for any set n , its successor set is defined as:

$$S(n) = n \cup \{n\}$$

Example 2.4.1 (Practical/Numerical Application). Write down the elements of the set 2. Count how many elements are inside this set, and explain how this relates to its numerical value.

Solution. By definition, the set 2 is:

$$2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}$$

Let us list the elements inside this set:

- (1) The first element is 0 (which is the empty set, $\emptyset$).
- (2) The second element is 1 (which is the set containing the empty set, $\{\emptyset\}$).

Counting these elements, we find exactly **two** distinct objects inside the brackets. Thus, the numerical value of the number 2 is exactly equal to the number of elements inside its set-theoretic representation. $\square$

Example 2.4.2 (Abstract Applied Mathematics Application). In nested database folders, a directory can contain files and other folders. Show how the von Neumann definition of 3 can be visualized as a folder structure.

Solution. Let us write the folder structure for $3 = \{0, 1, 2\}$:

- Folder 0 is empty: `Folder_0 = {}`
- Folder 1 contains Folder 0: `Folder_1 = { Folder_0 }`
- Folder 2 contains Folder 0 and Folder 1: `Folder_2 = { Folder_0, Folder_1 }`
- Folder 3 contains Folders 0, 1, and 2: `Folder_3 = { Folder_0, Folder_1, Folder_2 }`

This shows that each folder contains all its predecessor folders as elements. This mirrors how computers store nested directories. $\square$

Example 2.4.3 (Generalization to More Variables). Let n represent any von Neumann natural number. Show that the successor rule $S(n) = n \cup \{n\}$ always creates a set with exactly $n + 1$ elements.

Solution. Assume the set n contains exactly n elements:

$$n = \{0, 1, 2, \dots, n-1\}$$

Now, we calculate $S(n)$ using our successor rule:

$$S(n) = n \cup \{n\} = \{0, 1, \dots, n-1\} \cup \{n\}$$

Combining these two sets into a single union:

$$S(n) = \{0, 1, \dots, n-1, n\}$$

Since n is not an element of the set n (this is a basic rule of set theory, preventing self-containment), adding $\{n\}$ inserts exactly **one new element** into our set.

Therefore, the number of elements in $S(n)$ is exactly $n+1$. This proves our successor rule works for any variable n . $\square$

Example 2.4.4 (Pure Abstract Discussion). In this set representation of numbers, prove that the standard ordering relation " $x < y$ " is defined entirely by set membership " $x \in y$ ".

Solution. Let us test this beautiful concept with the numbers 1 and 2:

- We know numerically that $1 < 2$.
- Let us check the set membership: Is $1 \in 2$? Recall that $2 = \{0, 1\}$. Since 1 is listed inside the brackets, the statement $1 \in 2$ is True.

Let us test with 1 and 3:

- We know $1 < 3$.
- Check membership: Is $1 \in 3$? Since $3 = \{0, 1, 2\}$, 1 is indeed an element of 3.

In von Neumann ordinals, we do not need to invent a separate definition for "less than". We define:

$$x < y \iff x \in y$$

This unifies arithmetic and set theory into a single language. $\square$

Exercises 2.4

- (1) Write down the set representing the number 0.
- (2) Write down the set representing the number 1.
- (3) Write down the set representing the number 2.
- (4) Write down the set representing the number 3 using only brackets $\{$ and $\}$ and the empty set symbol $\emptyset$.
- (5) Compute $S(3)$ using the set rule $S(3) = 3 \cup \{3\}$.
- (6) Show that the number of elements in the set 4 is equal to 4.
- (7) True or False: $0 \in 1$. Explain.
- (8) True or False: $0 \subseteq 1$. Explain.
- (9) True or False: $1 \in 2$. Explain.

- (10) True or False: $1 \subseteq 2$. Explain.
- (11) Find the intersection of the sets 1 and 2.
- (12) Find the union of the sets 1 and 2.
- (13) Explain the difference between 0 (which is $\emptyset$) and $\{0\}$ (which is $\{\emptyset\}$).
- (14) Let a and b be von Neumann ordinals. If $a \in b$, show that $a \subseteq b$. (This means every von Neumann number is a transitive set).
- (15) Write down the power set of 1.
- (16) Is the power set of 1 equal to 2? Explain.
- (17) Why does the von Neumann definition satisfy Peano Axiom 4 ($S(x) \neq \emptyset$)?
- (18) Explain how we can construct all of mathematics starting from literally nothing ($\emptyset$).
- (19) Find the number of elements in the set $S(S(S(S(0))))$.
- (20) Discuss how set-theoretic numbers represent the ultimate level of abstract arithmetic.

2.5 The Backbone: Summary of Natural Number Constructions

We have now explored three completely different ways to view and build the natural numbers:

- (1) **The Naive Way**: Physical counting of items $(1, 2, 3, \dots)$ using visual tallies.
- (2) **The Axiomatic Way**: Using the five Peano Axioms starting from 0 and stepping forward using $S(n)$.
- (3) **The Set-Theoretic Way**: Building numbers out of pure sets where $n = \{0, 1, \dots, n-1\}$.

Despite their differences, all three models share the exact same structural skeleton. They represent a **linear path of assignments** starting from an origin and moving forward step-by-step. This structural backbone is what makes mathematical induction possible. If our numbers had loops, gaps, or branches, the domino effect of induction would fail.

Now that we have constructed our one-dimensional ladder of numbers, we are ready to learn how to climb it infinitely fast—this is the art of Mathematical Induction.

Exercises 2.5

- (1) Write a summary comparing the three constructions of natural numbers.
- (2) What is the common logical "backbone" shared by these three models?
- (3) Draw a flowchart connecting the origin 0 to the number 3 using the successor map.
- (4) Discuss why a system with a loop (such as $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$) cannot act as a backbone for induction.
- (5) Explain why having a "gap" (like missing the number 5) would break mathematical induction.
- (6) Explain why branching paths (where a number has two different successors) would break the linear backbone.
- (7) How does the axiomatic approach protect us from errors in our human visual intuition?
- (8) Discuss the famous quote by mathematician Leopold Kronecker: "God made the integers, all else is the work of man." How does this relate to Section 2?
- (9) Show that if we can index a list of files as $\{f_1, f_2, \dots, f_n\}$, we are using a map from a subset of $\mathbb{N}$ to our files.
- (10) Why is order the most fundamental property of the natural number system?
- (11) Could we do any algebra if we did not have a starting element like 0 or 1?
- (12) Show how the successor map can be thought of as a directed arrow in a network.
- (13) Explain why there cannot be a "last" natural number in any of our three models.
- (14) How does the set counter view unify the concepts of "numbers" and "groups of items"?
- (15) Let $J(n) = n + 2$ be a "jump successor". If we start at 0 and apply J repeatedly, why does this fail to generate all natural numbers?
- (16) How does a computer loop counter (like a `for` loop) use the linear backbone of natural numbers?
- (17) Why does learning natural numbers formally change your perspective on basic arithmetic?

- (18) Explain the relationship between the successor map and the physical action of "taking one step forward".
- (19) Why is the simplicity of the natural numbers' backbone both beautiful and powerful?
- (20) Write a short reflection on what you have learned in Section 2.

3 The Mathematical Induction

3.1 The Doubt of Infinite Process

When we state a mathematical claim about an infinite number of cases—such as "every odd integer squared is odd"—how can we be absolutely sure it holds true forever? We are finite humans with finite lifetimes. We can calculate $1^2 = 1$ (odd), $3^2 = 9$ (odd), $5^2 = 25$ (odd), and even check up to a billion. But a billion is still infinitely far from "all".

The core of logical deduction is not to physically test every case, which is impossible, but to establish ****certainty against arbitrary selection****. If we can prove that our claim holds for an ***unspecified, general case***, then our proof automatically holds for any specific case a skeptic might pick from our endless list.

Definition 3.1.1 (Certainty of Arbitrary Selection). To prove a statement $P(n)$ is true for all $n \in \mathbb{N}$, we do not demonstrate it for $1, 2, 3, \dots$ one-by-one. Instead, we show that if we select an arbitrary, unspecified integer $k \in \mathbb{N}$ from our infinite list, we have a rigorous logical mechanism that guarantees $P(k)$ must be True.

Example 3.1.1 (Practical/Numerical Application). Consider the statement $P(n)$: "The number $n^2 + n$ is always even." Let us show why proving this for an arbitrary integer k yields absolute certainty, whereas checking numerical examples up to $n = 3$ does not.

Solution. If we only check examples:

- For $n = 1$: $1^2 + 1 = 2$ (even).
- For $n = 2$: $2^2 + 2 = 6$ (even).
- For $n = 3$: $3^2 + 3 = 12$ (even).

This does not prove it for $n = 4$, or $n = 1000$.

To establish certainty, we select an ****arbitrary**** integer k . By the Trichotomy Law, k must be either even or odd:

- Case 1: If k is even, then $k = 2m$ for some integer m .

$$k^2 + k = (2m)^2 + 2m = 4m^2 + 2m = 2(2m^2 + m)$$

Since $2m^2 + m$ is an integer, $k^2 + k$ is even.

- Case 2: If k is odd, then $k = 2m + 1$ for some integer m .

$$k^2 + k = (2m + 1)^2 + (2m + 1) = (4m^2 + 4m + 1) + 2m + 1 = 4m^2 + 6m + 2 = 2(2m^2 + 3m + 1)$$

Since $2m^2 + 3m + 1$ is an integer, $k^2 + k$ is even.

Because k was an arbitrary choice, this proof instantly covers any number from 1 to infinity. □

Example 3.1.2 (Abstract Applied Mathematics Application). In network routing protocols, packets are forwarded from Node n to Node $n + 1$. To prevent data loss, engineers must prove that "for all nodes in an infinite linear network, every packet eventually reaches its destination." Explain how arbitrary selection applies here.

Solution. Testing the network node-by-node is impossible if the line is theoretically infinite.

Instead, we select an arbitrary node k along the path. If we can prove that "the packet arriving at Node k is guaranteed to be successfully transmitted to Node $k + 1$," then we have established a universal rule. The certainty of the network's reliability depends entirely on this arbitrary transition step being flawless. □

Example 3.1.3 (Generalization to More Variables). Suppose we have n variables $x_1, x_2, \dots, x_n$ representing real numbers in the interval $[0, 1]$. We want to show that their product $P = x_1 x_2 \dots x_n$ also lies within the interval $[0, 1]$.

Solution. If we test only for $n = 2$: since $0 \leq x_1 \leq 1$ and $0 \leq x_2 \leq 1$, multiplying a number between 0 and 1 by another number between 0 and 1 cannot exceed 1, so $0 \leq x_1 x_2 \leq 1$.

To generalize this to any number of variables n , we assume the property holds for an arbitrary group of k variables: $0 \leq \prod_{i=1}^k x_i \leq 1$.

Now, if we add one more variable x_{k+1} from our interval, we treat the product of the first k variables as a single consolidated variable $Y = \prod_{i=1}^k x_i$. Since $0 \leq Y \leq 1$ (by our arbitrary assumption) and $0 \leq x_{k+1} \leq 1$, our basic two-variable calculation applies:

$$0 \leq Y \cdot x_{k+1} \leq 1 \implies 0 \leq \prod_{i=1}^{k+1} x_i \leq 1$$

This shows how the behavior of $k + 1$ variables is easily resolved by collapsing them into an arbitrary k -variable state. $\square$

Example 3.1.4 (Pure Abstract Discussion). Discuss how the logical structure of "Universal Generalization" in formal logic allows mathematicians to conclude that a statement is true for all elements of an infinite set S .

Solution. In symbolic logic, the rule of Universal Generalization (UG) states:

$$\frac{P(c) \text{ for an arbitrary } c}{\forall x, P(x)}$$

The critical requirement for this rule to be valid is that the element c must be completely generic. We cannot assume c has any special properties other than being a member of the set S . If we successfully prove $P(c)$ under this restriction, our logic holds for all elements, and we can mathematically declare $\forall x, P(x)$. $\square$

Exercises 3.1

- (1) Why is checking a thousand numerical examples insufficient to prove a general statement about natural numbers?
- (2) Define what it means to choose an "arbitrary" element in a proof.
- (3) Let $P(n)$ be the statement " $n^2 - n + 41$ is prime." Calculate the value of $P(n)$ for $n = 1, 2$, and 3 . Are they prime?
- (4) Find a specific natural number n that acts as a counterexample to the statement in Exercise 3. (Hint: Try $n = 41$).
- (5) How does your result from Exercise 4 demonstrate the danger of relying on patterns?
- (6) Explain the difference between inductive reasoning in science and mathematical induction in deductive logic.
- (7) Let x be an arbitrary integer. Prove directly that if x is odd, then x^3 is odd.
- (8) Let x be an arbitrary integer. Show that $x(x + 1)$ is always even by testing the only two possible cases for x .
- (9) Show that the statement "Every natural number n is strictly less than 1,000,000" is False by finding a single counterexample.

- (10) Explain why we must assume no special properties (like being a multiple of 10) when picking an arbitrary variable.
- (11) If a proof holds for an arbitrary integer k , does it automatically hold for the number 5,829,102? Why?
- (12) Discuss how the concept of "infinity" makes empirical (experimental) verification impossible in mathematics.
- (13) Let $P(n)$ be a statement. If we know $P(k)$ is True for some k , does that tell us anything about $P(k+1)$? Explain.
- (14) Prove that the product of any three consecutive integers is divisible by 6 using case analysis on an arbitrary integer n .
- (15) Why does mathematics require absolute certainty rather than "highly probable" patterns?
- (16) Let x be an arbitrary positive real number. Show that $x + \frac{1}{x} \geq 2$. (Hint: Start from $(\sqrt{x} - 1/\sqrt{x})^2 \geq 0$).
- (17) Explain the logical flaw in the claim: "I proved the theorem for $n = 1, 2, 3$ and $n = 4$, so it must be true for all n ."
- (18) Write down the logical negation of the statement: "For all $n \in \mathbb{N}$, $P(n)$ is True."
- (19) How does the concept of arbitrary selection resolve our human inability to count to infinity?
- (20) Write a paragraph on the role of skepticism in mathematical proof.

3.2 The Domino Effect

The most intuitive way to visualize mathematical induction is the **domino effect**. Imagine an endless line of standing dominoes, numbered $1, 2, 3, 4, \dots$ to infinity.

To make every single domino fall, we only need to secure two things:

- (1) **The First Push**: We must physically push the first domino (Domino 1 falls).
- (2) **The Connection**: We must position the dominoes close enough so that if any domino (say, Domino k) falls, it is guaranteed to hit and knock down the very next domino (Domino $k + 1$).

If both conditions are met, then every domino in the infinite line will eventually fall!

Definition 3.2.1 (The Principle of Mathematical Induction). Let $P(n)$ be a mathematical statement that depends on a natural number $n \in \mathbb{N}$. To prove that $P(n)$ is True for all $n \in \mathbb{N}$, we must perform two steps:

- (1) **Base Case**: Prove that $P(1)$ is True.
- (2) **Inductive Step**: Assume that $P(k)$ is True for some arbitrary $k \in \mathbb{N}$ (this assumption is called the **Inductive Hypothesis**). Using this assumption, prove that $P(k + 1)$ must also be True.

Example 3.2.1 (Practical/Numerical Application). Prove that for all $n \in \mathbb{N}$, the sum of the first n positive integers is given by the formula:

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Solution. Let $P(n)$ be the statement: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Step 1: Base Case ($n = 1$): We test the formula for $n = 1$:

- Left Hand Side (LHS) of $P(1)$ is just the first term: 1.
- Right Hand Side (RHS) is: $\frac{1(1+1)}{2} = \frac{1(2)}{2} = 1$.

Since LHS = RHS, the base case $P(1)$ is **True**.

Step 2: Inductive Step: We assume the statement is True for an arbitrary integer $k \in \mathbb{N}$. That is, our **Inductive Hypothesis** is:

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

Now, we must prove that $P(k + 1)$ is True, which is the statement:

$$1 + 2 + 3 + \dots + k + (k + 1) = \frac{(k + 1)((k + 1) + 1)}{2} = \frac{(k + 1)(k + 2)}{2}$$

Let us write down the Left Hand Side of our target equation and group the first k terms:

$$\text{LHS} = \underbrace{(1 + 2 + 3 + \dots + k)}_{\text{Replace this using our Inductive Hypothesis}} + (k + 1)$$

Substituting our assumption into the bracket:

$$\text{LHS} = \frac{k(k+1)}{2} + (k + 1)$$

Now, we use basic algebra to simplify this expression. We factor out the common term $(k + 1)$:

$$\text{LHS} = (k + 1) \left(\frac{k}{2} + 1 \right)$$

Find a common denominator for the terms inside the parentheses:

$$\text{LHS} = (k+1) \left(\frac{k+2}{2} \right) = \frac{(k+1)(k+2)}{2}$$

This matches the Right Hand Side of $P(k+1)$ perfectly. Thus, $P(k) \implies P(k+1)$ is proven.

By the Principle of Mathematical Induction, the formula holds for all $n \in \mathbb{N}$. $\square$

Example 3.2.2 (Abstract Applied Mathematics Application). A network server uses a binary tree structure to index files. Prove that a binary tree of depth d has at most $2^d - 1$ total nodes.

Solution. Let $P(d)$ be the statement: "A binary tree of depth d has at most $2^d - 1$ nodes."

Base Case ($d = 1$): A tree of depth 1 consists of only a single root node.

- Physical count = 1 node.
- Formula gives: $2^1 - 1 = 1$ node.

Since $1 \leq 1$, $P(1)$ is True.

Inductive Step: Assume $P(k)$ is True: a binary tree of depth k has at most $2^k - 1$ nodes.

Now, consider a tree of depth $k+1$. A tree of depth $k+1$ is constructed by taking a root node and connecting it to at most two separate subtrees of depth k .

$$\text{Max nodes of depth } k+1 = 1 + 2 \times (\text{Max nodes in a subtree of depth } k)$$

Applying our inductive hypothesis to the subtrees:

$$\text{Max nodes of depth } k+1 \leq 1 + 2(2^k - 1)$$

Using basic expansion:

$$= 1 + 2 \cdot 2^k - 2 = 2^{k+1} - 1$$

This matches our target formula for depth $k+1$. Thus, $P(k) \implies P(k+1)$ is True.

By induction, the property holds for all depths $d \in \mathbb{N}$. $\square$

Example 3.2.3 (Generalization to More Variables). Let $x_1, x_2, \dots, x_n$ be n real numbers such that $x_i > 0$ for all i . Prove that for any $n \in \mathbb{N}$:

$$(1 + x_1)(1 + x_2) \dots (1 + x_n) \geq 1 + (x_1 + x_2 + \dots + x_n)$$

Solution. Let $P(n)$ be the statement.

Base Case ($n = 1$):

- LHS: $(1 + x_1) = 1 + x_1$.
- RHS: $1 + x_1$.

Since $\text{LHS} \geq \text{RHS}$ is True, $P(1)$ holds.

Inductive Step: Assume $P(k)$ is True for an arbitrary k :

$$(1 + x_1)(1 + x_2) \dots (1 + x_k) \geq 1 + \sum_{i=1}^k x_i$$

We must prove $P(k+1)$. Let us multiply both sides of our inductive assumption by the positive term $(1 + x_{k+1})$. Since $(1 + x_{k+1}) > 0$, the inequality sign remains unchanged:

$$((1 + x_1) \dots (1 + x_k))(1 + x_{k+1}) \geq \left(1 + \sum_{i=1}^k x_i \right) (1 + x_{k+1})$$

Let us expand the right-hand side using basic algebra:

$$= 1 + x_{k+1} + \sum_{i=1}^k x_i + \left(\sum_{i=1}^k x_i \right) x_{k+1}$$

Since all $x_i > 0$, the last product term $\left(\sum_{i=1}^k x_i \right) x_{k+1}$ is strictly positive. If we remove this positive term from the right side, the expression becomes even smaller:

$$\geq 1 + \sum_{i=1}^k x_i + x_{k+1} = 1 + \sum_{i=1}^{k+1} x_i$$

Thus, the statement holds for $k + 1$. By induction, it is True for all $n \in \mathbb{N}$. $\square$

Example 3.2.4 (Pure Abstract Discussion). Prove Bernoulli's Inequality: For any real number $x \geq -1$ and any natural number $n \in \mathbb{N}$, $(1 + x)^n \geq 1 + nx$.

Solution. Let $P(n)$ be the statement: $(1 + x)^n \geq 1 + nx$.

Base Case ($n = 1$):

- LHS: $(1 + x)^1 = 1 + x$.
- RHS: $1 + (1)x = 1 + x$.

Since LHS $\geq$ RHS is True, $P(1)$ is True.

Inductive Step: Assume $P(k)$ is True: $(1 + x)^k \geq 1 + kx$.

We must prove $P(k + 1)$: $(1 + x)^{k+1} \geq 1 + (k + 1)x$.

Multiply our assumption by $(1 + x)$. Since $x \geq -1$, we know $(1 + x) \geq 0$, which preserves the inequality sign:

$$(1 + x)^{k+1} \geq (1 + kx)(1 + x)$$

Now, expand the right-hand side:

$$(1 + kx)(1 + x) = 1 + x + kx + kx^2 = 1 + (k + 1)x + kx^2$$

Since k is a natural number ($k \geq 1$) and x^2 is a square ($x^2 \geq 0$), the term kx^2 is non-negative ($kx^2 \geq 0$). Dropping this term yields:

$$1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x$$

Therefore, $(1 + x)^{k+1} \geq 1 + (k + 1)x$, proving the inductive step.

By induction, Bernoulli's Inequality holds for all $n \in \mathbb{N}$. $\square$

Theorem 3.2.1. For all $n \in \mathbb{N}$, the sum of the first n odd integers is n^2 :

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

Example 3.2.5 (Bridging Verification for Theorem 3.2). Let us test the sum of the first n odd integers for small values of n :

- For $n = 1$: The sum is just 1. Formula gives $1^2 = 1$. (True)
- For $n = 2$: The sum is $1 + 3 = 4$. Formula gives $2^2 = 4$. (True)
- For $n = 3$: The sum is $1 + 3 + 5 = 9$. Formula gives $3^2 = 9$. (True)

Notice how adding the next odd number (7) to 9 gives 16, which is 4^2 .

Proof of Theorem 3.2. Let $P(n)$ be the statement: $\sum_{i=1}^n (2i - 1) = n^2$. **Base Case** ($n = 1$): LHS = $2(1) - 1 = 1$. RHS = $1^2 = 1$. Since LHS = RHS, $P(1)$ is True. **Inductive Step:** Assume $P(k)$ is True for an arbitrary integer $k \geq 1$:

$$1 + 3 + 5 + \cdots + (2k - 1) = k^2$$

We must prove $P(k + 1)$, which states that adding the $(k + 1)$ -th odd term yields $(k + 1)^2$:

$$\text{LHS of } P(k + 1) = \underbrace{1 + 3 + \cdots + (2k - 1)}_{\text{Substitute } k^2} + (2(k + 1) - 1)$$

Substituting our Inductive Hypothesis:

$$= k^2 + (2k + 2 - 1) = k^2 + 2k + 1$$

Factoring this perfect square trinomial gives $(k + 1)^2$, which matches the RHS of $P(k + 1)$. By the Principle of Mathematical Induction, $P(n)$ holds for all $n \in \mathbb{N}$. $\square$

Exercises 3.2

- (1) Explain the physical meaning of the "Base Case" and the "Inductive Step" in the domino analogy.
- (2) What happens to an induction proof if we forget to verify the Base Case? Show why we could "prove" False statements like $n = n + 1$.
- (3) What happens to an induction proof if we verify the base case but fail to prove the Inductive Step?
- (4) Prove that $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$ using induction.
- (5) Prove that $1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$ for all $n \in \mathbb{N}$ using induction.
- (6) Prove that $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for all $n \in \mathbb{N}$.
- (7) Prove that $n < 2^n$ for all $n \in \mathbb{N}$.
- (8) Prove that $2^{2n} - 1$ is divisible by 3 for all $n \in \mathbb{N}$.
- (9) Prove that $n^3 - n$ is divisible by 3 for all $n \in \mathbb{N}$.
- (10) Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$ for all $n \in \mathbb{N}$.
- (11) Prove that $2^n > n^2$ for all integers $n \geq 5$. (Note: Your base case should be $n = 5$).
- (12) Prove that the sum of the interior angles of a convex polygon with n sides is $(n - 2) \times 180^\circ$ for all $n \geq 3$.
- (13) Prove that $n! > 2^n$ for all integers $n \geq 4$.
- (14) Prove that $3^{2n-1} + 2^{2n-1}$ is divisible by 5 for all $n \in \mathbb{N}$.
- (15) Prove that the number of subsets of a set with n elements is 2^n for all $n \in \mathbb{N}$.
- (16) Prove that $(1 + x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$ when $x > -1$.
- (17) Prove that $7^n - 1$ is divisible by 6 for all $n \in \mathbb{N}$.
- (18) Prove that $\sum_{i=1}^n \frac{1}{2^i} = 1 - \frac{1}{2^n}$ for all $n \in \mathbb{N}$.
- (19) Identify the algebraic error in the following inductive step trying to prove all horses are of the same color.
- (20) Why is mathematical induction considered a foundational proof technique in discrete mathematics?

3.3 From Peano Axioms to Induction

Students often accept the "domino effect" as an intuitive picture, but intuition is not a rigorous proof. Why does mathematical induction actually work under the hood?

The answer lies in our fundamental system of numbers. Recall Peano **Axiom 5** from Section 2: "If a subset K of $\mathbb{N}_0$ contains 0, and contains $S(n)$ whenever it contains n , then $K = \mathbb{N}_0$."

This axiom is not just a rule of arithmetic; it is the mathematical engine that validates induction. If our proposition acts like a sequence, we can map the truth values directly to a Peano set.

Definition 3.3.1 (Peano Mapping of Induction). Let $P(n)$ be a statement. We define a truth set T containing all natural numbers for which our statement is True:

$$T = \{n \in \mathbb{N} : P(n) \text{ is True}\}$$

The Principle of Mathematical Induction is simply the statement that $T = \mathbb{N}$, which is proven by showing that T satisfies the conditions of Peano Axiom 5.

Example 3.3.1 (Practical/Numerical Application). Let $P(n)$ be the statement " $2n$ is even." Show how the truth set T of this statement maps to the Peano framework to conclude $T = \mathbb{N}$.

Solution. Let $T = \{n \in \mathbb{N} : 2n \text{ is even}\}$. We must verify the two Peano conditions:

- **Base condition**: Is $1 \in T$? We calculate $2(1) = 2$, which is even. Thus, $1 \in T$ is True.
- **Successor condition**: We assume $k \in T$ (meaning $2k$ is even $\implies 2k = 2m$ for some integer m). We must check if the successor $S(k) = k + 1 \in T$.

$$2(k + 1) = 2k + 2$$

Substituting our assumption:

$$= 2m + 2 = 2(m + 1)$$

Since $m + 1$ is an integer, $2(k + 1)$ is even. Thus, $k \in T \implies S(k) \in T$.

By Peano Axiom 5, the truth set T must contain all natural numbers ($T = \mathbb{N}$). □

Example 3.3.2 (Abstract Applied Mathematics Application). In computerized circuit logic, a clock signal updates a register state from $R(t)$ to $R(t + 1)$. If we model the register's correct operation as a subset of system states, show how Peano's induction validates continuous error-free operations.

Solution. Let T be the set of clock ticks $t \in \mathbb{N}_0$ where the register contains correct data.

- At boot time ($t = 0$), we verify correct register values ($0 \in T$).
- The hardware design ensures that if the register is correct at tick k ($k \in T$), the physical gating preserves correctness at the next tick $S(k) = k + 1$ ($S(k) \in T$).

By Peano's Axiom 5, the set of correct states T equals all possible operational times $\mathbb{N}_0$. The register remains error-free forever. □

Example 3.3.3 (Generalization to More Variables). Let $A_1, A_2, \dots, A_n$ be a series of sets. Prove that the size of their Cartesian product satisfies $|A_1 \times A_2 \times \dots \times A_n| = |A_1| \times |A_2| \times \dots \times |A_n|$ for any $n \in \mathbb{N}$, using Peano-based induction.

Solution. Let $T = \{n \in \mathbb{N} : \text{The product size formula holds for any } n \text{ sets}\}$.

Base Case ($1 \in T$): For $n = 1$, $|A_1| = |A_1|$, which is True.

Inductive Step: Assume $k \in T$. We must prove $S(k) = k + 1 \in T$. Let us group the first k sets as a single composite set $B = A_1 \times \dots \times A_k$. The product of $k + 1$ sets can be written as:

$$A_1 \times \dots \times A_k \times A_{k+1} = B \times A_{k+1}$$

Since this is a simple product of two sets, we know from basic set theory that $|B \times A_{k+1}| = |B| \times |A_{k+1}|$.

Applying our assumption $k \in T$ to substitute $|B|$:

$$|B| \times |A_{k+1}| = (|A_1| \times \cdots \times |A_k|) \times |A_{k+1}|$$

Thus, the formula holds for $k + 1$ sets, so $S(k) \in T$.

By Peano Axiom 5, $T = \mathbb{N}$. □

Example 3.3.4 (Pure Abstract Discussion). If we defined natural numbers without Axiom 5, we saw in Section 2 that a system could contain extra disconnected elements like loops $\{\alpha, \beta\}$. Explain why mathematical induction would fail on such a system.

Solution. Suppose we try to prove a property P on the system $X = \mathbb{N}_0 \cup \{\alpha, \beta\}$ where $S(\alpha) = \beta$ and $S(\beta) = \alpha$. Let us assume we successfully prove the two standard induction steps:

- (1) $P(0)$ is True.
- (2) If $P(n)$ is True, then $P(S(n))$ is True.

Using step 1 and repeatedly applying step 2, we prove P is True for $1, 2, 3, \dots$ and all standard natural numbers.

However, because the loop $\{\alpha, \beta\}$ is completely disconnected from 0, we can never reach α or β by starting at 0 and taking successor steps. As a result, the statement P might be completely False for α and β .

Without Peano Axiom 5, we cannot declare that P is true for "all" elements of our system. Induction fails because the "domino chain" never reaches the disconnected parts. □

Theorem 3.3.1. Every natural number $n \in \mathbb{N}$ is either 1, or it can be written as $S(m)$ for some $m \in \mathbb{N}_0$.

Example 3.3.5 (Bridging Verification for Theorem 3.3). Let us check our counting numbers $\mathbb{N} = \{1, 2, 3, \dots\}$:

- The first number is 1, which fits the first option of our theorem.
- Pick any other number, say 4. Can it be written as a successor? Yes, $4 = S(3)$, and $3 \in \mathbb{N}_0$.
- Pick 100. We have $100 = S(99)$, and $99 \in \mathbb{N}_0$.

Every number except the starting block 1 is one step ahead of some predecessor.

Proof of Theorem 3.3. Let $T = \{n \in \mathbb{N} : n = 1 \vee \exists m \in \mathbb{N}_0 \text{ such that } n = S(m)\}$. We prove $T = \mathbb{N}$ by Peano induction. **Base Case** ($n = 1$): The statement $1 \in T$ is True because it trivially satisfies the first disjunctive clause ($n = 1$). **Inductive Step:** Assume $k \in T$. We must prove $S(k) \in T$. Observe that $S(k)$ is explicitly written in the form $S(m)$ where $m = k \in \mathbb{N}_0$. Therefore, $S(k)$ satisfies the second disjunctive clause, so $S(k) \in T$. By Peano Axiom 5, $T = \mathbb{N}$. □

Exercises 3.3

- (1) Explain how Peano's 5th Axiom mathematically defines the Principle of Mathematical Induction.
- (2) Define the "truth set" T of a mathematical statement $P(n)$.
- (3) If a truth set $T \subseteq \mathbb{N}$ satisfies $1 \in T$ and $k \in T \implies k + 1 \in T$, what is T ? State the axiom that guarantees this.
- (4) Show that if we define $S(n) = n + 1$, then the step $k \implies k + 1$ is exactly the successor step $k \implies S(k)$.

- (5) Why does induction fail to prove a statement for a set that contains negative numbers, like $\mathbb{Z}$?
- (6) Let $T = \{n \in \mathbb{N} : n + 5 > 5\}$. Use the Peano induction framework to prove $T = \mathbb{N}$.
- (7) Let $T = \{n \in \mathbb{N} : 3^n > n\}$. Use the Peano induction framework to prove $T = \mathbb{N}$.
- (8) Suppose a system violates Axiom 3 (injectivity). Let $S(0) = 1$, $S(1) = 2$, and $S(2) = 1$ (a loop). If $P(1)$ is True, does induction prove $P(n)$ is True for all elements?
- (9) What is the logical difference between an intuitive explanation (dominoes) and an axiomatic proof (Peano)?
- (10) Let $P(n)$ be the statement: "The sum of the first n integers is positive." Express its truth set T and show $1 \in T$.
- (11) Show that the set of even natural numbers $\{2, 4, 6, \dots\}$ is closed under the successor-by-two operation.
- (12) Why are the Peano Axioms considered the "under-the-hood" engine of arithmetic?
- (13) Let T be a subset of $\mathbb{N}_0$ satisfying: $0 \in T$ and $k \in T \implies S(k) \in T$. Write this in the notation of Peano Axiom 5.
- (14) Show that if $T = \mathbb{N}$, then there is no natural number n that lies outside our truth list.
- (15) If we have a sequence a_n and we prove a property for a_1 and show $a_k \implies a_{k+1}$, why does this prove the property for any term in the sequence?
- (16) Discuss why Peano Axioms are called "axioms" rather than "theorems".
- (17) Show that the set of rational numbers $\mathbb{Q}$ cannot satisfy the Peano induction framework because it lacks a discrete successor map.
- (18) Explain how Peano Axiom 4 ($S(x) \neq 0$) guarantees we never loop back to the beginning of our induction.
- (19) Write down a formal proof showing that the set $T = \{n \in \mathbb{N} : n \geq 1\}$ is equal to $\mathbb{N}$.
- (20) Summary: Why is Peano's 5th Axiom the backbone of mathematical induction?

3.4 The Weak Induction

The standard form of mathematical induction that we have used so far is often called **Weak Induction**. The word "weak" does not mean it is fragile or incorrect; it simply refers to the amount of information we are allowed to assume in our inductive hypothesis.

In weak induction, when we try to prove $P(k+1)$, we are only allowed to assume that the single, immediately preceding case $P(k)$ is True.

Definition 3.4.1 (Weak Induction Set Formulation). A set $S \subseteq \mathbb{N}$ is defined to be **inductive** if:

- (1) $1 \in S$.
- (2) For any $k \in \mathbb{N}$, if $k \in S$, then $k+1 \in S$.

If a set is inductive, then $S = \mathbb{N}$. This is the standard "weak" template of induction.

Example 3.4.1 (Practical/Numerical Application). Prove that for all $n \in \mathbb{N}$, the number $8^n - 3^n$ is divisible by 5.

Solution. Let $S = \{n \in \mathbb{N} : 8^n - 3^n \text{ is divisible by } 5\}$. We prove S is inductive.

Base Case ($1 \in S$): For $n = 1$, we evaluate:

$$8^1 - 3^1 = 8 - 3 = 5$$

Since 5 is divisible by 5, $1 \in S$ is True.

Inductive Step: We assume $k \in S$. This means $8^k - 3^k$ is divisible by 5. Thus, there exists an integer M such that:

$$8^k - 3^k = 5M \implies 8^k = 3^k + 5M \quad (\text{Inductive Hypothesis})$$

We must show that $k+1 \in S$, which means $8^{k+1} - 3^{k+1}$ is divisible by 5. Let us write the expression for $k+1$:

$$8^{k+1} - 3^{k+1} = 8 \cdot 8^k - 3 \cdot 3^{k+1-1} = 8 \cdot 8^k - 3 \cdot 3^k$$

Now, substitute our Inductive Hypothesis $8^k = 3^k + 5M$ into the expression:

$$= 8(3^k + 5M) - 3 \cdot 3^k$$

Expand using basic algebra:

$$= 8 \cdot 3^k + 40M - 3 \cdot 3^k$$

Group the terms containing 3^k :

$$= (8 - 3) \cdot 3^k + 40M = 5 \cdot 3^k + 40M$$

Factor out the common number 5:

$$= 5(3^k + 8M)$$

Since $3^k + 8M$ is an integer, the expression is a multiple of 5. Thus, $k+1 \in S$.

Since S is inductive, $S = \mathbb{N}$. □

Example 3.4.2 (Abstract Applied Mathematics Application). In designing digital security keys, an algorithm generates security hashes of length L bits. Prove that the total number of unique security hashes of length L is 2^L .

Solution. Let $S = \{L \in \mathbb{N} : \text{Number of unique hashes of length } L \text{ is } 2^L\}$.

Base Case ($1 \in S$): A hash of length 1 has only 1 bit, which can be either 0 or 1.

- Total unique hashes = 2.

- Formula gives: $2^1 = 2$.

Thus, $1 \in S$ is True.

Inductive Step: Assume $k \in S$. This means a hash of length k has 2^k unique configurations.

To make a hash of length $k + 1$, we take any hash of length k and append a single new bit to the end. Since this new bit can be either 0 or 1, every original hash branches into two unique longer hashes:

$$\text{Total hashes of length } k + 1 = 2 \times (\text{Total hashes of length } k)$$

Applying our inductive hypothesis:

$$\text{Total hashes of length } k + 1 = 2 \times 2^k = 2^{k+1}$$

This proves that $k + 1 \in S$.

Therefore, S is inductive, so $S = \mathbb{N}$. □

Example 3.4.3 (Generalization to More Variables). Prove that for any n variables $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ satisfying $a_i \leq b_i$ for all i , we have:

$$\sum_{i=1}^n a_i \leq \sum_{i=1}^n b_i$$

Solution. Let $S = \{n \in \mathbb{N} : \text{The summation inequality holds for } n \text{ variables}\}$.

Base Case ($1 \in S$): For $n = 1$, $a_1 \leq b_1$, which is given. True.

Inductive Step: Assume $k \in S$. This means:

$$\sum_{i=1}^k a_i \leq \sum_{i=1}^k b_i$$

We must prove $k + 1 \in S$. Let us write the sum of $k + 1$ variables:

$$\sum_{i=1}^{k+1} a_i = \left(\sum_{i=1}^k a_i \right) + a_{k+1}$$

Applying our inductive assumption to the first part:

$$\leq \left(\sum_{i=1}^k b_i \right) + a_{k+1}$$

Now, we use our given condition that $a_{k+1} \leq b_{k+1}$ to substitute the last term:

$$\leq \left(\sum_{i=1}^k b_i \right) + b_{k+1} = \sum_{i=1}^{k+1} b_i$$

This proves $k + 1 \in S$.

By weak induction, the inequality holds for all $n \in \mathbb{N}$. □

Example 3.4.4 (Pure Abstract Discussion). Prove that the derivative of x^n is nx^{n-1} for all $n \in \mathbb{N}$ using weak induction and the product rule of calculus.

Solution. Let $P(n)$ be the statement: $\frac{d}{dx}(x^n) = nx^{n-1}$.

Base Case ($n = 1$):

- LHS: $\frac{d}{dx}(x^1) = 1$.

- RHS: $1 \cdot x^{1-1} = 1 \cdot x^0 = 1$.

LHS = RHS, so $P(1)$ is True.

Inductive Step: Assume $P(k)$ is True: $\frac{d}{dx}(x^k) = kx^{k-1}$.

We must prove $P(k+1)$: $\frac{d}{dx}(x^{k+1}) = (k+1)x^k$.

We write x^{k+1} as a product: $x^{k+1} = x^k \cdot x$. Now, we apply the derivative product rule:

$$\frac{d}{dx}(x^{k+1}) = \frac{d}{dx}(x^k \cdot x) = \frac{d}{dx}(x^k) \cdot x + x^k \cdot \frac{d}{dx}(x)$$

Substituting our inductive hypothesis and the base case:

$$= (kx^{k-1}) \cdot x + x^k \cdot (1)$$

Simplify using basic exponent rules:

$$= kx^k + x^k$$

Factor out x^k :

$$= (k+1)x^k$$

This is exactly the statement $P(k+1)$. By weak induction, the power rule of calculus is proven for all $n \in \mathbb{N}$. $\square$

Exercises 3.4

- (1) Define what makes a set $S \subseteq \mathbb{N}$ "inductive".
- (2) Why is this method called "weak" induction?
- (3) Prove that $9^n - 1$ is divisible by 8 for all $n \in \mathbb{N}$ using weak induction.
- (4) Prove that $5^{2n} - 1$ is divisible by 24 for all $n \in \mathbb{N}$ using weak induction.
- (5) Prove that $10^n - 1$ is divisible by 9 for all $n \in \mathbb{N}$ using weak induction.
- (6) Prove that $2^{3n} - 1$ is divisible by 7 for all $n \in \mathbb{N}$ using weak induction.
- (7) Let $f(n) = 3^{2n} + 7$. Prove that $f(n)$ is divisible by 8 for all $n \in \mathbb{N}$.
- (8) Prove that the sum of the first n terms of an arithmetic progression $a + (a + d) + \cdots + [a + (n - 1)d]$ is given by $\frac{n}{2}[2a + (n - 1)d]$ for all $n \in \mathbb{N}$.
- (9) Prove that $\sum_{i=1}^n i(i + 1) = \frac{n(n+1)(n+2)}{3}$ for all $n \in \mathbb{N}$.
- (10) Prove that $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$ for all $n \in \mathbb{N}$.
- (11) Prove that $(2n)! \geq 2^n(n!)^2$ for all $n \in \mathbb{N}$.
- (12) Prove that for all $n \in \mathbb{N}$, $4^n + 15n - 1$ is divisible by 9.
- (13) Prove that for all $n \in \mathbb{N}$, $1 \cdot 2 + 2 \cdot 3 + \cdots + n(n + 1) = \frac{n(n+1)(n+2)}{3}$.
- (14) Prove that for all $n \in \mathbb{N}$, $\frac{1}{2} \cdot \frac{3}{4} \cdot \cdots \cdot \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}$.
- (15) Let a_n be a sequence defined by $a_1 = 1$ and $a_{n+1} = \sqrt{a_n + 2}$. Prove that $a_n < 2$ for all $n \in \mathbb{N}$.
- (16) Show that the set of odd integers $\{1, 3, 5, \dots\}$ is NOT an inductive set because it violates condition 2.
- (17) If $1 \in S$ and $k \in S \implies k + 2 \in S$, what is S ? Does $S = \mathbb{N}$? Explain.

- (18) Show that $\sum_{i=1}^n \frac{1}{i^2} \leq 2 - \frac{1}{n}$ for all $n \in \mathbb{N}$.
- (19) Why does weak induction require the step $k \implies k + 1$ to work for any generic k ?
- (20) Write down the standard checklist of writing a weak induction proof for a secondary school exam.

3.5 The Unsatisfactoriness of Weak Induction

Weak induction is a beautiful tool, but it has a major limitation. Because the inductive hypothesis **only** allows us to assume the truth of the single immediately preceding state $P(k)$, it fails when we analyze complex systems where the next state depends on a broader history.

Think of this like climbing a mountain: if you have to jump a crevasse, knowing you made it safely to your current step k is not enough; you might need the momentum or data from multiple steps behind you $(k-1, k-2, \dots)$ to safely land on step $k+1$.

Definition 3.5.1 (Historical Dependency). A mathematical statement $P(n)$ has **historical dependency** if proving the truth of $P(k+1)$ requires knowing that $P(i)$ is True for multiple previous states $i < k+1$, rather than just the single state $i = k$. Weak induction is unsatisfactory for these systems.

Example 3.5.1 (Practical/Numerical Application). Consider the famous **Fibonacci Sequence**, which is defined recursively by adding the two previous terms:

$$F_1 = 1, \quad F_2 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 3$$

Suppose we want to prove that the n -th term is given by Binet's formula: $F_n \leq 2^n$. Explain why weak induction fails.

Solution. Let us attempt to use weak induction:

- Base case $n = 1$: $F_1 = 1 \leq 2^1 = 2$. (True).
- Inductive step: We assume $P(k)$ is True: $F_k \leq 2^k$.
- We want to prove $P(k+1)$: $F_{k+1} \leq 2^{k+1}$.

Let us write down the formula for F_{k+1} using its definition:

$$F_{k+1} = F_k + F_{k-1}$$

Now, we try to apply our inductive hypothesis. We can substitute $F_k \leq 2^k$:

$$F_{k+1} \leq 2^k + F_{k-1}$$

But now we are stuck! We have no information about the term F_{k-1} because weak induction only allowed us to assume the truth of the single step $P(k)$. We cannot simplify further. This shows why weak induction is unsatisfactory for historical-dependent systems. $\square$

Example 3.5.2 (Abstract Applied Mathematics Application). In simultaneous game theory, two players take turns. A winning strategy for Turn $T+1$ depends on analyzing all possible moves made in every previous turn $1, 2, \dots, T$.

Solution. If we try to prove a theorem "Player 1 always has a winning strategy at Turn N ," our inductive step must assume the strategy worked not just at Turn $N-1$, but was maintained across all preceding rounds. If the opponent made a wild move at Turn $N-3$, a weak induction model (which only remembers Turn $N-1$) cannot calculate the correct winning response. $\square$

Example 3.5.3 (Generalization to More Variables). Suppose we have a system of n linear equations where each variable x_{k+1} is defined by a weighted sum of all its predecessors:

$$x_{k+1} = \frac{x_1 + x_2 + \dots + x_k}{k}$$

Show why verifying a property of x_{k+1} requires a multi-variable history.

Solution. To evaluate any claim about x_{k+1} , our formula explicitly contains every single predecessor variable $x_1, x_2, \dots, x_k$.

If we only assume our property holds for x_k , we cannot say anything about $x_{k-1}, x_{k-2}, \dots, x_1$. The equation forces us to possess truth values for the entire list of variables. Weak induction leaves us completely unable to evaluate this formula. $\square$

Example 3.5.4 (Pure Abstract Discussion). Discuss how the Fundamental Theorem of Arithmetic (every integer $n > 1$ can be written as a product of prime numbers) highlights the limitation of weak induction.

Solution. Let us try to prove that every integer $n \geq 2$ is either prime or a product of primes, using weak induction:

- Base Case ($n = 2$): 2 is prime. (True).
- Inductive Step: Assume the integer k is prime or a product of primes.
- We want to prove that $k + 1$ is prime or a product of primes.

If $k + 1$ is prime, we are done. If $k + 1$ is composite, then by definition, it can be factored into two smaller integers:

$$k + 1 = a \times b \quad \text{where } 2 \leq a, b < k + 1$$

We want to argue that a and b can be written as prime products.

However, our weak inductive hypothesis only told us that the single number k is a prime product. It gave us absolutely zero information about a or b , because a and b are smaller integers, not k . Thus, the proof fails under weak induction. $\square$

Exercises 3.5

- (1) Why does weak induction fail when analyzing the Fibonacci sequence?
- (2) What does it mean for a mathematical sequence to have "historical dependency"?
- (3) Consider the recurrence $a_n = a_{n-1} + 2a_{n-2}$. Show why calculating a_{k+1} requires two previous terms.
- (4) Try to prove that the terms of a_n in Exercise 3 satisfy $a_n = 2^n$ (given $a_1 = 2, a_2 = 4$). Where does weak induction get stuck?
- (5) Explain the mountain climber analogy of complete induction in your own words.
- (6) Let $x_{n+1} = x_1 + x_2 + \dots + x_n$. Write down the formula for x_5 .
- (7) In the factorization of a polynomial $x^n - 1$, why does finding the factors of $x^{k+1} - 1$ often depend on the factors of smaller powers like $x^d - 1$ where $d \mid (k + 1)$?
- (8) Why does the factorization of composite numbers require knowing the properties of multiple smaller integers?
- (9) If a sequence is defined by $a_n = a_{n-1} \cdot a_{n-2}$, find a_4 if $a_1 = 2$ and $a_2 = 3$.
- (10) Show that to prove a_n in Exercise 9 is always of the form $2^x 3^y$, you need information about both a_{n-1} and a_{n-2} .
- (11) Discuss the logical limitation of a memory system that can only remember "the last tick".
- (12) Explain why simultaneous equations are difficult to solve with weak induction.

- (13) Write down the composite factors of 12. Show that none of these factors are equal to 11 (which would be k if $k + 1 = 12$).
- (14) Why is the term "Complete Induction" another common name for complete complete complete complete complete completed strong induction?
- (15) Let $P(n)$ be the statement: " n can be written as a sum of distinct powers of 2." Explain why breaking down a number like 13 into powers of 2 depends on the binary representation of smaller numbers, not just 12.
- (16) Discuss why weak induction is unsatisfactory for game trees in chess.
- (17) Show that for a sequence where $y_n = y_{n-1} + y_{n-2} + y_{n-3}$, we need three base cases to start the sequence.
- (18) Why is a single base case like $P(1)$ insufficient for sequences with multi-step recurrences?
- (19) Explain how weak induction is like a single-step ladder, whereas some mathematical structures act like a lattice grid.
- (20) Summary: Identify three mathematical topics where weak induction is completely insufficient.

3.6 The Strong Induction

To overcome the limitations of weak induction, we introduce **Strong Induction** (also known as Complete Induction).

Think of this as a memory upgrade: instead of only assuming the truth of the single immediately preceding step $P(k)$, we are now allowed to assume that **every single step** from our starting point up to k is True! This powerful change gives us the entire history of our system to prove the next step $P(k+1)$.

Definition 3.6.1 (Strong Induction Template). Let $P(n)$ be a mathematical statement. To prove $P(n)$ is True for all $n \in \mathbb{N}$ using Strong Induction:

- (1) **Base Case:** Prove that $P(1)$ is True (and sometimes other initial cases $P(2), \dots, P(m)$ if required by the recurrence).
- (2) **Inductive Step:** Assume that $P(i)$ is True for **all** integers i satisfying $1 \leq i \leq k$. Using this complete historical assumption, prove that $P(k+1)$ is True.

Example 3.6.1 (Practical/Numerical Application). Prove that any integer $n \geq 2$ is either a prime number or can be written as a product of prime numbers, using Strong Induction.

Solution. Let $P(n)$ be the statement: " n is prime or a product of primes."

Base Case ($n = 2$): Since 2 is a prime number, $P(2)$ is True.

Inductive Step: We assume that $P(i)$ is True for **all** integers i from 2 up to k . That is, our **Strong Inductive Hypothesis** is:

Every integer $i \in \{2, 3, \dots, k\}$ is prime or a product of primes.

We must prove that $P(k+1)$ is True. Let us analyze the number $k+1$:

- Case 1: If $k+1$ is prime, then the statement $P(k+1)$ is immediately True.
- Case 2: If $k+1$ is not prime (it is composite), then by definition of composite numbers, we can factor it into two smaller positive integers:

$$k+1 = a \times b \quad \text{where } 2 \leq a, b \leq k$$

Now, look at our factors a and b . Because both a and b lie in the interval $[2, k]$, they are covered by our **Strong Inductive Hypothesis**!

Therefore, we can write both a and b as products of prime numbers:

$$a = p_1 p_2 \dots p_r, \quad b = q_1 q_2 \dots q_s \quad (\text{where } p_j, q_y \text{ are primes})$$

Now, we substitute these prime representations back into the factors of $k+1$:

$$k+1 = a \times b = (p_1 p_2 \dots p_r) \times (q_1 q_2 \dots q_s)$$

Since this is a multiplication of only prime numbers, $k+1$ is a product of primes. Thus, $P(k+1)$ is True.

By the Principle of Strong Induction, the theorem is True for all integers $n \geq 2$. □

Example 3.6.2 (Abstract Applied Mathematics Application). In computer chip design, a power distribution grid divides power using cascading switches. A grid of size n can be split into two smaller grids of sizes a and b (where $a+b=n$). The total cost to split a grid of size n is exactly $a \times b$ dollars.

Prove that no matter how you cascadingly split a grid of size n down to individual units of size 1, the total cost is always $\frac{n(n-1)}{2}$ dollars.

Solution. Let $P(n)$ be the statement: "The cost to split a grid of size n is $\frac{n(n-1)}{2}$."

Base Case ($n = 1$): A grid of size 1 is already fully split.

- Physical cost = 0.
- Formula gives: $\frac{1(1-1)}{2} = 0$.

Thus, $P(1)$ is True.

Inductive Step: Assume $P(i)$ is True for all $1 \leq i \leq k$. This means any grid of size $i \leq k$ costs $\frac{i(i-1)}{2}$ to split.

We must prove $P(k+1)$ is True: a grid of size $k+1$ costs $\frac{(k+1)k}{2}$ to split.

Suppose we perform the first split on our grid of size $k+1$. It splits into two smaller grids of sizes a and b , where:

$$a + b = k + 1 \implies b = k + 1 - a$$

The initial split cost is $a \times b$. The remaining cost is the cost to split the subgrids of sizes a and b . Since $a, b \leq k$, we can apply our Strong Inductive Hypothesis to both subgrids:

$$\text{Total Cost} = (a \times b) + \text{Cost}(a) + \text{Cost}(b)$$

Substituting our formulas:

$$\text{Total Cost} = ab + \frac{a(a-1)}{2} + \frac{b(b-1)}{2}$$

Combine these terms under a common denominator of 2:

$$\text{Total Cost} = \frac{2ab + a^2 - a + b^2 - b}{2} = \frac{(a^2 + 2ab + b^2) - (a + b)}{2}$$

Recognize the perfect square trinomial $a^2 + 2ab + b^2 = (a + b)^2$:

$$\text{Total Cost} = \frac{(a + b)^2 - (a + b)}{2}$$

Now, substitute our original split size relation $a + b = k + 1$:

$$\text{Total Cost} = \frac{(k + 1)^2 - (k + 1)}{2}$$

Factor out the common term $(k + 1)$:

$$\text{Total Cost} = \frac{(k + 1)((k + 1) - 1)}{2} = \frac{(k + 1)k}{2}$$

This matches our target cost formula for $P(k + 1)$ perfectly. Thus, the inductive step holds.

By Strong Induction, the cost is always $\frac{n(n-1)}{2}$ dollars. □

Example 3.6.3 (Generalization to More Variables). Let a_n be a sequence of variables defined by $a_1 = 1$, $a_2 = 8$, and for all $n \geq 3$:

$$a_n = a_{n-1} + 2a_{n-2}$$

Prove that for any variable $n \in \mathbb{N}$, the terms of the sequence are given by the formula:

$$a_n = 3 \cdot 2^{n-1} + 2 \cdot (-1)^n$$

Solution. Let $P(n)$ be the statement. Since the formula relies on a two-step recurrence, we must verify two base cases.

Base Cases:

- For $n = 1$: Formula gives $3 \cdot 2^{1-1} + 2 \cdot (-1)^1 = 3(1) - 2 = 1$. (Matches $a_1 = 1$). True.
- For $n = 2$: Formula gives $3 \cdot 2^{2-1} + 2 \cdot (-1)^2 = 3(2) + 2(1) = 8$. (Matches $a_2 = 8$). True.

Inductive Step: Assume $P(i)$ is True for all $1 \leq i \leq k$ (where $k \geq 2$). We must prove $P(k+1)$:

$$a_{k+1} = 3 \cdot 2^k + 2 \cdot (-1)^{k+1}$$

Using the definition of our sequence:

$$a_{k+1} = a_k + 2a_{k-1}$$

Since k and $k-1$ are both strictly less than $k+1$, we apply our Strong Inductive Hypothesis to both terms:

$$a_{k+1} = \left(3 \cdot 2^{k-1} + 2 \cdot (-1)^k\right) + 2\left(3 \cdot 2^{k-2} + 2 \cdot (-1)^{k-1}\right)$$

Expand and group terms containing powers of 2, and terms containing negative powers:

$$a_{k+1} = \left(3 \cdot 2^{k-1} + 6 \cdot 2^{k-2}\right) + \left(2 \cdot (-1)^k + 4 \cdot (-1)^{k-1}\right)$$

Simplify the first group:

$$3 \cdot 2^{k-1} + 3 \cdot 2 \cdot 2^{k-2} = 3 \cdot 2^{k-1} + 3 \cdot 2^{k-1} = 2 \cdot (3 \cdot 2^{k-1}) = 3 \cdot 2^k$$

Simplify the second group. Note that $(-1)^{k-1} = -(-1)^k$:

$$2 \cdot (-1)^k + 4 \cdot (-(-1)^k) = 2 \cdot (-1)^k - 4 \cdot (-1)^k = -2 \cdot (-1)^k = 2 \cdot (-1)^{k+1}$$

Combining our simplified parts:

$$a_{k+1} = 3 \cdot 2^k + 2 \cdot (-1)^{k+1}$$

This matches our target formula for $k+1$.

By Strong Induction, the formula holds for all $n \in \mathbb{N}$. □

Example 3.6.4 (Pure Abstract Discussion). Discuss why Strong Induction is logically equivalent to Weak Induction. Prove that any theorem proven by Strong Induction could theoretically be proven by Weak Induction using a consolidated history set.

Solution. Suppose we have a strong induction proof where $P(1)$ is True, and $(\forall i \leq k, P(i)) \implies P(k+1)$.

To prove this same theorem using weak induction, we define a new, consolidated statement $Q(n)$:

$$Q(n) : \text{"The statement } P(i) \text{ is True for all } 1 \leq i \leq n\text{"}$$

Let us apply weak induction to $Q(n)$:

- ****Base Case**:** $Q(1)$ means $P(1)$ is True, which is given by our strong base case.
- ****Inductive Step**:** Assume $Q(k)$ is True. This means $P(1), P(2), \dots, P(k)$ are all True. Using our strong induction step, this history guarantees that $P(k+1)$ is True. Combining $Q(k)$ and $P(k+1)$, we conclude that $P(1), \dots, P(k), P(k+1)$ are all True, which is exactly the statement $Q(k+1)$.

Thus, $Q(k) \implies Q(k+1)$ holds. By weak induction, $Q(n)$ is True for all $n \in \mathbb{N}$, which means $P(n)$ is True for all $n \in \mathbb{N}$.

This proves that Strong Induction does not create "new" truths; it is simply a more convenient, consolidated way to write weak induction over historical sets. □

Exercises 3.6

- (1) State the difference between the induction hypotheses of weak induction and strong induction.
- (2) Why does a two-step recurrence (like Fibonacci) require two base cases to be verified?
- (3) Let a_n be a sequence defined by $a_1 = 1, a_2 = 3$, and $a_n = a_{n-1} + a_{n-2}$. Prove that $a_n < 2^n$ for all $n \in \mathbb{N}$ using strong induction.
- (4) Let x_n be a sequence defined by $x_1 = 1, x_2 = 2, x_3 = 3$, and $x_n = x_{n-1} + x_{n-2} + x_{n-3}$ for $n \geq 4$. Show that $x_n \leq 2^n$ for all $n \in \mathbb{N}$.
- (5) Prove that any amount of postage of 12 cents or more can be formed using only 4-cent and 5-cent stamps. (Hint: Use strong induction with base cases for 12, 13, 14, and 15 cents).
- (6) Prove that any amount of postage of 8 cents or more can be formed using only 3-cent and 5-cent stamps.
- (7) A sequence is defined by $y_1 = 1$, and $y_n = \sum_{i=1}^{n-1} y_i$ for $n \geq 2$. Prove that $y_n = 2^{n-2}$ for all $n \geq 2$.
- (8) Prove that the Fibonacci numbers satisfy Binet's Formula: $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$, where $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$.
- (9) In a game, a pile of n matches is on a table. Two players take turns removing either 1, 2, or 3 matches. The player who takes the last match wins. Prove that if n is a multiple of 4, the second player always has a winning strategy.
- (10) Let $a_1 = 3, a_2 = 5$, and $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \geq 3$. Prove that $a_n = 2^n + 1$ for all $n \in \mathbb{N}$.
- (11) Let $a_1 = 1, a_2 = 4$, and $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 3$. Prove that $a_n = 3n - 2$ for all $n \in \mathbb{N}$.
- (12) Let $x_1 = 1$ and $x_n = x_1 + x_2 + \cdots + x_{n-1}$ for all $n \geq 2$. Show that $x_n = 2^{n-2}$ for $n \geq 2$.
- (13) Why is the postage stamp problem difficult to solve with weak induction? Where does weak induction fail?
- (14) Explain the equivalence of Strong Induction and Weak Induction using the history statement $Q(n)$.
- (15) Prove that the number of diagonals in a convex n -gon is $\frac{n(n-3)}{2}$ for all $n \geq 3$.
- (16) Let F_n be the Fibonacci sequence. Prove that $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$ for all $n \in \mathbb{N}$.
- (17) Prove that F_{3n} is even for all $n \in \mathbb{N}$.
- (18) Prove that F_{5n} is divisible by 5 for all $n \in \mathbb{N}$.
- (19) Discuss why Strong Induction is often preferred when analyzing complex computer algorithms.
- (20) Summary: Write a short guide on how to decide whether to use weak or strong induction to solve a given problem.

4 The Product Sets

4.1 Cartesian Product

To transition from counting along a single straight line to navigating a flat page, we must learn how to combine sets. For a student with weak algebra skills, think of this like a coordinate grid. If you have a set of horizontal labels (like letters A, B, C) and a set of vertical labels (like numbers 1, 2, 3), you can combine them to locate any square on a chessboard (such as A1, B3, etc.).

This systematic pairing of elements from two different buckets is called the **Cartesian product**.

Definition 4.1.1 (Cartesian Product). Let A and B be two sets. The **Cartesian product**, denoted by $A \times B$ (read as " A cross B "), is the set of all possible ordered pairs (a, b) such that the first element a belongs to set A , and the second element b belongs to set B :

$$A \times B = \{(a, b) : a \in A \wedge b \in B\}$$

The pair (a, b) is called an **ordered pair** because the order matters: $(a, b) \neq (b, a)$ unless $a = b$.

Example 4.1.1 (Practical/Numerical Application). Let $A = \{1, 2\}$ and $B = \{3, 4, 5\}$. Write down the complete set representing the Cartesian product $A \times B$, and verify the total number of ordered pairs.

Solution. To find $A \times B$, we systematically match each element of A with every element of B :

- Start with $1 \in A$: Pair it with $3, 4, 5 \in B \implies (1, 3), (1, 4), (1, 5)$.
- Next with $2 \in A$: Pair it with $3, 4, 5 \in B \implies (2, 3), (2, 4), (2, 5)$.

Putting these together inside set brackets:

$$A \times B = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}$$

Counting the elements, we find exactly 6 unique pairs. Since A has 2 elements and B has 3 elements, the total number of elements in the product set is $2 \times 3 = 6$. This matches our arithmetic expectation. $\square$

Example 4.1.2 (Abstract Applied Mathematics Application). In computerized graphics, a small display screen consists of a grid of pixels. Let the set of horizontal screen coordinates be $X = \{0, 1, 2\}$ and the vertical coordinates be $Y = \{0, 1\}$. Write the set of all screen pixels, and show how a graphics card locates a pixel.

Solution. The set of all possible pixel locations is the Cartesian product $X \times Y$:

$$X \times Y = \{(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1)\}$$

If the graphics card wants to illuminate the pixel at the second column, first row, it addresses the unique ordered pair coordinate $(1, 0) \in X \times Y$. By treating screen spaces as Cartesian products, computers can store image buffers in linear memory channels. $\square$

Example 4.1.3 (Generalization to More Variables). We can extend the Cartesian product to a list of n sets: $A_1, A_2, \dots, A_n$. Define this generalized product set, and show how a coordinate in n -dimensional space is represented.

Solution. The generalized Cartesian product, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ where each coordinate a_i belongs to its respective set A_i :

$$\prod_{i=1}^n A_i = \{(a_1, a_2, \dots, a_n) : \forall i \in \{1, 2, \dots, n\}, \quad a_i \in A_i\}$$

For example, if we model 3D space using real numbers $\mathbb{R}$, we write the product of three sets:

$$\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

This allows us to describe physical points using three independent numerical variables. $\square$

Example 4.1.4 (Pure Abstract Discussion). Let A, B , and C be sets. Prove that the Cartesian product distributes over the intersection of sets:

$$(A \cap B) \times C = (A \times C) \cap (B \times C)$$

Solution. We prove this by showing that an arbitrary ordered pair (x, y) belongs to the left-hand set if and only if it belongs to the right-hand set:

$$\begin{aligned} (x, y) \in (A \cap B) \times C &\iff x \in (A \cap B) \wedge y \in C && \text{(by product definition)} \\ &\iff (x \in A \wedge x \in B) \wedge y \in C && \text{(by intersection definition)} \\ &\iff (x \in A \wedge y \in C) \wedge (x \in B \wedge y \in C) && \text{(by logical regrouping)} \\ &\iff (x, y) \in (A \times C) \wedge (x, y) \in (B \times C) && \text{(by product definition)} \\ &\iff (x, y) \in (A \times C) \cap (B \times C) && \text{(by intersection definition)} \end{aligned}$$

Since each step is connected by logical equivalence ($\iff$), the two sets contain the exact same elements. Thus, $(A \cap B) \times C = (A \times C) \cap (B \times C)$. $\square$

Theorem 4.1.1. Let A, B , and C be sets. The Cartesian product distributes over the union of sets:

$$(A \cup B) \times C = (A \times C) \cup (B \times C)$$

Example 4.1.5 (Bridging Verification for Theorem 4.1). Let $A = \{1\}$, $B = \{2\}$, and $C = \{a, b\}$.

- The union is $A \cup B = \{1, 2\}$.
- The left side product is $(A \cup B) \times C = \{(1, a), (1, b), (2, a), (2, b)\}$.
- Calculate separate products: $A \times C = \{(1, a), (1, b)\}$ and $B \times C = \{(2, a), (2, b)\}$.
- Their union is $\{(1, a), (1, b)\} \cup \{(2, a), (2, b)\} = \{(1, a), (1, b), (2, a), (2, b)\}$.

Both methods yield the exact same 4 ordered pairs.

Proof of Theorem 4.1. We prove set equality by showing bidirectional logical equivalence for an arbitrary ordered pair (x, y) :

$$\begin{aligned} (x, y) \in (A \cup B) \times C &\iff x \in (A \cup B) \wedge y \in C && \text{(by definition of Cartesian product)} \\ &\iff (x \in A \vee x \in B) \wedge y \in C && \text{(by definition of union)} \\ &\iff (x \in A \wedge y \in C) \vee (x \in B \wedge y \in C) && \text{(by logical distribution of } \wedge \text{ over } \vee) \\ &\iff (x, y) \in (A \times C) \vee (x, y) \in (B \times C) && \text{(by definition of Cartesian product)} \\ &\iff (x, y) \in (A \times C) \cup (B \times C) && \text{(by definition of union)} \end{aligned}$$

Since the logical statements are equivalent at every step, $(A \cup B) \times C = (A \times C) \cup (B \times C)$. $\square$

Exercises 4.1

- (1) Let $A = \{a, b\}$ and $B = \{1, 2\}$. Find the Cartesian product $A \times B$.
- (2) Let $B \times A$ be the product set for the sets in Exercise 1. Find $B \times A$.
- (3) Based on Exercises 1 and 2, does $A \times B = B \times A$? Explain why or why not.
- (4) Let $S = \{1, 2\}$. Find $S \times S$ (commonly written as S^2).
- (5) Let $A = \{3, 4\}$, $B = \emptyset$. Find $A \times B$.
- (6) If a set A has m elements and a set B has n elements, how many elements are in $A \times B$?

- (7) Let $A = \{1, 2\}$, $B = \{3, 4\}$, and $C = \{5\}$. Find the three-set product $A \times B \times C$.
- (8) True or False: $(2, 3) = (3, 2)$. Explain using the definition of ordered pairs.
- (9) Let $A = \{x \in \mathbb{N} : x \leq 2\}$ and $B = \{y \in \mathbb{N} : y^2 \leq 9\}$. Write $A \times B$ in roster form.
- (10) If $A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2)\}$, reconstruct the original sets A and B .
- (11) Let $A \subseteq B$. Prove that for any set C , $A \times C \subseteq B \times C$.
- (12) Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ using logical steps.
- (13) Simplify the expression $(A \times B) \cap (A^c \times B)$ using set-theoretic distribution.
- (14) Let $S \times T = \emptyset$. What can you conclude about the sets S and T ?
- (15) Draw a geometric diagram of $\{1, 2, 3\} \times \{1, 2\}$ as points on a 2D Cartesian plane.
- (16) Let $A = [1, 3]$ and $B = [2, 4]$ be intervals on the real number line. What geometric shape does $A \times B$ represent on the plane?
- (17) Why does the order of elements in a pair (a, b) matter, whereas the order of elements in a set $\{a, b\}$ does not?
- (18) Let A, B, C, D be sets. Prove that $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.
- (19) Let S have k elements. Show that S^n has k^n elements.
- (20) Discuss how the Cartesian product provides the logical bridge from arithmetic of single values to coordinate algebra.

4.2 Integer Set

Up to this point, our natural number system $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ only allowed us to count forward. If we try to perform a subtraction like $2 - 5$, we are stuck; there is no natural number that can represent this state.

To help students with weak algebra skills, think of this like a bank account. If you have \$2 and spend \$5, you are in **debt** by \$3. To formalize this rigorously without inventing "magic" negative numbers, we define the set of integers $\mathbb{Z}$ as pairs of natural numbers, representing our assets and liabilities.

Definition 4.2.1 (The Integers as Equivalence Classes). We define the set of integers $\mathbb{Z}$ using the Cartesian product of natural numbers, $\mathbb{N}_0 \times \mathbb{N}_0$. An ordered pair (a, b) represents the abstract quantity " $a - b$ ".

Since many different pairs can represent the same difference (for example, $5 - 2 = 3$ and $4 - 1 = 3$), we define an **equivalence relation** $\sim$ on $\mathbb{N}_0 \times \mathbb{N}_0$:

$$(a, b) \sim (c, d) \iff a + d = b + c$$

The set of **integers** $\mathbb{Z}$ is the collection of all equivalence classes of these pairs:

$$\mathbb{Z} = \{[(a, b)] : a, b \in \mathbb{N}_0\}$$

We denote positive integers as $[(a, 0)]$, negative integers as $[(0, b)]$, and zero as $[(0, 0)]$.

Example 4.2.1 (Practical/Numerical Application). Show that the pairs $(5, 2)$ and $(4, 1)$ represent the same integer in $\mathbb{Z}$ using our formal equivalence relation.

Solution. Let $(a, b) = (5, 2)$ and $(c, d) = (4, 1)$. We test the equivalence relation condition $a + d = b + c$:

- Left-hand calculation: $a + d = 5 + 1 = 6$.
- Right-hand calculation: $b + c = 2 + 4 = 6$.

Since $6 = 6$, the relation $(5, 2) \sim (4, 1)$ is True. Both pairs belong to the same equivalence class, which represents the standard integer 3. $\square$

Example 4.2.2 (Abstract Applied Mathematics Application). In a video game, a player's score offset is updated by +10 points for collecting a coin, and -15 points for hitting an obstacle. Show how a computer database can record this integer state using only natural numbers.

Solution. To avoid negative numbers in basic binary memory registers, the game database represents the score offset as an ordered pair $(a, b) \in \mathbb{N}_0 \times \mathbb{N}_0$, where a is the total positive points earned, and b is the total penalty points incurred.

The current score status is:

$$(a, b) = (10, 15)$$

This pair represents the equivalence class $[(10, 15)]$, which corresponds to the integer -5. This demonstrates how integer calculations are executed on raw silicon architectures. $\square$

Example 4.2.3 (Generalization to More Variables). If we can perform mathematical induction forward along the positive integers, can we also run induction backward into the negative numbers? Introduce **Backward Induction** (or Cauchy Induction) and show its structure.

Solution. Yes. Because $\mathbb{Z}$ extends in both directions, we can formulate a backward induction template on integers.

To prove a statement $P(n)$ is True for all integers $n \leq 0$:

- (1) **Base Case**: Prove $P(0)$ is True.

- (2) ****Backward Inductive Step****: Assume $P(k)$ is True for some arbitrary integer $k \leq 0$. Prove that $P(k - 1)$ is True.

By running the successor step in reverse, the "domino chain" moves backward along the negative integers, proving the statement for the entire non-positive set $\mathbb{Z}_{\leq 0}$. $\square$

Example 4.2.4 (Pure Abstract Discussion). Define the formal addition of integers using our equivalence classes of natural numbers, and prove that this addition is well-defined.

Solution. We define the addition of two integers $[(a, b)]$ and $[(c, d)]$ as:

$$[(a, b)] + [(c, d)] = [(a + c, b + d)]$$

To prove this is ****well-defined****, we must show that if we pick different representatives for our classes, say $(a_1, b_1) \sim (a_2, b_2)$ and $(c_1, d_1) \sim (c_2, d_2)$, the sum yields the same equivalence class:

$$\begin{aligned} (a_1, b_1) \sim (a_2, b_2) &\iff a_1 + b_2 = b_1 + a_2 \\ (c_1, d_1) \sim (c_2, d_2) &\iff c_1 + d_2 = d_1 + c_2 \end{aligned}$$

Adding these two equations of natural numbers:

$$(a_1 + c_1) + (b_2 + d_2) = (b_1 + d_1) + (a_2 + c_2)$$

By definition of our equivalence relation, this means:

$$(a_1 + c_1, b_1 + d_1) \sim (a_2 + c_2, b_2 + d_2)$$

Thus, the sum $[(a_1 + c_1, b_1 + d_1)] = [(a_2 + c_2, b_2 + d_2)]$, proving that addition does not depend on our choice of representative pairs. $\square$

Theorem 4.2.1 (Principle of Integers Induction). *Let $P(n)$ be a statement about an integer $n \in \mathbb{Z}$. If $P(0)$ is True, and for all $k \in \mathbb{Z}$, $P(k) \implies P(k + 1)$ and $P(k) \implies P(k - 1)$ are True, then $P(n)$ is True for all $n \in \mathbb{Z}$.*

Example 4.2.5 (Bridging Verification for Theorem 4.2). Imagine a two-way train track with station 0 at the center:

- You start by proving the light is ON at station 0 ($P(0)$ is True).
- The rule $P(k) \implies P(k + 1)$ pushes the light forward to stations $1, 2, 3, \dots$ ($\mathbb{N}_0$).
- The rule $P(k) \implies P(k - 1)$ pushes the light backward to stations $-1, -2, -3, \dots$ ($\mathbb{Z}^-$).

With both directions active, every integer station along the track is illuminated.

Proof of Theorem 4.2. We split the set of integers into two parts: $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and $\mathbb{Z}^- = \{-1, -2, -3, \dots\}$. **Part 1 (Non-negative integers)**: We are given $P(0)$ is True, and $\forall k \in \mathbb{N}_0, P(k) \implies P(k + 1)$. By standard Peano induction, $P(n)$ holds for all $n \in \mathbb{N}_0$. **Part 2 (Negative integers)**: We define an auxiliary statement $Q(m) := P(-m)$ for natural numbers $m \in \mathbb{N}_0$.

- Base Case: $Q(0) = P(0)$, which is True.
- Inductive Step: Assume $Q(k)$ is True, meaning $P(-k)$ is True. Using our given backward step $P(x) \implies P(x - 1)$ for $x = -k$, we get $P(-k - 1)$ is True. Since $-k - 1 = -(k + 1)$, this means $Q(k + 1)$ is True.

By induction on m , $Q(m)$ is True for all $m \in \mathbb{N}_0$, which means $P(n)$ is True for all negative integers $n \leq 0$. Combining Part 1 and Part 2, $P(n)$ is True for all $n \in \mathbb{Z}$. $\square$

Exercises 4.2

- (1) Determine if $(7, 3) \sim (5, 1)$ is True under our integer equivalence relation.
- (2) Determine if $(2, 6) \sim (1, 4)$ is True under our integer equivalence relation.
- (3) Write down the standard integer represented by the equivalence class $[(8, 2)]$.
- (4) Write down the standard integer represented by the equivalence class $[(1, 6)]$.
- (5) Calculate the sum $[(4, 1)] + [(2, 5)]$ using the formal definition of integer addition.
- (6) Let multiplication of integers be defined as $[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)]$. Calculate $[(3, 1)] \cdot [(2, 5)]$.
- (7) Prove that the integer multiplication defined in Exercise 6 is well-defined.
- (8) Show that $[(0, 0)]$ acts as the additive identity element of $\mathbb{Z}$.
- (9) Show that $[(1, 0)]$ acts as the multiplicative identity element of $\mathbb{Z}$.
- (10) Find the additive inverse of $[(a, b)]$ in our set of integers.
- (11) Define the ordering relation $\leq$ on $\mathbb{Z}$ using natural number arithmetic: $[(a, b)] \leq [(c, d)] \iff a + d \leq b + c$. Show that $[(2, 5)] \leq [(1, 1)]$.
- (12) Using Exercise 11, show that $[(0, 0)] \leq [(a, b)] \iff a \geq b$.
- (13) Explain how the equivalence class system allows us to define "subtraction" as simply adding an additive inverse.
- (14) Write down the three steps of proving a statement $P(n)$ for all $n \in \mathbb{Z}$ using combined forward and backward induction.
- (15) Let a_n be a sequence defined for all integers $n \in \mathbb{Z}$. If $a_0 = 5$, and $a_{n+1} = a_n + 2$ for all $n \in \mathbb{Z}$, calculate a_2 and a_{-2} .
- (16) Show that the set of negative integers $\mathbb{Z}^- = \{-1, -2, -3, \dots\}$ is not inductive under the standard definition (since adding 1 to -1 yields 0, which escapes the set).
- (17) Why did mathematicians require a formal construction of integers rather than just declaring "negative numbers exist"?
- (18) Prove that if $[(a, b)] \cdot [(c, d)] = [(0, 0)]$, then either $[(a, b)] = [(0, 0)]$ or $[(c, d)] = [(0, 0)]$. (Integral domain property).
- (19) Show how the integers $\mathbb{Z}$ can be viewed geometrically as a symmetric extension of $\mathbb{N}_0$ mirrored across the origin.
- (20) Discuss the importance of proving that operations on equivalence classes are "well-defined."

4.3 Rational Numbers

Just as subtraction forced us to expand our numbers from $\mathbb{N}_0$ to $\mathbb{Z}$, the operation of division forces another expansion. If you have 3 pizzas and want to distribute them equally among 4 people, there is no integer that can represent the size of each slice.

To help students with weak algebra skills, think of this as **sharing**. We represent a fraction as a pair of integers, (p, q) , where p is the number of parts we have, and q is the total number of cuts in a whole.

Definition 4.3.1 (Rational Numbers as Equivalence Classes). We define the set of rational numbers $\mathbb{Q}$ using the Cartesian product $\mathbb{Z} \times \mathbb{N}$, where an ordered pair (p, q) represents the fraction $\frac{p}{q}$.

Since many different fractions represent the same ratio (for example, $\frac{1}{2} = \frac{2}{4}$), we define an **equivalence relation** $\sim$ on $\mathbb{Z} \times \mathbb{N}$:

$$(p, q) \sim (r, s) \iff p \times s = q \times r$$

The set of **rational numbers** $\mathbb{Q}$ is the collection of all equivalence classes of these pairs:

$$\mathbb{Q} = \{[(p, q)] : p \in \mathbb{Z} \wedge q \in \mathbb{N}\}$$

Example 4.3.1 (Practical/Numerical Application). Show that the pairs $(1, 2)$ and $(3, 6)$ represent the same rational number in $\mathbb{Q}$ using our formal relation.

Solution. Let $(p, q) = (1, 2)$ and $(r, s) = (3, 6)$. We test the equivalence condition $p \times s = q \times r$:

- Left-hand calculation: $p \times s = 1 \times 6 = 6$.
- Right-hand calculation: $q \times r = 2 \times 3 = 6$.

Since $6 = 6$, the relation $(1, 2) \sim (3, 6)$ is True. Both pairs represent the same rational number, which corresponds to the standard fraction $\frac{1}{2}$. $\square$

Example 4.3.2 (Abstract Applied Mathematics Application). In computer vector graphics, images are scaled using fraction coefficients. If an image is scaled by a factor of 1.5, show how the computer's rendering engine stores this scale factor using only integers.

Solution. Floating point numbers can introduce rounding errors. To maintain perfect scaling precision, the rendering engine represents the scale factor 1.5 as an equivalence class of integer pairs:

$$1.5 \rightarrow [(3, 2)] \in \mathbb{Q}$$

When multiplying coordinate variables, the graphics processor calculates the scale coordinates by executing integer multiplication on the numerator and denominator separately. This prevents fractional pixel drift. $\square$

Example 4.3.3 (Generalization to More Variables). Is the set of rational numbers $\mathbb{Q}$ inductive? Discuss how the inductivity of $\mathbb{Z}$ helps us show that $\mathbb{Q}$ is inductive.

Solution. A set is inductive if we can construct a successor path. Because every rational number $r \in \mathbb{Q}$ can be mapped to an ordered pair of integers $(p, q) \in \mathbb{Z} \times \mathbb{N}$, and the integers are inductive (as shown in Subsection 4.2), we can establish inductive steps over $\mathbb{Q}$.

Specifically, we can define a grid pathway over the product set $\mathbb{Z} \times \mathbb{N}$. Since we can run induction along the horizontal paths (numerator p) and vertical paths (denominator q) using the inductivity of $\mathbb{Z}$ and $\mathbb{N}$ respectively, the rational set $\mathbb{Q}$ inherits this multi-dimensional inductive backbone. $\square$

Example 4.3.4 (Pure Abstract Discussion). Define the formal multiplication of rational numbers, and prove that this operation is well-defined.

Solution. We define the multiplication of two rational numbers $[(p, q)]$ and $[(r, s)]$ as:

$$[(p, q)] \cdot [(r, s)] = [(p \cdot r, q \cdot s)]$$

To prove this is ****well-defined****, we must show that if we choose different representatives $(p_1, q_1) \sim (p_2, q_2)$ and $(r_1, s_1) \sim (r_2, s_2)$, the product yields the same equivalence class:

$$\begin{aligned} (p_1, q_1) \sim (p_2, q_2) &\iff p_1 \cdot q_2 = q_1 \cdot p_2 \\ (r_1, s_1) \sim (r_2, s_2) &\iff r_1 \cdot s_2 = s_1 \cdot r_2 \end{aligned}$$

Multiplying these two equations of integers together:

$$(p_1 \cdot r_1) \cdot (q_2 \cdot s_2) = (q_1 \cdot s_1) \cdot (p_2 \cdot r_2)$$

By the definition of our rational equivalence relation, this means:

$$(p_1 \cdot r_1, q_1 \cdot s_1) \sim (p_2 \cdot r_2, q_2 \cdot s_2)$$

Thus, $[(p_1 \cdot r_1, q_1 \cdot s_1)] = [(p_2 \cdot r_2, q_2 \cdot s_2)]$, proving that multiplication is independent of our choice of representative pairs. $\square$

Theorem 4.3.1. *There exists a well-defined projection map $\pi : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ that is surjective (onto).*

Proof. $\square$

Exercises 4.3

- (1) Determine if $(2, 3) \sim (4, 6)$ is True under our rational equivalence relation.
- (2) Determine if $(-1, 3) \sim (3, -9)$ is True under our rational equivalence relation. (Note: denominator must be in $\mathbb{N}$).
- (3) Write down the fraction represented by the equivalence class $[(5, 8)]$.
- (4) Write down the fraction represented by the equivalence class $[(-2, 7)]$.
- (5) Calculate the product $[(1, 3)] \cdot [(2, 5)]$ using our formal definition of rational multiplication.
- (6) Define addition of rational numbers as $[(p, q)] + [(r, s)] = [(ps + qr, qs)]$. Calculate $[(1, 2)] + [(1, 3)]$.
- (7) Prove that the rational addition defined in Exercise 6 is well-defined.
- (8) Show that $[(0, 1)]$ acts as the additive identity element of $\mathbb{Q}$.
- (9) Show that $[(1, 1)]$ acts as the multiplicative identity element of $\mathbb{Q}$.
- (10) Find the multiplicative inverse of a non-zero rational number $[(p, q)]$.
- (11) Prove that there are no "zero divisors" in $\mathbb{Q}$: if $a \cdot b = 0$ in $\mathbb{Q}$, then either $a = 0$ or $b = 0$.
- (12) Define the ordering relation $\leq$ on $\mathbb{Q}$ using integer arithmetic: $[(p, q)] \leq [(r, s)] \iff p \cdot s \leq q \cdot r$. Show that $[(1, 3)] \leq [(1, 2)]$.
- (13) Using Exercise 12, prove that the rational numbers are dense: between any two distinct rational numbers, there is another rational number.
- (14) Why does the density of $\mathbb{Q}$ (shown in Exercise 13) make it impossible to define a "successor" by simply adding a fixed smallest fraction?

- (15) Show how we can still run induction over $\mathbb{Q}$ by instead inducting on the components (numerator and denominator) inside the product set $\mathbb{Z} \times \mathbb{N}$.
- (16) Explain the logical difference between the set of integers $\mathbb{Z}$ (where every number has a next consecutive number) and the set of rationals $\mathbb{Q}$ (where there is no next consecutive number).
- (17) Let $S = \{[(p, q)] \in \mathbb{Q} : p \in \mathbb{N} \wedge q = 1\}$. Show that S is isomorphic to the set of natural numbers $\mathbb{N}$.
- (18) Show that any rational number can be written in a unique "simplest form" where $\gcd(p, q) = 1$.
- (19) Why must we restrict the denominator of our rational representatives to the natural numbers $\mathbb{N}$ (excluding 0)?
- (20) Summarize how equivalence classes allow us to construct fractions from raw integers.

4.4 Generalized Induction

Now that we understand how Cartesian products combine sets, we can expand our induction framework to multi-dimensional coordinate spaces.

To help students with weak algebra backgrounds, think of this like a grid of light bulbs. If you want to turn on every bulb on a flat table, you do not just walk in a single line. You start at the corner $(1, 1)$, and show two rules: how to move one step right ($x \rightarrow x + 1$), and how to move one step up ($y \rightarrow y + 1$).

By generalizing our linear backbone to product sets like $\mathbb{N} \times \mathbb{N}$ (or $\mathbb{N}^k$), we can perform induction over grids.

Definition 4.4.1 (Isomorphic Sets). Two sets X and Y are *isomorphic* (denoted $X \cong Y$) in an algebraic sense if there exists a one-to-one and onto function (a bijection) $f : X \rightarrow Y$ that preserves the structural operations (like addition or successor steps).

Example 4.4.1 (Practical/Numerical Application). Suppose we want to prove a statement $P(x, y)$ is True for all grid points $(x, y) \in \mathbb{N} \times \mathbb{N}$. Write down the multi-variable induction template required to complete this proof.

Solution. To prove $P(x, y)$ holds for all $(x, y) \in \mathbb{N} \times \mathbb{N}$, we use the following template:

- (1) **Base Case**: Prove $P(1, 1)$ is True.
- (2) **Horizontal Inductive Step**: Assume $P(k, y)$ is True for all $y \in \mathbb{N}$. Prove that $P(k + 1, y)$ is True.
- (3) **Vertical Inductive Step**: Assume $P(x, m)$ is True for all $x \in \mathbb{N}$. Prove that $P(x, m + 1)$ is True.

By proving these grid transition steps, the truth value spreads across the entire two-dimensional surface. $\square$

Example 4.4.2 (Abstract Applied Mathematics Application). In computer matrix operations, a pathfinding algorithm searches a 2D array of size $M \times N$. Prove that the coordinate pointer can visit every cell using a nested induction loop.

Solution. Let $P(i, j)$ be the statement: "The cell at row i , column j is reachable."

- Base Case: The algorithm starts at the top-left cell, so $P(1, 1)$ is True.
- Step 1: If the cell at (i, j) is reachable, the program has a rule to step right, making $(i, j + 1)$ reachable.
- Step 2: If an entire row i is verified, the program has a rule to step down to the next row, making $(i + 1, 1)$ reachable.

By modeling the array as the Cartesian product $\{1, \dots, M\} \times \{1, \dots, N\}$, the nested loops mathematically span the entire matrix without skipping any cells. $\square$

Example 4.4.3 (Generalization to More Variables). Suppose we want to perform induction in k -dimensional space, $\mathbb{N}^k = \mathbb{N} \times \mathbb{N} \times \dots \times \mathbb{N}$. Write down the general statement of inductivity for k -tuples.

Solution. A statement $P(n_1, n_2, \dots, n_k)$ is True for all $(n_1, n_2, \dots, n_k) \in \mathbb{N}^k$ if:

- (1) **Base Case**: $P(1, 1, \dots, 1)$ is True.
- (2) **Inductive Steps**: For each coordinate axis $i \in \{1, 2, \dots, k\}$, if the statement holds for $(n_1, \dots, n_i, \dots, n_k)$, it must hold for the successor coordinate along that axis:

$$P(n_1, \dots, n_i, \dots, n_k) \implies P(n_1, \dots, n_i + 1, \dots, n_k)$$

Since $\mathbb{N}^k$ is constructed from finite Cartesian products of inductive sets, this multi-axis coordinate step guarantees that the statement holds across the entire k -dimensional space. $\square$

Example 4.4.4 (Pure Abstract Discussion). Discuss how lexicographical ordering (dictionary order) allows us to define a linear path through the product set $\mathbb{N} \times \mathbb{N}$, turning a 2D induction back into a 1D induction.

Solution. We can define a total order relation $\prec$ on $\mathbb{N} \times \mathbb{N}$ as follows:

$$(a, b) \prec (c, d) \iff (a < c) \vee (a = c \wedge b < d)$$

This orders our pairs just like words in a dictionary:

$$(1, 1) \prec (1, 2) \prec (1, 3) \prec \dots \prec (2, 1) \prec (2, 2) \prec \dots$$

Because this lexicographical relation is a well-ordering (every non-empty subset has a least element), we can map the elements of $\mathbb{N} \times \mathbb{N}$ to a single linear path. This isomorphism allows us to prove statements on 2D grids using standard, one-dimensional mathematical induction. $\square$

Theorem 4.4.1. *If $S_1, S_2, \dots, S_k$ are inductive sets, then their Cartesian product $S_1 \times S_2 \times \dots \times S_k$ is inductive under coordinate-wise progression.*

Example 4.4.5 (Bridging Verification for Theorem 4.4). Consider the 2D grid $S_1 \times S_2 = \mathbb{N} \times \mathbb{N}$. Let us check if we can step from $(2, 3)$ along each coordinate axis:

- Taking a step along the first axis gives $(S(2), 3) = (3, 3) \in \mathbb{N} \times \mathbb{N}$.
- Taking a step along the second axis gives $(2, S(3)) = (2, 4) \in \mathbb{N} \times \mathbb{N}$.

Since individual steps stay inside the grid, the entire product space preserves inductivity.

Proof of Theorem 4.4. We prove this by induction on the number of sets k . **Base Case** ($k = 1$): A single inductive set S_1 is inductive by definition. **Inductive Step:** Assume that for k inductive sets, the Cartesian product $P_k = S_1 \times \dots \times S_k$ is inductive under coordinate-wise progression. Now consider $k + 1$ sets, forming $P_{k+1} = P_k \times S_{k+1}$. Let $(u, v) \in P_k \times S_{k+1}$, where $u = (n_1, \dots, n_k) \in P_k$ and $v = n_{k+1} \in S_{k+1}$.

- For any axis $i \leq k$, applying the successor step shifts the i -th coordinate of u to $S(n_i)$. Since P_k is inductive, the new tuple $u' \in P_k$, so $(u', v) \in P_{k+1}$.
- For axis $k + 1$, applying the successor step shifts v to $S(v)$. Since S_{k+1} is inductive, $S(v) \in S_{k+1}$, so $(u, S(v)) \in P_{k+1}$.

Thus, coordinate-wise progression is closed in P_{k+1} , proving the theorem for all $k \in \mathbb{N}$. $\square$

Exercises 4.4

- (1) Define what it means for two sets to be "isomorphic."
- (2) Write down the first five elements of $\mathbb{N} \times \mathbb{N}$ under lexicographical ordering.
- (3) Let $P(x, y)$ be a statement on grid coordinates. If $P(1, 1)$ is True, and $P(x, y) \implies P(x + 1, y)$ is True, can we conclude $P(2, 3)$ is True? Why or why not?
- (4) Prove that the set $\{2n : n \in \mathbb{N}\}$ is isomorphic to $\mathbb{N}$.
- (5) Let S be the grid $\mathbb{N} \times \mathbb{N}$. Show that S is closed under the coordinate successor steps $S_x(x, y) = (x + 1, y)$ and $S_y(x, y) = (x, y + 1)$.

- (6) Write down a coordinate pathway that connects $(1, 1)$ to $(3, 4)$ using only successor steps.
- (7) Explain the "grid of light bulbs" analogy for 2D induction in your own words.
- (8) Let $P(n_1, n_2)$ be a statement. If we prove $P(1, 1)$ is True, and $P(n_1, n_2) \implies P(n_1 + 1, n_2 + 1)$, does this prove the statement for $(1, 2)$? Explain.
- (9) Show that lexicographical ordering satisfies the Trichotomy Law for any two distinct pairs (a, b) and (c, d) .
- (10) Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be defined by $f(n) = \frac{n}{2}$ if n is even, and $f(n) = -\frac{n-1}{2}$ if n is odd. Show that f is a bijection, proving $\mathbb{N} \cong \mathbb{Z}$.
- (11) Based on Exercise 10, explain why induction can be performed over the entire set of integers $\mathbb{Z}$.
- (12) Prove that if $A \cong B$ and $B \cong C$, then $A \cong C$ (transitivity of isomorphism).
- (13) Let $P(x, y)$ be a statement. If we use the lexicographical pathway, write down the inductive hypothesis.
- (14) Explain why branching structures are more difficult to index than grid structures.
- (15) Let $x_{i,j}$ be a 2D sequence where $x_{1,1} = 2$, $x_{i+1,j} = 2x_{i,j}$, and $x_{i,j+1} = 3x_{i,j}$. Find $x_{2,2}$.
- (16) Using induction, prove that the terms of the sequence in Exercise 15 are given by $x_{i,j} = 2^i 3^{j-1}$.
- (17) Show that $\mathbb{N} \times \mathbb{N}$ has the same cardinality (size) as $\mathbb{N}$ by describing a zigzag mapping pathway.
- (18) Why does the existence of multi-dimensional induction prove that our coordinate math is logically consistent?
- (19) Let $P(x, y, z)$ be a statement on $\mathbb{N}^3$. Write down the complete list of coordinate steps required for an inductive step.
- (20) Summary: Why does the inductivity of $\mathbb{N}$ guarantee that any product set $\mathbb{N}^k$ is also inductive?

5 The Countability of Reals

5.1 The Countability of $\mathbb{Q}$

When we say a set of items is "countable," we intuitively mean that we can list them one by one in a systematic order ($1^{\text{st}}, 2^{\text{nd}}, 3^{\text{rd}}, \dots$) so that every single item eventually gets a label. This matches our backbone of natural numbers $\mathbb{N}$.

To help students who are weak at algebra, think of this like ticket numbering at a bakery: as long as everyone waiting gets a unique queue ticket, the customer count is orderly and reachable. Even if the shop queue is infinite, if the ticket machine never jams and every customer eventually gets served, the crowd is **"countable"**. Surprisingly, even though the rational numbers $\mathbb{Q}$ (fractions) are packed incredibly dense on the number line, we can still queue them up perfectly!

Definition 5.1.1 (Countable and Uncountable Sets). We define the sizes of infinite sets as follows:

- A set S is **"countable"** if it is finite, or if there exists a one-to-one and onto function (a bijection) $f : \mathbb{N} \rightarrow S$. This bijection essentially numbers the elements of S as $\{f(1), f(2), f(3), \dots\}$. Any infinite countable set is called **"countably infinite"**.
- A set S is **"uncountable"** if it is infinite but there is no possible way to build a bijection between $\mathbb{N}$ and S . It is "too large" to be placed into a single linear queue.

Example 5.1.1 (Practical/Numerical Application). Let $S = \{2, 4, 6, 8, \dots\}$ be the set of positive even integers. Show that S is countable by constructing an explicit queue numbering system.

Solution. To prove S is countable, we must find a bijection $f : \mathbb{N} \rightarrow S$. Let us define the formula:

$$f(n) = 2n$$

Let us verify the first few steps of our queue:

- Ticket $n = 1 \implies f(1) = 2 \times 1 = 2$.
- Ticket $n = 2 \implies f(2) = 2 \times 2 = 4$.
- Ticket $n = 3 \implies f(3) = 2 \times 3 = 6$.

Since every positive even integer $2k$ will eventually receive ticket number k , the function is injective and surjective. Thus, S is countably infinite. $\square$

Example 5.1.2 (Abstract Applied Mathematics Application). In computer networking, an infinite stream of data packets is sent from various port numbers. Let A be the set of active packet IDs. If $A = \mathbb{Z}$ (all positive and negative integers), show how a network router can index these packets using only positive natural numbers $\mathbb{N}$ to keep its memory indexing simple.

Solution. The router must map the set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ to the positive counting numbers $\mathbb{N}$. It can arrange $\mathbb{Z}$ in a zigzag queue:

$$0, \quad 1, \quad -1, \quad 2, \quad -2, \quad 3, \quad -3, \quad \dots$$

We write this bijection $g : \mathbb{N} \rightarrow \mathbb{Z}$ as:

$$g(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd} \end{cases}$$

Let us test this rule:

- $n = 1$ (odd) $\implies g(1) = -\frac{1-1}{2} = 0$.

- $n = 2$ (even) $\implies g(2) = \frac{2}{2} = 1$.
- $n = 3$ (odd) $\implies g(3) = -\frac{3-1}{2} = -1$.
- $n = 4$ (even) $\implies g(4) = \frac{4}{2} = 2$.

Every single positive and negative integer gets a unique indexing position, showing that $\mathbb{Z}$ is countable. $\square$

Example 5.1.3 (Generalization to More Variables). The set of all positive fractions (rational numbers $\mathbb{Q}^+$) is represented by coordinate pairs $(p, q) \in \mathbb{N} \times \mathbb{N}$, where each pair means $\frac{p}{q}$. Show how we can queue up these two variables to prove $\mathbb{Q}^+$ is countable.

Solution. If we try to write the fractions in order of size, we get stuck because there is no "smallest" fraction to start with.

Instead, we organize them in a 2D grid where the row represents the numerator p and the column represents the denominator q :

$$\begin{array}{ccccc} 1/1 & 1/2 & 1/3 & 1/4 & \dots \\ 2/1 & 2/2 & 2/3 & 2/4 & \dots \\ 3/1 & 3/2 & 3/3 & 3/4 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Now, we draw a diagonal zigzag path starting from the top left corner:

$$\frac{1}{1} \rightarrow \frac{1}{2} \rightarrow \frac{2}{1} \rightarrow \frac{3}{1} \rightarrow \frac{2}{2} \rightarrow \frac{1}{3} \rightarrow \frac{1}{4} \dots$$

If we encounter a fraction that can be simplified (like $\frac{2}{2} = 1$, which is the same as $\frac{1}{1}$), we simply skip it.

Because every possible positive fraction $\frac{p}{q}$ lies somewhere in our grid at row p and column q , our zigzag line is guaranteed to cross it eventually. This diagonal queuing system establishes a bijection with $\mathbb{N}$, proving that $\mathbb{Q}^+$ is countable. $\square$

Example 5.1.4 (Pure Abstract Discussion). Prove that any subset A of a countable set B must also be countable.

Solution. If A is finite, it is countable by definition.

If A is infinite, since B is countable, we can list the elements of B as a sequence:

$$B = \{b_1, b_2, b_3, \dots\}$$

To build a queue for A , we search through this list of B from left to right. We let a_1 be the first element in the sequence that belongs to A . We let a_2 be the next element that belongs to A , and so on.

Because A is infinite, this process never terminates, and we generate an infinite list:

$$A = \{a_1, a_2, a_3, \dots\}$$

Since every element of A is some b_i , and i is a finite index, every element of A will eventually be reached and numbered. Thus, we have constructed a bijection with $\mathbb{N}$, proving A is countable. $\square$

Theorem 5.1.1. *The set of all rational numbers $\mathbb{Q}$ is countable.*

Example 5.1.5 (Bridging Verification for Theorem 5.1). Let us trace how Cantor's queue assigns unique integer tickets to positive fractions without missing any:

- Ticket 1 is given to $\frac{1}{1}$ (Sum of numerator and denominator is 2).
- Ticket 2 is given to $\frac{1}{2}$, Ticket 3 to $\frac{2}{1}$ (Sum is 3).

- Ticket 4 is given to $\frac{3}{1}$, Ticket 5 to $\frac{1}{3}$ (skipping $\frac{2}{2}$ since it equals $\frac{1}{1}$).

Every fraction $\frac{p}{q}$ has a finite coordinate sum $p + q$, so it is guaranteed to receive a finite ticket number!

Proof of Theorem 5.1. We first show that $\mathbb{Q}^+$ (positive rationals) is countable. Define a mapping $\phi : \mathbb{Q}^+ \rightarrow \mathbb{N} \times \mathbb{N}$ by sending each fraction in simplest form $r = \frac{p}{q}$ (where $\gcd(p, q) = 1$) to the ordered pair (p, q) . Since ϕ is injective, $\mathbb{Q}^+$ is isomorphic to a subset of $\mathbb{N} \times \mathbb{N}$. In Section 4.4, we established that $\mathbb{N} \times \mathbb{N}$ is countable; since any subset of a countable set is countable (proven in 5.1), $\mathbb{Q}^+$ is countable. Now, the entire set of rationals can be written as the union of three sets:

$$\mathbb{Q} = \mathbb{Q}^+ \cup \{0\} \cup \mathbb{Q}^-$$

Since $\mathbb{Q}^-$ is trivially isomorphic to $\mathbb{Q}^+$ via the bijection $f(x) = -x$, $\mathbb{Q}^-$ is also countable. Because the finite union of countable sets is countable, $\mathbb{Q}$ is countable. $\square$

Exercises 5.1

- (1) Define what it means for an infinite set to be "countable."
- (2) Show that the set of positive multiples of 3, $S = \{3, 6, 9, 12, \dots\}$, is countable.
- (3) Show that the set of odd natural numbers $T = \{1, 3, 5, 7, \dots\}$ is countable.
- (4) Is the set of prime numbers countable? Explain using the subset rule.
- (5) Find the first 7 fractions in the zigzag diagonal queue for $\mathbb{Q}^+$ (remember to skip duplicates).
- (6) If A and B are disjoint countable sets, prove that their union $A \cup B$ is countable. (Hint: Alternately select elements from A and B).
- (7) Show that the set of all integers $\mathbb{Z}$ is countable without using the piecewise formula.
- (8) Explain the "bakery queue" analogy for countability in your own words.
- (9) Let S be a countable set. If we add a single new element x to S , is the new set $S \cup \{x\}$ still countable? Explain.
- (10) Show that the Cartesian product $\mathbb{N} \times \mathbb{N}$ is countable.
- (11) Why is it impossible to list the rational numbers in order of size (increasing order)?
- (12) Explain how the skipping of duplicates in Cantor's zigzag argument ensures the mapping is injective.
- (13) Let A be a countable set, and let $f : A \rightarrow B$ be a surjective map. Is B guaranteed to be countable? Explain.
- (14) Prove that the set of all powers of 2, $P = \{2^0, 2^1, 2^2, 2^3, \dots\}$, is countable.
- (15) Show that the set of all perfect squares is countable.
- (16) Discuss why the concept of "countable infinity" is denoted by the Hebrew letter aleph-null, $\aleph_0$.
- (17) Let $A_1, A_2, A_3, \dots$ be a countable collection of countable sets. Show that their union $\bigcup_{i=1}^{\infty} A_i$ is countable.
- (18) If a set S has a bijection with $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$, is it countable?
- (19) Can we perform standard mathematical induction over any countably infinite set? Discuss why or why not.
- (20) Summary: Why does the zigzag path prove that a 2D grid is no "larger" than a 1D line of numbers?

5.2 The Existence of Irrationals

In secondary school, we are taught that any real number can be written as a decimal. Some decimals terminate (like $0.25 = \frac{1}{4}$), and some repeat forever (like $0.333\ldots = \frac{1}{3}$). It is highly tempting for a weak algebra student to believe that **all** decimal numbers can eventually be converted into fractions of whole numbers.

However, this is not true. There are numbers on the real line that can never, under any circumstances, be written as a ratio of two integers. These are called **irrational numbers**.

Definition 5.2.1 (Irrational Numbers ($\mathbb{Q}^c$)). The set of **irrational numbers**, denoted by $\mathbb{Q}^c$ (the complement of the rational numbers), is the set of all real numbers that cannot be expressed as a fraction $\frac{p}{q}$ for any integers p and q ($q \neq 0$):

$$\mathbb{Q}^c = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}$$

Example 5.2.1 (Practical/Numerical Application). Prove that the number $\sqrt{2}$ cannot be a rational number.

Solution. We write a classic **Proof by Contradiction**:

(1) Assume $\sqrt{2}$ is rational. This means we can write:

$$\sqrt{2} = \frac{p}{q} \quad \text{for some integers } p, q \in \mathbb{Z} \quad (q \neq 0)$$

(2) We assume this fraction is in simplest form, meaning p and q share no common factors ($\gcd(p, q) = 1$).

(3) Square both sides of the equation:

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2$$

(4) Since p^2 is equal to 2 times an integer, p^2 must be even. By our results in Section 1, if p^2 is even, then p must also be even. Thus, we can write $p = 2k$ for some integer k .

(5) Substitute $p = 2k$ back into our equation:

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies q^2 = 2k^2$$

(6) Since q^2 is equal to 2 times an integer, q^2 must be even, which implies q is also even.

(7) If both p and q are even, they both share a common factor of 2.

This directly contradicts our assumption that $\gcd(p, q) = 1$. Thus, our starting assumption must be False. We conclude that $\sqrt{2}$ is irrational. $\square$

Example 5.2.2 (Abstract Applied Mathematics Application). In designing geometric components like a square solar panel of side length 1 meter, the diagonal distance is exactly $\sqrt{2}$ meters. Show why a computer-aided design (CAD) tool cannot represent this diagonal length with absolute, perfect fractional precision.

Solution. The diagonal length d satisfies the Pythagorean theorem:

$$d^2 = 1^2 + 1^2 = 2 \implies d = \sqrt{2}$$

Because $\sqrt{2}$ is irrational (as proven above), it cannot be written as any fraction $\frac{p}{q}$.

No matter how high we set the resolution of the CAD software (even if we use a fraction with a trillion in the denominator), the computer can only store a rational approximation. This mathematical limitation means physical manufacturing must always allow for a tolerance offset ($\pm\epsilon$) when cutting diagonals. $\square$

Example 5.2.3 (Generalization to More Variables). Let us generalize our surd proof to more variables. Prove that if p is any prime number, then $\sqrt{p}$ is irrational.

Solution. We use the same proof by contradiction structure for any prime variable p :

- (1) Assume $\sqrt{p} = \frac{a}{b}$ with $\gcd(a, b) = 1$.
- (2) Square both sides: $p = \frac{a^2}{b^2} \implies a^2 = pb^2$.
- (3) This means a^2 is a multiple of the prime p . A fundamental lemma in number theory (Euclid's Lemma) states that if a prime p divides a^2 , then p must divide a . Thus, $a = p \cdot k$ for some integer k .
- (4) Substitute back:

$$(pk)^2 = pb^2 \implies p^2k^2 = pb^2 \implies b^2 = pk^2$$

- (5) This implies b^2 is a multiple of prime p , which means p must divide b .
- (6) Both a and b share the common prime factor p , contradicting $\gcd(a, b) = 1$.

Thus, $\sqrt{p}$ is irrational for any prime p . This generalizes our simple $\sqrt{2}$ proof to an infinite collection of irrational square roots. $\square$

Example 5.2.4 (Pure Abstract Discussion). Discuss how the discovery of irrational numbers by the Pythagoreans in ancient Greece shattered their philosophy that "all things are numbers" (which to them meant whole-number ratios).

Solution. The ancient Greeks believed that any two physical lengths L_1 and L_2 were commensurable, meaning they could find some tiny unit of length u that fit perfectly into both lengths a whole number of times. This is equivalent to saying that the ratio $\frac{L_1}{L_2}$ is a rational number.

When Hippasus proved that the diagonal of a unit square is incommensurable with its side (meaning their ratio $\sqrt{2}$ is irrational), it proved that geometry cannot be reduced to simple whole-number counts. Legend says the Pythagoreans were so disturbed by this logical reality that they drowned Hippasus at sea to protect their mathematical philosophy. $\square$

Theorem 5.2.1. *Let r be a rational number and s be an irrational number. Then their sum $r + s$ is irrational.*

Example 5.2.5 (Bridging Verification for Theorem 5.2). Let us test what happens if we add the rational number $r = 5$ to the irrational number $s = \sqrt{2}$:

- Suppose their sum $5 + \sqrt{2}$ was a simple fraction, say x .
- Then we could rearrange the equation: $\sqrt{2} = x - 5$.
- Since subtracting an integer from a fraction always gives another fraction, this would force $\sqrt{2}$ to be rational!

Since $\sqrt{2}$ is strictly irrational, our assumption is impossible. Thus, $5 + \sqrt{2}$ must be irrational.

Proof of Theorem 5.2. We proceed by contradiction. Let $r \in \mathbb{Q}$ be a rational number, and let $s \in \mathbb{Q}^c$ be an irrational number. Assume that their sum $x = r + s$ is a rational number ($x \in \mathbb{Q}$). By basic algebraic manipulation, we can isolate s :

$$s = x - r$$

We know that the set of rational numbers $\mathbb{Q}$ is a field, which means it is closed under subtraction. Since both x and r are rational, their difference $x - r$ must also be rational. This implies that $s \in \mathbb{Q}$, which directly contradicts our initial given premise that s is irrational ($s \notin \mathbb{Q}$). Therefore, our assumption is False, and the sum $r + s$ must be irrational. $\square$

Exercises 5.2

- (1) Explain the difference between a rational number and an irrational number.
- (2) Prove that $\sqrt{3}$ is irrational.
- (3) Prove that $\sqrt{5}$ is irrational.
- (4) Try to apply the proof of $\sqrt{2}$'s irrationality to $\sqrt{4}$. At which exact step does the logic break down?
- (5) Let r be rational and s be irrational. Prove by contradiction that $r \times s$ is irrational (assume $r \neq 0$).
- (6) Is the sum of two irrational numbers always irrational? Find a counterexample.
- (7) Is the product of two irrational numbers always irrational? Find a counterexample.
- (8) Classify the following numbers as rational or irrational: $\frac{22}{7}$, $\sqrt{8}$, 3.14159, π .
- (9) If a is a rational number, is a^2 guaranteed to be rational? Explain.
- (10) If b is an irrational number, is b^2 guaranteed to be irrational? Find a counterexample.
- (11) Show that $\sqrt{2} + \sqrt{3}$ is irrational.
- (12) Why is it impossible for a computer screen with discrete pixels to draw a perfect straight diagonal line of a unit square?
- (13) If $\gcd(p, q) = 1$ means a fraction is in simplest form, explain why any fraction can be written in simplest form.
- (14) Let p and q be primes. Show that $\sqrt{pq}$ is irrational.
- (15) Explain Euclid's Lemma: "If a prime p divides ab , then $p \mid a$ or $p \mid b$." How is this used in our proofs?
- (16) Show that $2 + \sqrt{2}$ is irrational.
- (17) Show that $3\sqrt{2}$ is irrational.
- (18) What is the decimal expansion behavior of an irrational number compared to a rational number?
- (19) Why does the existence of irrational numbers prove that our counting numbers $\mathbb{N}$ are not enough to fully describe continuous physical lengths?
- (20) Summary: State three different geometric scenarios where irrational lengths arise naturally.

5.3 The Density of Irrationals

A fascinating property of numbers is how they are arranged on the line. If you pick any two integers, say 3 and 4, there are no other integers between them. The integers are discrete and separated by empty gaps.

However, rational and irrational numbers do not have gaps. To help students with weak algebra backgrounds, think of this like sand on a beach: no matter how closely you zoom in on two grains of sand, there are always more grains packed between them. This property is called **density**.

Definition 5.3.1 (Density). A subset $S \subseteq \mathbb{R}$ is **dense** in $\mathbb{R}$ if for any two real numbers a and b satisfying $a < b$, there exists an element $s \in S$ such that:

$$a < s < b$$

This means that within any interval, no matter how microscopically small, we can always find an element of S .

Example 5.3.1 (Practical/Numerical Application). Let $a = 0.5$ and $b = 0.51$. Find a rational number and an irrational number that lie strictly between a and b .

Solution. We seek numbers x such that $0.5 < x < 0.51$:

- To find a rational number, we can compute the midpoint:

$$r = \frac{a+b}{2} = \frac{0.5+0.51}{2} = \frac{1.01}{2} = 0.505$$

Since $0.505 = \frac{505}{1000}$ is a fraction of integers, $r \in \mathbb{Q}$ and lies strictly between a and b .

- To find an irrational number, we can add a tiny irrational offset to a . Let:

$$s = 0.5 + \frac{\sqrt{2}}{1000}$$

Since $\sqrt{2} \approx 1.414$, we have $\frac{\sqrt{2}}{1000} \approx 0.001414$. Thus, $s \approx 0.501414$. Since $0.5 < 0.501414 < 0.51$, and adding a rational and irrational yields an irrational (proven in 5.2), $s \in \mathbb{Q}^c$ and lies strictly between a and b .

□

Example 5.3.2 (Abstract Applied Mathematics Application). In computer floating-point calculations, the system uses binary decimals. Show why the density of rational numbers guarantees that we can always increase numerical precision, but also explains why floating-point errors are inevitable.

Solution. Because the rational numbers are dense, if we have two different measurements a and b , we can always find a more precise value $c = \frac{a+b}{2}$ between them.

However, because the irrational numbers are **also** dense, any real physical quantity (like a temperature or speed) is highly likely to be an irrational value. Since a computer can only represent rational binary fractions, the dense packing of irrationals means there is always an unrepresentable irrational state infinitesimally close to our computer's best approximation. Thus, tiny rounding errors are mathematically guaranteed. □

Example 5.3.3 (Generalization to More Variables). Suppose we have n intervals on the real line: $I_1, I_2, \dots, I_n$. Show how density guarantees we can select a sequence of n distinct irrational numbers $s_1, s_2, \dots, s_n$ such that each $s_i \in I_i$.

Solution. Let each interval be $I_i = (a_i, b_i)$ where $a_i < b_i$.

By the density of the irrational numbers, for the first interval (a_1, b_1) , there exists some irrational number s_1 such that $a_1 < s_1 < b_1$.

Similarly, for any general interval $I_i = (a_i, b_i)$, we can select an irrational s_i satisfying $a_i < s_i < b_i$. Since the irrational numbers are dense, we can easily select these s_i to be completely distinct from each other. This shows how density allows us to populate any finite grid of intervals with unique irrational variables. $\square$

Example 5.3.4 (Pure Abstract Discussion). Prove that between any two rational numbers $x < y$, there exists an irrational number.

Solution. Since $x < y$, we subtract x from both sides to get $0 < y - x$.

Since $\sqrt{2} > 1$, we know that $\frac{1}{\sqrt{2}} < 1$. Thus:

$$0 < \frac{y - x}{\sqrt{2}} < y - x$$

Now, let us add x to all parts of the inequality:

$$x < x + \frac{y - x}{\sqrt{2}} < y$$

Let $z = x + \frac{y - x}{\sqrt{2}}$. We check the algebraic nature of z :

- z can be rewritten as: $x(1 - \frac{1}{\sqrt{2}}) + \frac{y}{\sqrt{2}}$.
- Because x, y are rational, and $\sqrt{2}$ is irrational, this linear combination yields an irrational value (by our rules from 5.2).

Thus, z is an irrational number, and it lies strictly between x and y . This proves the density of irrationals between any two rational boundaries. $\square$

Theorem 5.3.1. *Between any two distinct real numbers $a < b$, there exists a rational number r such that $a < r < b$.*

Example 5.3.5 (Bridging Verification for Theorem 5.3). Let $a = 3.1415$ and $b = 3.1416$. Let us find a rational fraction strictly between them:

- The gap between them is $b - a = 0.0001 = \frac{1}{10000}$.
- If we choose a denominator larger than 10000, say $q = 100000$, the step size $\frac{1}{100000}$ is smaller than the gap.
- Taking 314155 steps of size $\frac{1}{100000}$ lands us at $r = \frac{314155}{100000} = 3.14155$.

Since $3.1415 < 3.14155 < 3.1416$, we have trapped a rational number inside the interval!

Proof of Theorem 5.3. Let $a, b \in \mathbb{R}$ with $a < b$. Then the length of the interval is strictly positive: $b - a > 0$. By the Archimedean Property of real numbers, there exists a natural number $q \in \mathbb{N}$ large enough such that:

$$\frac{1}{q} < b - a \implies 1 < q(b - a) \implies qa + 1 < qb$$

Now consider the real number qa . By the Well-Ordering Principle of integers, there exists a unique integer $p \in \mathbb{Z}$ that is the smallest integer strictly greater than qa :

$$p - 1 \leq qa < p \implies qa < p \leq qa + 1$$

Combining our two inequalities ($p \leq qa + 1$ and $qa + 1 < qb$), we establish a strict chain:

$$qa < p < qb$$

Since $q > 0$, we divide all parts of the inequality by q :

$$a < \frac{p}{q} < b$$

Letting $r = \frac{p}{q} \in \mathbb{Q}$, we have proven that a rational number exists strictly between a and b . $\square$

Exercises 5.3

- (1) State the definition of a set S being "dense" in $\mathbb{R}$.
- (2) Find a rational number between $\frac{1}{3}$ and $\frac{1}{2}$.
- (3) Find an irrational number between $\frac{1}{3}$ and $\frac{1}{2}$.
- (4) Let $a = 1.414$ and $b = 1.415$. Find a rational number between them.
- (5) Let $a = 1.414$ and $b = 1.415$. Find an irrational number between them.
- (6) If we have two rational numbers x and y , show that their average $\frac{x+y}{2}$ is always rational.
- (7) Why does the result of Exercise 6 prove that we can find infinitely many rational numbers between any two rationals?
- (8) Is the set of integers $\mathbb{Z}$ dense in $\mathbb{R}$? Find a counterexample interval.
- (9) Is the set of natural numbers $\mathbb{N}$ dense in $\mathbb{R}$? Explain.
- (10) Prove that if r is a non-zero rational, then $r\sqrt{2}$ is irrational.
- (11) Let $a < b$. Show that there exists an integer n such that $n(b-a) > 1$ (this relies on the Archimedean Property).
- (12) Using Exercise 11, explain how we can always find a fraction of the form $\frac{m}{n}$ between a and b .
- (13) Explain the "sand on a beach" analogy for density in your own words.
- (14) Show that between any two irrational numbers, there is a rational number.
- (15) Show that between any two irrational numbers, there is another irrational number.
- (16) If a set is dense in $\mathbb{R}$, is it guaranteed to be infinite? Explain.
- (17) Let S be a dense set in $\mathbb{R}$. If we remove a single element x from S , is the remaining set still dense?
- (18) Discuss why the density of numbers makes drawing a continuous line on a computer screen challenging.
- (19) Prove that the set of all decimals that end in 5 is dense in $\mathbb{R}$.
- (20) Summary: Contrast the spacing of integers $\mathbb{Z}$ with the spacing of rational numbers $\mathbb{Q}$.

5.4 The Uncountability of $\mathbb{Q}^c$

We have proven two remarkable facts: the set of rational numbers $\mathbb{Q}$ is countable (meaning we can list them in a single queue), and the set of irrational numbers $\mathbb{Q}^c$ is packed densely between them.

Now, we must ask: is the set of irrational numbers $\mathbb{Q}^c$ also countable? Can we place all irrationals into a queue? The answer is a resounding **no**. The irrationals are so incredibly numerous that they can never be placed into any list. They represent a larger, deeper level of infinity.

Definition 5.4.1 (Uncountability of Irrationals). The set of irrational numbers $\mathbb{Q}^c$ is **uncountable**. There is no possible bijection between $\mathbb{N}$ and $\mathbb{Q}^c$.

Example 5.4.1 (Practical/Numerical Application). Suppose we have a set A that is countable, and we know that the union $A \cup B$ is uncountable. Prove by contradiction that the set B must be uncountable.

Solution. We write a formal **Proof by Contradiction**:

- (1) Assume B is countable.
- (2) We are given that A is countable.
- (3) In Subsection 5.1 (Exercise 6), we established that the union of two countable sets is always countable.
- (4) Therefore, the union $A \cup B$ must be countable.
- (5) However, we are explicitly given that $A \cup B$ is uncountable. This is a direct contradiction!

Since our starting assumption leads to a contradiction, the assumption must be False. We conclude that B must be uncountable. $\square$

Example 5.4.2 (Abstract Applied Mathematics Application). In quantum computing and advanced physical simulations, system states are modeled using continuous wave functions. Show why these physical systems cannot be fully simulated step-by-step on a digital computer.

Solution. A digital computer is a discrete machine. Its set of all possible operational steps and memory states is countable (since computer memory is constructed from finite bit combinations).

However, a physical continuous wave function exists across a continuum of states, which relies on the uncountable set of irrational numbers. Because the set of physical states is uncountable ($\mathbb{Q}^c$), while the computer's simulation steps are countable ($\mathbb{N}$), no digital computer can ever compute a perfect, step-by-step simulation of the physical system without omitting an infinite number of intermediate states. $\square$

Example 5.4.3 (Generalization to More Variables). Let us define a multi-dimensional coordinate space where the coordinates are irrational numbers: $\mathbb{Q}^c \times \mathbb{Q}^c$. Show that this 2D space is uncountable.

Solution. Suppose $\mathbb{Q}^c \times \mathbb{Q}^c$ was countable.

If the Cartesian product of two sets is countable, then their individual projections onto the coordinate axes must also be countable (since the projection of a countable product is a subset, which is countable).

But the projection of $\mathbb{Q}^c \times \mathbb{Q}^c$ onto the x-axis is exactly the set of irrational numbers $\mathbb{Q}^c$, which we know is uncountable.

This contradiction proves that the multi-variable space $\mathbb{Q}^c \times \mathbb{Q}^c$ must be uncountable. This generalizes the uncountability of a 1D line to higher-dimensional spaces. $\square$

Example 5.4.4 (Pure Abstract Discussion). Use the uncountability of the real numbers $\mathbb{R}$ (which we will prove in 5.5) and the countability of the rational numbers $\mathbb{Q}$ to prove that the set of irrationals $\mathbb{Q}^c$ is uncountable.

Solution. We write a formal proof using our set relations:

- (1) Recall that the set of real numbers $\mathbb{R}$ is the union of the rational and irrational numbers:

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$$

- (2) We assume for the sake of contradiction that the irrationals $\mathbb{Q}^c$ is countable.
- (3) We already proved in Subsection 5.1 that the rational numbers $\mathbb{Q}$ is countable.
- (4) If both $\mathbb{Q}$ and $\mathbb{Q}^c$ were countable, then their union $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$ would be the union of two countable sets.
- (5) Therefore, the set of real numbers $\mathbb{R}$ would be countable.
- (6) But this contradicts the fundamental theorem that $\mathbb{R}$ is uncountable.

Thus, our assumption must be False. The set of irrational numbers $\mathbb{Q}^c$ is uncountable. $\square$

Theorem 5.4.1. *The set of irrational numbers $\mathbb{Q}^c$ has a strictly larger cardinality (size) than the set of rational numbers $\mathbb{Q}$.*

Example 5.4.5 (Bridging Verification for Theorem 5.4). Imagine a stadium ($\mathbb{R}$) filled with fans. We know two facts:

- (1) The stadium has an uncountable number of total seats (as we will prove in 5.5).
- (2) The fans wearing blue shirts ($\mathbb{Q}$) can all be seated in a countable bakery queue.

If the fans wearing red shirts ($\mathbb{Q}^c$) were also countable, combining two queues would mean the entire stadium is countable! Since the stadium is uncountable, the red-shirted irrational fans must vastly outnumber the rational ones.

Proof of Theorem 5.4. We prove this by comparing set cardinalities. The real number line is the disjoint union of rational and irrational numbers: $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$. In Subsection 5.1, we proved that $\mathbb{Q}$ is countably infinite (cardinality $\aleph_0$). In Subsection 5.5, we prove via Cantor's Diagonal Argument that $\mathbb{R}$ is uncountable (cardinality $c = 2^{\aleph_0}$). Assume for the sake of contradiction that $\mathbb{Q}^c$ is countable. By the standard theorem of set algebra, the union of two countable sets is countable. Thus, $\mathbb{Q} \cup \mathbb{Q}^c = \mathbb{R}$ would be countable. This contradicts the established uncountability of $\mathbb{R}$. Therefore, $\mathbb{Q}^c$ must be uncountable, meaning its cardinality strictly exceeds that of $\mathbb{Q}$ ($|\mathbb{Q}^c| > |\mathbb{Q}|$). $\square$

Exercises 5.4

- (1) State the difference between countably infinite and uncountably infinite.
- (2) If S is uncountable and $S \subseteq T$, what can you say about the countability of T ? Prove your answer.
- (3) If T is countable and $S \subseteq T$, what can you say about the countability of S ?
- (4) Use the algebraic union rule to explain why the real numbers cannot be made of only rational fractions.
- (5) Let A be the set of all transcendental numbers (numbers like π and e that are not roots of algebraic equations). Given that the algebraic numbers are countable, what does this imply about the countability of transcendental numbers?
- (6) Why is it incorrect to say that "since both $\mathbb{Q}$ and $\mathbb{Q}^c$ are infinite, they must be the same size"?
- (7) Let $S = \{x \in \mathbb{Q}^c : 0 < x < 1\}$. Is S countable or uncountable? Explain.
- (8) Show that if we add a countable set to an uncountable set, the union remains uncountable.

- (9) If we remove a countable set of numbers from the real line, why does the remaining set still contain almost all the original numbers?
- (10) Discuss the concept of "measure zero" in simple terms: why do the rational numbers take up "zero width" on the number line compared to the irrationals?
- (11) Let $f : A \rightarrow B$ be an injective function. If A is uncountable, prove that B must be uncountable.
- (12) Let $A \times B$ be uncountable. Is it guaranteed that at least one of A or B is uncountable? Explain.
- (13) Show that the set of all points in a 2D plane with at least one irrational coordinate is uncountable.
- (14) Explain why we cannot use Cantor's diagonal zigzag path (from 5.1) on the set of irrational numbers.
- (15) Why does the uncountability of $\mathbb{Q}^c$ suggest that the vast majority of real numbers are decimals that never repeat or terminate?
- (16) If we select a random real number on the number line, what is the probability that we select a rational fraction? (Hint: Think about their relative sizes).
- (17) Let $A \subseteq \mathbb{R}$. If $\mathbb{R} \setminus A$ is countable, prove that A is uncountable.
- (18) Discuss the philosophical impact of Cantor's discovery that there are different "sizes" of infinity.
- (19) Show that the set of irrational numbers cannot be written as a sequence $s_1, s_2, s_3, \dots$.
- (20) Summary: Explain why the countability of $\mathbb{Q}$ and the uncountability of $\mathbb{Q}^c$ means that fractions are just tiny islands in an endless ocean of irrational numbers.

5.5 The Uncountability of Reals

We now arrive at the grand conclusion of our transition course. The set of all real numbers $\mathbb{R}$ is the union of the rational fractions and the irrational decimals:

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$$

In this final section, we will prove that the interval of real numbers $[0, 1]$ is uncountable. This proof, called **Cantor's Diagonal Argument**, is one of the most beautiful and celebrated demonstrations in all of mathematics.

Finally, we will discuss a deep, critical limit of our primary tool: because the real numbers are uncountable, **mathematical induction can never be used to span across the continuous real line**. Induction requires a discrete, step-by-step backbone ($\mathbb{N}$), which is completely absent in a continuous, uncountable space ($\mathbb{R}$).

Definition 5.5.1 (Cantor's Diagonal Argument). A proof method developed by Georg Cantor in 1891. It proves that any attempt to list the real decimals in a sequence:

$$r_1, r_2, r_3, r_4, \dots$$

is guaranteed to fail, because we can always construct a brand new decimal that is missing from our list.

Example 5.5.1 (Practical/Numerical Application). Suppose a student claims to have listed all real decimals between 0 and 1 in a table. Let the first four numbers in their list be:

$$r_1 = 0.1359\dots$$

$$r_2 = 0.2468\dots$$

$$r_3 = 0.8888\dots$$

$$r_4 = 0.1234\dots$$

Construct a new decimal $D = 0.d_1d_2d_3d_4\dots$ that is guaranteed to be missing from the student's list.

Solution. To guarantee that our new decimal D is different from every number in the list, we make sure that the i -th digit of D is different from the i -th digit of the i -th number in the list (the diagonal digits):

- Compare with r_1 : The first digit of r_1 is **1**. We choose our first digit d_1 to be different, say $d_1 = 5$. Thus, $D \neq r_1$.
- Compare with r_2 : The second digit of r_2 is **4**. We choose our second digit d_2 to be different, say $d_2 = 5$. Thus, $D \neq r_2$.
- Compare with r_3 : The third digit of r_3 is **8**. We choose our third digit d_3 to be different, say $d_3 = 5$. Thus, $D \neq r_3$.
- Compare with r_4 : The fourth digit of r_4 is **4**. We choose our fourth digit d_4 to be different, say $d_4 = 5$. Thus, $D \neq r_4$.

Our new decimal is:

$$D = 0.5555\dots$$

Because D differs from r_1 in the first decimal place, from r_2 in the second, and so on, D cannot be equal to any number on the list. The student's list is incomplete. $\square$

Example 5.5.2 (Abstract Applied Mathematics Application). In computer network graphics, an analog signal varies continuously over time. Show why this analog signal possesses an infinite degree of information that can never be fully captured by any digital sampler, regardless of how fast we sample.

Solution. An analog signal is modeled as a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$.

Any digital recording device captures samples at discrete time steps (such as 44,100 times per second). The set of all these digital sample ticks is countable (since they can be indexed $1, 2, 3, \dots$).

However, because the time domain $\mathbb{R}$ is uncountable, between any two sample ticks (no matter how microscopically close), there lies an uncountable continuum of real time states. Because the continuous time domain is uncountable, the digital sampler is mathematically forced to omit an infinite, uncountable amount of physical signal information. Analog-to-digital conversion is always an approximation. $\square$

Example 5.5.3 (Generalization to More Variables). Suppose we have a multidimensional continuous space $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$. Prove that any mathematical proposition that is isomorphic to $\mathbb{R}^n$ must not be inductive.

Solution. A mathematical structure is inductive if it can be mapped to the discrete, step-by-step backbone of the natural numbers $\mathbb{N}$.

For a space to be inductive, its elements must be countable, so we can define a sequential successor path $S(n) = n + 1$ that visits every state.

However, since $\mathbb{R}$ is uncountable, any product space $\mathbb{R}^n$ containing $\mathbb{R}$ as a coordinate axis must also be uncountable.

Because $\mathbb{R}^n$ is uncountable, there is no possible bijection between $\mathbb{N}$ and $\mathbb{R}^n$. There is no way to queue the coordinates, meaning we cannot define any step-by-step successor map that spans the space. Therefore, any proposition depending on a continuous variable in $\mathbb{R}^n$ is completely unreachable by mathematical induction. $\square$

Example 5.5.4 (Pure Abstract Discussion). Discuss the logical consequence of the Continuum Hypothesis: is there any infinite set size that lies strictly between the countable integers $\aleph_0$ and the uncountable reals $c = 2^{\aleph_0}$?

Solution. The Continuum Hypothesis (CH), formulated by Georg Cantor, asserts that there is no set whose size lies strictly between the size of the integers and the size of the real numbers.

In 1963, Paul Cohen proved a mind-bending result in mathematical logic: the Continuum Hypothesis is ****independent**** of the standard axioms of set theory (ZFC). This means we can neither prove CH nor disprove it using our basic rules.

We can choose to assume CH is True as a new axiom, or assume it is False, and both choices lead to completely consistent, valid systems of mathematics. This highlights how modern logic explores multiple parallel mathematical universes. $\square$

Theorem 5.5.1 (Cantor's Theorem). *The interval of real numbers $[0, 1]$ is uncountable.*

Example 5.5.5 (Bridging Verification for Theorem 5.5). Let us visualize why a "complete list" of decimals in $[0, 1]$ always fails. Imagine a computer prints a list of 3 decimals:

$$r_1 = 0.\mathbf{3}14\dots$$

$$r_2 = 0.8\mathbf{6}7\dots$$

$$r_3 = 0.50\mathbf{2}\dots$$

We build a new decimal $D = 0.d_1d_2d_3\dots$ by looking at the diagonal digits (**3, 6, 2**) and changing each one to 5:

$$D = 0.555\dots$$

Why is D missing from the list? It differs from r_1 at digit 1 ($5 \neq 3$), from r_2 at digit 2 ($5 \neq 6$), and from r_3 at digit 3 ($5 \neq 2$). No matter how infinite the list is, the diagonal trick always creates an unlisted decimal!

Proof of Theorem 5.5. We proceed by contradiction. Assume the interval $[0, 1]$ is countably infinite. Then there exists a bijection $f : \mathbb{N} \rightarrow [0, 1]$ that allows us to list every real number in the interval as a sequential queue:

$$\begin{aligned} f(1) &= 0.d_{11}d_{12}d_{13}d_{14}\dots \\ f(2) &= 0.d_{21}d_{22}d_{23}d_{24}\dots \\ f(3) &= 0.d_{31}d_{32}d_{33}d_{34}\dots \\ &\vdots \end{aligned}$$

where each digit $d_{ij} \in \{0, 1, \dots, 9\}$. To prevent dual representations (e.g., $0.1999\dots = 0.2000\dots$), we enforce the standard convention that terminating decimals are written with trailing zeros rather than trailing nines. We now construct a new real number $y = 0.y_1y_2y_3\dots$ by defining its n -th decimal digit y_n based on the n -th diagonal digit d_{nn} of our list:

$$y_n = \begin{cases} 4 & \text{if } d_{nn} \neq 4 \\ 5 & \text{if } d_{nn} = 4 \end{cases}$$

By construction, for every natural number $n \in \mathbb{N}$, the number y differs from the listed number $f(n)$ at the n -th decimal place ($y_n \neq d_{nn}$). Furthermore, since y consists entirely of digits 4 and 5, it contains no trailing nines or zeros, avoiding dual representation ambiguity. Therefore, $y \in [0, 1]$, but $y \neq f(n)$ for any $n \in \mathbb{N}$. This means y is missing from our queue, contradicting the assumption that f is surjective. Thus, no bijection with $\mathbb{N}$ exists, and $[0, 1]$ is uncountable. $\square$

Exercises 5.5

- (1) Explain the core logic of Cantor's Diagonal Argument.
- (2) Let a list of decimals be:

$$\begin{aligned} r_1 &= 0.2500\dots \\ r_2 &= 0.1732\dots \\ r_3 &= 0.9999\dots \end{aligned}$$

Construct a decimal $D = 0.d_1d_2d_3\dots$ that is not on this list.

- (3) Why does the existence of the missing decimal D prove that $[0, 1]$ cannot be countable?
- (4) Show that the set of all real numbers $\mathbb{R}$ is uncountable because it contains the uncountable interval $[0, 1]$.
- (5) Why is it impossible to perform mathematical induction directly over the interval $[0, 1]$?
- (6) Explain why the successor map $S(x) = x + 1$ fails to step through the real numbers. What numbers does it skip?
- (7) If we try to define a real successor as "the next largest real number," explain why this violates the density property of $\mathbb{R}$ (from 5.3).
- (8) Show that the set of all infinite binary sequences (sequences of 0s and 1s) is uncountable using a diagonal argument.
- (9) Let $P(x)$ be a statement about a real variable $x \in [0, 1]$. If $P(0)$ is True, and $P(x) \implies P(x + 0.01)$ is True, why does this fail to prove $P(x)$ for all real numbers in the interval?
- (10) What is the cardinality (size) of the real numbers denoted by in set theory? (Hint: c or $2^{\aleph_0}$).

- (11) Prove that the open interval $(0, 1)$ is isomorphic to the entire real line $\mathbb{R}$ by using the tangent function $f(x) = \tan(\pi(x - 1/2))$.
- (12) Based on Exercise 11, discuss how a tiny finite line segment can contain the exact same number of points as the entire infinite universe.
- (13) Let A be a countable set of real numbers. Why is the complement $\mathbb{R} \setminus A$ guaranteed to be dense and uncountable?
- (14) Why does any digital audio format (like MP3 or WAV) always lose some sound details compared to physical analog sound waves?
- (15) Explain why the real number system $\mathbb{R}$ behaves like a continuous "continuum," while the natural numbers $\mathbb{N}$ behave like discrete "islands."
- (16) If we cannot use induction on the real numbers, how do mathematicians prove theorems in real analysis? (Hint: Limits, continuity, and topology).
- (17) Show that the set of all subsets of natural numbers, $\mathcal{P}(\mathbb{N})$, is uncountable.
- (18) Discuss the Continuum Hypothesis in your own words.
- (19) Why is the journey from basic counting (Section 2.1) to Cantor's diagonal argument (Section 5.5) considered one of the greatest intellectual achievements of humanity?
- (20) Summary: Explain why mathematical induction is an incredibly powerful tool for discrete structures, but is completely powerless against continuous systems.